ECONOMIC AND SOCIAL COMMISSION FOR ASIA AND THE PACIFIC

ATLAS OF MINERAL RESOURCES OF THE ESCAP REGION

VOLUME 10

CAMBODIA

EXPLANATORY BROCHURE

UNITED NATIONS

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ห้องสมุดกรมทรัพยากรธรณี
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FOREWORD

Geological studies and mineral investigations carried out in the latter half of the nineteenth century and then more intensively in the 1950s, 1960s and early 1970s have indicated potential for the existance of significant mineral deposits in Cambodia. These indications include sapphires, rubies, gold, base metals and silver, silica sands, lateritic bauxite and manganese, kaolin, pagodite, phosphorite, coal, peat, ornamental stones, construction materials and other minerals. Despite the findings and encouraging results of these studies and investigations, the mining industry in Cambodia remains undeveloped, geological archives destroyed and lost and an overall information on the geology and mineral endowement of the country in English is missing.

Volume 10 of the ESCAP Mineral Atlas Series represents the results of the intensive literature research and compilation work, which has been undertaken by the Economic and Social Commission for Asia and the Pacific (ESCAP) secretariat to support the efforts of the Government of Cambodia in evaluating the possibilities of the development and reconstruction of mineral-based industries of the country through inventoring and reassessing the general mineral abundance of Cambodia, and the geographical distribution of mineral potential and geologic potential for target minerals.

As part of its efforts to support the reconstruction process in Cambodia, the United Nations Transitional Authority in Cambodia (UNTAC) in cooperation with the ESCAP Natural Resources Division and the Department of Geology and Mines of Cambodia organized in 1993 a series of three Expert Working Group Meetings in Phnom Penh on Assessment of the Mineral Resources of Cambodia on 5-6 May 1993, on Assessment of Hydrocarbon Potential of Cambodia on 17-18 June 1993 and on Institutional and Legal Aspects of Exploration, Mining and Exploitation of Mineral and Hydrocarbon Resources in Cambodia on 21-22 July 1993. These meetings have provided an opportunity for Cambodian geologists, petroleum companies and other interested parties to review and update knowledge on new mineral occurrences and to reassess potential in the light of current conditions. The list of major technical papers presented at these meetings appears as appendix 1 of this volume.

Like earlier volumes of the ESCAP Mineral Atlas series – Viet Nam (Volume 6) and Lao People's Democratic Republic (Volume 7) issued in 1990 this publication aims at furthering the understanding of the geology, tectonic setting, magmatism and metallogeny of Indochina through an overview of relevant geoscientific sections in Cambodia.

The publication consists of this explanatory brochure and two atlas sheets, a geological and a mineral resources map at a scale of 1:1,500,000. The explanatory brochure contains a review of geological and mineral exploration work, an overview of major tectonic and structural units of Cambodia and geological descriptions of the stratigraphy, magmatism and associated metallogeny.

The review of mineral deposits and occurrences, either metallic or non-metallic, is given irrespective of their current economic significance and provides information on their distribution patterns, geographic location, contained commodities, their relation to geologic environment, tectonic and magmatic control of mineralization, chemical and mineralogical composition, genesis of deposits and other characteristics. The chapter is followed by the index of mineral deposits and occurrences and extensive bibliographic references to guide the interested professional reader.

The ESCAP secretariat views this publication as one of the initial and necessary steps in generating interest among geoscientists, the international mining community and financial institutions in further research on the geology and mineral resources of Indochina and investment promotion in the development of mineral-based industries of Cambodia.

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I. INTRODUCTION

Cambodia occupies the south-central region of the Indochina peninsula. The country covers an area of 176,500 square kilometres and it is bounded on the north by Thailand and the Lao People's Democratic Republic, on the southeast and south by Viet Nam, and on the west by Thailand. On its southwest flank Cambodia has a coastline of approximately 250 km facing the Gulf of Thailand.

Of the country's surface, approximately 30,500 square kilometres (17.3 per cent) is arable, and 133,700 square kilometres (75.7 per cent) is forest and jungle. The remainder (7 per cent) is covered by lakes, marshes and rivers (5).

The central parts of Cambodia are occupied by the valley of the Mekong River which flows south from the Lao People's Democratic Republic in the north, entering Cambodia at the Khone Falls. The mature, broad river then traverses Cambodia and exits southward to its delta region in Viet Nam. The west central region is dominated by the broad valley of the Tonle Sap drainage system, including the great lake of Tonle Sap and the Tonle Sap river which joins the Mekong near Phnom Penh.

The topography is marked by regions of notable highlands, especially in the western region and in northern and eastern border areas. In the west, the Cardamom Highlands, also known as the Kravanh Range, extend approximately from the latitude of Pailin southward to Kampot, and from the region of Pursat west to the coast, and reach heights of over 1500 metres. The northern border with Thailand is marked by the watershed of the Dangrek range, a linear escarpment rising locally to heights over 700 metres which marks the southern edge of the Khorat plateau of eastern Thailand. In the northeast, the hills and high plateau of the Kontum massif of central Viet Nam extend into Cambodia, reaching heights of over 1500 metres above the level of the Mekong plains. In the east, some high basalt plateaus locally achieve heights over 1000 metres near the Vietnamese border.

The country's climate is tropical, dominated by the seasonal swing between the effects of the wet southeast monsoon and a dry north-east monsoon season. The result is that the western highlands offer a rainforest climate with a relatively short dry season, and the remainder of the country has a tropical savannah climate, similar to eastern Thailand and the Lao People's Democratic Republic, with a longer dry season in winter.

Principal towns, places and transportation routes are shown on Figure 1.

For a better understanding of certain words, which appear frequently in the text their meanings are provided below:

(a) O = creak

(b) Stoeng = river

(c) Phnom = mountain

(d) Phum = village

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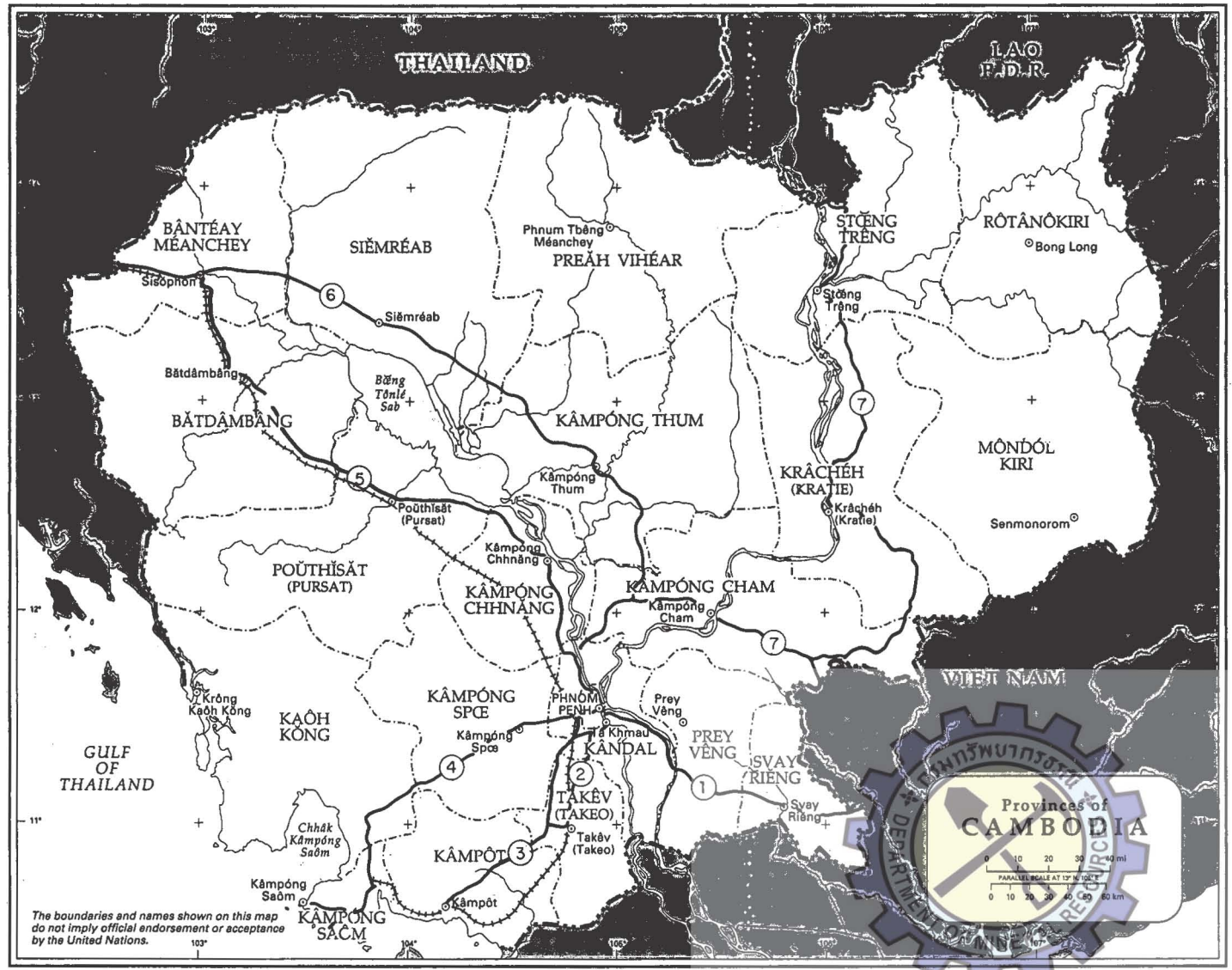

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Figure 1. Provinces and principal transportation routes of Cambodia

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II. REVIEW OF GEOLOGICAL WORK IN CAMBODIA

The history of geological research and publication in Indochina dates from the latter half of the nineteenth century, beginning in various irregularly published papers on special topics in paleontology and mineralogy. The first geological map of Indochina is said to be that of E. Fuchs and E. Saladin, published at a scale of 1:400 000 in 1882. This was complemented by A. Petiton's "Geologie de l'Indochine", the first general work on the geology of the peninsula, published in Paris in 1895, some twenty years after its completion (16).

In 1898 the Service Geologique de l'Indochine was founded and geological work in the region proceeded in a more organized manner, although efforts for many years were limited to the coastal regions in what is now Viet Nam. A notable worker of the early decades of this century was H. Mansuy, who wrote extensively on the paleontology of Indochina and established the geochronological framework for future geologists (16). In the 1920s organized field work spread into the Lao People's Democratic Republic and Cambodia under the directorship of Ch. Jacob (34). From 1925 onwards a commitment was made to the completion of a geological map of Indochina at 1:500 000 scale and this work though interrupted by war was almost completed in 1945. All sheets were finally published by 1963. In this task, which absorbed the efforts of a large number of French geologists over decades of time, the work of E. Saurin in central and southern Viet Nam and eastern Cambodia and of J. Gubler in western Cambodia are of special note. Their reports were both published in 1935. Other notable workers during this period were J. Blondel who replaced Jacob as director and who published much research on the tectonics of Indochina, ancient volcanism, and on a variety of mineral deposits, and J. Fromaget whose interest was in the stratigraphy, structure and tectonics of the peninsula. The understanding of the structural and geological evolution of Indochina, though modified by later workers to various degrees, owes its founding to Fromaget's seminal efforts. This work was brought together in his monograph "l'Indochine francais, sa structure geologique, ses roches, ses mines, et leurs relations possible avec la tectonique", published in 1941. This paper, Fromaget's geological map of Indochina at 1:2 000 000 scale (1956) and E. Saurin's 140-page chapter "Indochine" in the "Lexique Stratigraphique Internationale" (1956) perhaps represent the final synthesis of the geological understanding of Indochina as generated during the French colonial period (34).

The geological literature of Indochina up to 1950 is largely the product of the efforts of the Service Geologique de l'Indochine, published in its own Bulletin and Memoir series, but also in the Bulletin de la Societe Geologique de France, in the Comptes Rendus de L'Academie des Sciences, and in other publications (16). A bibliography compiled by H. Fontaine in 1973 lists 186 pages of references on the geology of Indochina up to 1972, most of which was compiled during the French era (15).

Commencing in 1953, the newly independent countries of Indochina founded their own national geological services. In Cambodia, new reconnaissance mapping was carried out in cooperation with French geologists of the Bureau de Recherches Geologiques et Minieres, during the years 1966-1970. Among these workers, the names of J . P. Contri, O . Dottin, and P. Fleuriot de Langle are perhaps most prominent. The mapping was published in 14 sheets at a scale of 1:200 000 with accompanying notes in 1972-1973, but these unfortunately have never been widely released for public use. A mineral resources map (1:1 000 000 scale) was produced at the same time, with a report on mineral occurrences (21).

Other prominent geologists working in Cambodia during the post-independence period have included J.P. Carbonnel (Quaternary geology and geochronology), H. Fontaine (paleontology, general geology), the team of R. Strzetelsky and A. Tokarski (petroleum prospection), and E. Saurin (paleontology, economic geology). Details of their bibliographies are cited by Fontaine (15) and Fontaine and Workman (16).

A team of Chinese geologists in 1960-1962 conducted studies of potential mineral resources in the Rovieng, Battambang, Kompong Thom areas (10). Their reports are not widely available, however.

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In the post-independence period the General Department of Geology and Mines of Viet Nam has been an active force in the region. The first major effort of this organization was the completion of the 1:500 000 geological map of North Viet Nam in 1965, incorporating new work as well as inherited but unpublished work of the former Service Geologique de l'Indochine. Meanwhile, work in South Viet Nam, the Lao People's Democratic Republic and Cambodia was continued by French and other foreign geologists, mainly through technical assistance programmes. After 1975 Vietnamese geologists continued work in the Lao People's Democratic Republic, Cambodia and southern Viet Nam in cooperative programmes with local geologists of these countries; this work culminated in 1988 with the publication of the 1:1 000 000 scale Geological Map of Cambodia, the Lao People's Democratic Republic and Viet Nam (33) compiled by Phan Cu Tien and a large body of co-workers. This map, which is the best current summation of the geology of the region is accompanied by the volume "Geology of Cambodia, Laos and Viet Nam (1988), revised 1991, edited by Tien (34).

Important review papers for those interested in Cambodian geology and mineral resources are "Review of the Geology and Mineral Resources of Kampuchea, Laos and Viet Nam" by H. Fontaine and D.R. Workman (1978) (16), and "Mineral Resources of the Lower Mekong Basin and Adjacent Areas of Khmer Republic, Laos, Thailand and Viet Nam", by D.R. Workman (1972) (41).

The Department of Geology and Mines of Cambodia is the agency currently responsible for geological work in Cambodia. Its efforts in the recent past have been mainly devoted to the geology of potential useful resources including gemstones, kaolin, phosphate, groundwater and other materials of value for the requirements of the national reconstruction process.

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III. GENERAL GEOLOGY OF CAMBODIA

The general geology of Cambodia is presented in the Geological Map of Scale 1:1500 000, which is adapted from the Geological Map of Kampuchea, Laos and Vietnam, Scale 1:1 000 000 (1988) of Phan Cu Tien and others. In this section the stratigraphic units represented in the geologic succession are described in geochronological order, followed by a section on the intrusive igneous units. In subsequent sections, these stratigraphic units and igneous members are placed in a tectono-metallogenic framework so that the mineral resources data can be understood within the metallogenetic history of the country.

The distribution of major stratigraphic – structural groups is shown in Figure 2, which is adapted from H. Fontaine's and D.R. Workman's (1978) "Review of the Geology and Mineral Resources of Kampuchea, Laos and Viet Nam". This map also indicates by name some of the major geologic and geographic features referred to in this Atlas review. Figure 3 summarizes the major geologic-geographic provinces of Cambodia.

A. Stratigraphy

The stratigraphy of Cambodia is represented by sedimentary formations, metamorphic units and igneous bodies extending from the Pre-cambrian through to Quaternary in age. The following summary, synthesized from a number of available references (16, 19, 33, 34, 43), presents the major units and their historical interpretation in geochronological order.

1. Precambrian (PR)

The Precambrian is not known directly in Cambodia, although there are numerous observations of Pre-Indosinian (pre-220 Ma) metamorphic basement rocks which are ascribed to the Proterozoic (PR). These include the Bo Kham Series crystalline limestones and schists (43) in the north-east part of the country, which are observed along the route from Bo Kham to Pleiku, Viet Nam. Moreover, much of the Pre-Indosinian folded basement formerly considered Precambrian is now placed in the Lower Paleozoic. The dating problem of these rocks is complicated by the fact that the pervasive metamorphism of the Indosinian orogenic period apparently has re-set the geochronologic clocks of most pre-existing metamorphic and igneous rocks.

Such metamorphic rocks comprise large areas of the Kontum massif of central Viet Nam, which extends into the extreme north-east of Cambodia. Granodiorite, granites and migmatites in the valley of the Se San river and small areas of biotite-sillimanite gneisses and schists, along with other high-grade metamorphic rocks along the Lao and Viet Nam borders are now assigned to the Proterozoic.

Elsewhere in Cambodia, similar rocks are observed in a 40 km-long belt extending in a southeasterly trending direction from Pailin, near the western border of Cambodia, and have also been designated as Proterozoic. The Pailin crystalline rocks comprise quartzo-feldspathic gneisses generally of granodioritic composition (oligoclase andesine plagioclases), mylonites, amphibolites and migmatites (19). There is no notable mineralization associated with rocks of this age.

2. Cambrian

Cambrian argillites outcrop in a small area northwest of Stung Treng (16) dated by worm tracks and a trilobite specimen. This is the only fossil-fixed Cambrian occurrence in southern Indochina according to these writers.

Other units of schist and quartzites in the southwest and northeast of the country have been assigned a Cambro-Silurian age (see section below) by virtue of stratigraphic position and general similarityto the Dalat Series of Viet Nam. Some scattered occurrences assigned to the Cambrian have been seen 30-40 km north of Kampot, and in the region of the Laos-Cambodia border (34).

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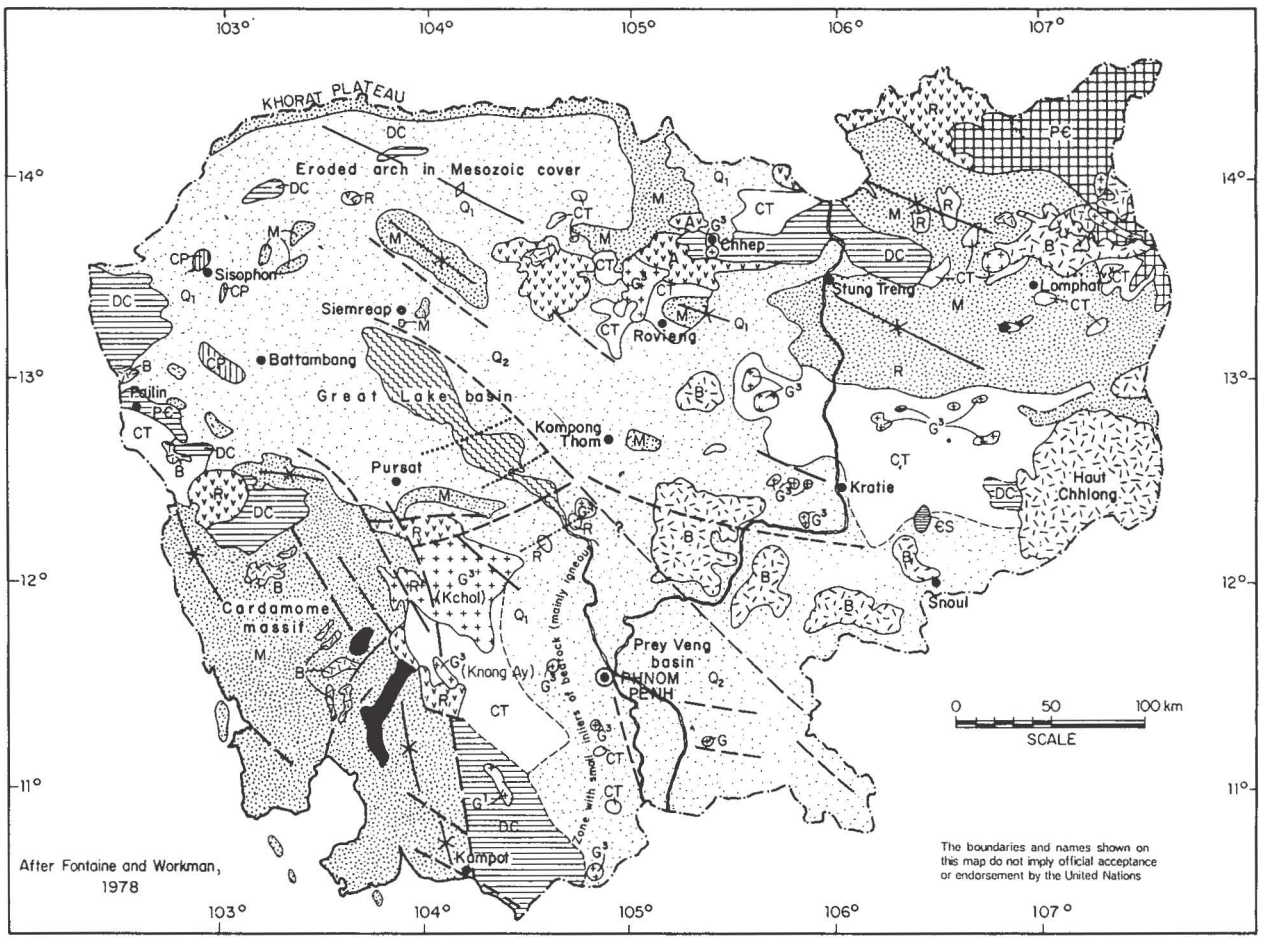

LEGEND

TERRANES

UNDEFORMED OR GENTLY FOLDED COVER STRATA

- Q — Quaternary sedimentary rocks and unconsolidated sediments. 1: Pleistocene; 2: Holocene. Includes some small Neogene basins
- B — Neogene – Quaternary platform basaltic rocks
- M — Mesozoic sedimentary units (upper Triassic-Cretaceous)
- Volcano-sedimentary units (mainly Triassic, some Paleozoic). A: andesitic; R: rhyolitic.
- C-P — Paleozoic sedimentary units (mainly Carboniferous-Permian).

ZONES OF INDOSINIAN FOLDING

- CT — Synclinorial zones in Indosinian fold-belts (mainly Carboniferous-Triassic).
- DC — Anticlinorial zones in Indosinian fold-belts (Precambrian-Silurian medium to high-grade metamorphics; Devonian-Carboniferous rocks, deformed and slightly metamorphosed).

EXPOSED BASEMENT ROCKS OF THE KONTUM MASSIF

- PC — Precambrian-Early Paleozoic granites and high-grade metamorphics.

INTRUSIVE ROCK GROUPS

- G — Acid – Intermediate intrusive suites
  - G_1 – Pre-Carboniferous
  - G_2 – Carboniferous
  - G_3 – Triassic-Jurassic
  - G_4 – Cretaceous

STRUCTURE SYMBOLS

- Regional faults, known and inferred
- Geographical lineament
- Axes of Cenozoic epeirogenic folding
- Limits of terranes, known and inferred
- Axis of swell in buried pre-Tertiary basement
- Intramontane grabens (Neogene)

Figure 2. Principal features of the geological structure of Cambodia.

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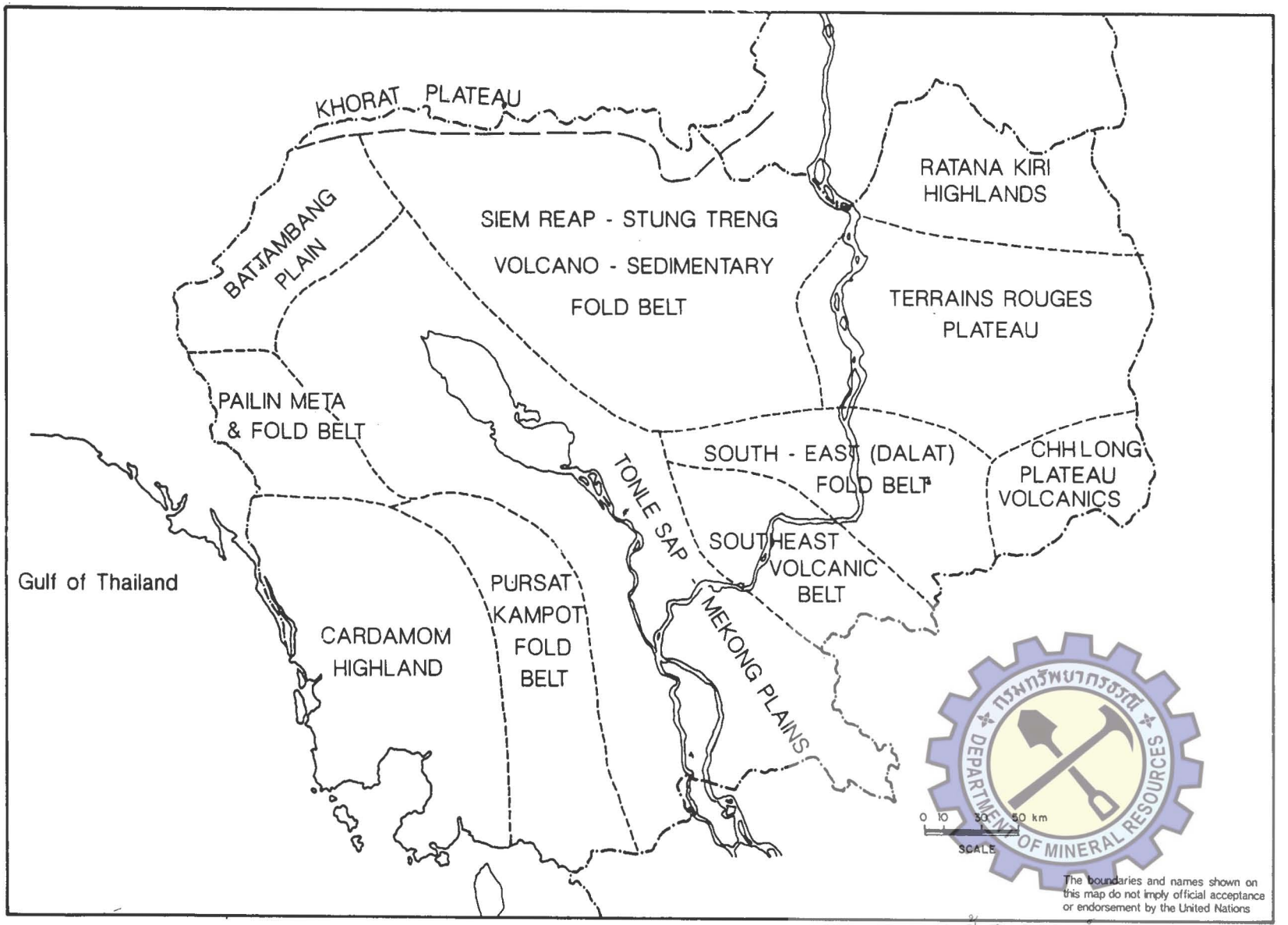

Figure 3. Geological provinces of Cambodia

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3. Ordovician – Silurian (0-S)

Rocks of definite Ordovician and Silurian ages have not been so identified in Cambodia, except as part of the general classification "Cambro-Ordovician" (see below).

4. Cambro-Ordovician

This term includes a variety of units of generally metasedimentary character, exhibiting low- to medium-grade metamorphism and weak to moderate deformation.

White to pink quartzites and associated calc-silicate rocks associated with this designation exist in the Pailin area, according to Fontaine and Workman (16) (although such do not figure on Tien's (1988) map of Indochina), who cite that these are also abundantly developed in the south of the country. Gubler (17) included a series of quartzites and schists in the southwest part of Cambodia in this group, which are probably those referred to by Fontaine and Workman, and are shown by Tien to outcrop in the hill country north and east of Kampot.

In the extreme northeast of the country, on the border with the Lao People's Democratic Republic and near the junction of the three frontiers, Saurin (28) mapped moderately metamorphosed terranes as Cambro-Silurian, considering them to be the metamorphosed equivalents of the so-called Dalat Series of Viet Nam (19). These rocks include spotted schists, micaceous schists, quartzites, and related rocks, some tourmalinized.

5. Devonian – Carboniferous (D-C)

The Devonian of Cambodia includes mainly shales, cherty shales, sandstones, marls and silicified shales; limestones are rare, and confined to the upper part of the succession. In western Cambodia, south of Pailin and in the vicinity of Pe Am Thu, the succession consists of 200-300 m of flinty cherts over which lie beds of sandy limestones of Lower Carboniferous age, as determined by foraminiferal remains. Further south, in the Cheom-Kampot area cherts are less in evidence, and the Devonian is represented by a non-fossiliferous sequence of sandstones, marls and impure limestones of discontinuous but widespread outcrop extending through Kampot province and southward into Viet Nam.

In northern Cambodia a similar sequence extends over a broad area east and west of the Mekong valley north of Stung Treng, and extending to the west of Chhep, but here the series contains some tuffs and ignimbrites. West of Sisophon there is a small area of Carboniferous sandstones, and a similar area occurs east of Kratie, on the west edge of the Haut Chhlong plateau.

The Devonian was named the "series schisto-greseuse" by Gubler (17) in respect of its characteristic sandstone-grit-shale lithologies.

The Lower Carboniferous (C) is generally separated from the Devonian by a discordance. It appears that rocks of this age represent only minor and temporary marine transgressions in the southern part of the peninsula during early Carboniferous time, as evidenced by angular unconformities within the Lower Carboniferous.

The Carboniferous (C) is seen as sandy limestone in the western part of the country near Peam Prous, overlying the Devonian (see above). Similar limestones outcrop northeast of Kampot, locally attaining thicknesses of 70 m (16); definitive dating remains incomplete, and on stratigraphic grounds these rocks may indeed be younger. Limestone is the most characteristic rock type of the Middle and Upper Carboniferous.

6. Upper Carboniferous-Permian (C_3-P)

Limestone of this age occurs as gray-black organogenic (fusulinid) carbonate rocks, often outcropping as isolated buttes rising from the surrounding plains; notably in the Rovieng – Stung Trep region. North of Stung Treng a 4 km long ridge of white-cream limestone forms a prominent exposure of this age-group.

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(19). Dolomitic limestones, locally recrystallized and containing a number of interstratified volcanic flows (19) and green carbonate beds containing unidentified volcanic rock materials are reported near Chhep (34), and the latter author also reports volcanic-bearing carbonate rocks of this age in the Kampot area.

7. Permian (P)

The marine Permian is well represented in the Pursat, Sisophon and Kampot areas by relatively pure limestones of white to grey-beige colour. Oolitic and organo-detrital varieties are especially abundant. Some intercalations of shales, marls and argillaceous grits are known but the units are essentially calcareous. Marine transgression began in mid- Permian time so these folmations are limited to the Middle and Upper Permian. The units are often deposited on the Devono-Carboniferous grits.

Near Sisophon the sequence includes, in upward order, andesitic tuffs, fossiliferous limestones, clastic beds including marls, with ferruginous and bauxitic beds and shell-rich horizons, passing upward into reefal limestones. The upper parts become more cherty in aspect.

Rhyolitic volcanism, noted especially in lower Permian (see above) apparently continued sporadically throughout the Permian, as Permian limestones are often (but not everywhere) intercalated by such volcanics (16).

Morphologically, these limestone often present a karstic topographic surface. Lateral erosion often produces a characteristic "organ-pipe" weathering pattern visible at great distances (19).

8. Triassic (T)

The Triassic in the southern part of the Indochina peninsula was marked by marine transgressions and regressions of varying length and extent. There is a general tendency for the early Triassic to be more marine in character, changing to increased continental sedimentation in the late Triassic. The Triassic was a time of widespread rhyolitic volcanism on the peninsula, especially so in the middle and late Triassic time. The late Triassic is also characterized by numerous granitic intrusions.

In western Cambodia, the Triassic sequence consists of interbeds of coarse conglomerates, blue, gray and green sedimentary breccias, sandstones, mudstones and rhyolitic tuffs. The sequence covers a large area east of Pailin, where it is up to 1000 m thick (T). Other areas of outcrop are seen near Poipet, Pursat and Kompong Speu. These beds are mostly if not exclusively continental or paralic in origin. In the extreme west, the Middle Triassic is represented by limestones grading upwards into sandstones and siltstone (Honnghe Formation, T_2).

In eastern Cambodia the Triassic marine units are more common. They have been found near Rovieng (T_2), in the Rovieng-Stung Treng belt (T_{1-2}), in the Kratie area (T, T_{1-2}) and in several areas of the north-east near the Vietnamese border (T_2). Greenish sandy shales, calcareous sandstones, green shales and marls are characteristic of the marine facies. Continental phases (sandstones, shales) intercalated with marine units have been identified and dated on plant fossil evidence in the Rovieng, Kratie, Chhlong and Phum Laak districts (T_{1-2}).

The Triassic continental sequence is host to black shales with coaly horizons in the region northwest of Stung Treng, and in Kompong Thom province. See section on Fuel Minerals.

The lower and middle Triassic (T_{1-2}) in eastern Cambodia is marked by volcanic deposits. These are most commonly rhyolites or dacites, plus their equivalent tuffs, often intercalated with marls. Locally they may predominate in the succession.

The Triassic grades upwards into the continental red beds of the "Terrain Rouge"; this is usually considered to be early to middle Jurassic although some authors (Kun and Say, 1986) have considered it to begin in mid-to late-Triassic time.

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9. Jurassic (J)

Volcano-sedimentary rocks of inferred Jurassic age occur in western Cambodia forming foothills of the Cardamom Highland in the region west of Pursat. The formation consists of light gray rhyolites, dacites, tuffs, and interbedded volcaniclastic sediments (J).

The main characteristic unit of this age is the so-called Terrain Rouge (J_{1-2}). This unit covers large areas in Mondolkiri and Ratanakiri provinces, extending westward into Stung Treng and Preah Vihear provinces, disappearing progressively towards the west, and appearing last as isolated outcrops north and west of Pursat.

The Terrain Rouge (red terrain) is so named for the abundance of red continental clastic units (red beds) comprising conglomerates, sandstones and siltstones of continental origin, which attain stratigraphic thicknesses up to, or locally exceeding 2000 m. The series is of near-shore or lagoonal origin, as evidenced by occasional intercalations of marine sediments, abundant silicified wood fossils and less common coaly strata.

The time-span represented by the series is problematic. It was considered Triassic by Fontaine and Workman (16) but Kun and Say (19) consider that it extends from post-Carnian (Upper Triassic) through the Lias (Lower Jurassic). In this report the series is assigned to the lower to middle Jurassic.

The marine units of this age are confined to the eastern limits of the country and consist of calcareous sands, silts and marls, grading up into shales, silts and sandstone, i.e. a regressive sequence, up to 1200 m in thickness (Bandon Formation).

These units are correlative with the Lower Khorat Group of Thailand.

10. Upper Jurassic-Cretaceous (J_3-K)

The term "gres superieures" (upper sandstones) is reserved for the assemblages of sandstones, grits and conglomerates of this age which constitute the broad subhorizontal highlands of western Cambodia (Cardamom Highlands, Kravanh hills, Bokor hills, etc.), northern Cambodia (the Dangrek and Koulen ranges) and numerous other tablelands and mesas rising from the western and central plains. The rocks are correlatives of the Khorat Group of Thailand (the Dangrek Range represents the south margin of the Khorat Plateau). The sequence exceeds 2000 m in thickness. The series is of continental origin and is poor in fossils, but its upper members have been dated as Lower Cretaceous on fossil pollen evidence, and it is seen to overlie the Liassic Terrain Rouge (19).

In western Cambodia, the Phu Quoc formation (for which Phu Quoc Island is the type area) is dominant as the main unit in the Cardomom, Bokor, Kravanh and Elephant highlands, comprising subtabular red coloured sandstones, siltstones and claystones. Peaks in this area rise from sea level to above 1700 m. The units have a stratigraphic thickness exceeding 2000 metres.

The Dangrek Hills which form the northern border of Cambodia with Thailand are of the same age and petrology. The most prominent members of the Khorat Group along this border are the Upper Jurassic Sao Khua and Phra Wihan Formations.

In eastern Cambodia, the Cato Formation was deposited at this time, comprising conglomerates and red sandstones, grading upward to andesites and tuffs with some dacite. Isolated outcrop areas of this formation are seen east of Stung Treng and Kratie.

11. Neogene-Pleistocene (N_2-Q)

There is little Paleocene or Miocene, if any, in Cambodia. Thus the whole of the Tertiary is represented only by Pliocene sedimentation. The Neogene-early Pleistocene is seen in large basins in eastern Cambodia (see map) where epeirogenic movements and adjustments created broad loci of Neogene -Early Quaternary sedimentary deposition. These are represented by the Bamieu Formation, comprising

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claystones and siltstones usually laid upon well-developed conglomerate horizons, indicating a paralic and coastal plain depositional facies.

Large volumes of this material fill the broad lowland grabens of the Mekong Valley and the Tonle Sap region, but are overlain by younger alluvial materials.

The deposits of this age are often referred to as the "alluvions anciens", the "Older Alluvium"; these sediments generally form terrains and plains in the levels 25-150 metres above sea level. The upper levels locally are strongly laterized and this has been used as building material (for example, at Angkor Wat). Their age is confirmed as greater than 650 000 years by the fact that the younger basalts of this age overlie the deposits. This places the Older Alluvium in the Lower Quaternary and possibly Upper Pliocene, i.e., 1.5-2.0 My.

A prominent unit of this period is the Pliocene – Quaternary plateau basalts (N_2-Q_1) which form broad dissected plateaus in several areas of the southern Indochina peninsula. In Cambodia, the most prominent areas of this formation are in the south-east (Haute Chhlong), north of Kompong Chom, and on the western highlands. Other smaller areas exist, mainly in the eastern sectors of the country. These rocks consist of basalts, alkaline basalts and high alumina basalts generally of tholeiitic affinities, with stratigraphic thicknesses of the order of 100 to 140 metres. This basalt is of interest because the weathered crusts have formed bauxites of potential economic grade and volume in some areas. They vary in age from 1.77 Ma to 2.6 Ma (Late Pliocene – Early Pleistocene).

The westernmost outcrops in Cambodia are a number of small (up to 3 km diameter) areas around Pailin, where the basalts are the host rock-source rock for rubies and sapphires (see later section on gems).

12. Quaternary

Quaternary deposits are widespread in Cambodia, especially in the broad central plains of the Mekong and Tonle Sap river systems and across the northern uplands (see map), generally occupying levels from 0 to 40 m above sea level (34).

The Quaternary occupies grabens and depressions on this broad area, with intercalation of continental and marine rocks revealing a history of periods of transgression and regression.

The Lower Quaternary (Q_1) consists of sands, silts and claystones of both fluvial and marine origin. It is here combined with the Pleistocene deposits (unit N_2-Q) for graphic convenience but is a separable unit in the subsurface. The Middle Quaternary (Q_2) Of the plains comprises red sandy sediments occupying terraces above 15 metres.

The Middle-Upper Quaternary (Q_{2-3}) is widespread in the north, southeastern and northwest sectors of Cambodia, where it is known as the Battambang Formation, composed of grits, pebbles, sands and clays outcropping on higher relief levels on the outer parts of the central plains. This unit is about 200 metres thick in the Battambang area. On the Mekong Plain the Upper Quaternary (Q_3) is recognized as the Mochoa formation, occupying the 10-15 metres terraces in areas southeast of Phnom Penh.

The Holocene (Q_4) is represented on the coastal and interior plains, and in small upland valleys by recent deposits of fluvial, lacustrine, paralic and shallow-sea origin. The central valley of Cambodia is occupied by the Tonlesap Formation, comprising gray silts, sands and clays containing organic remains. Upland valleys are marked by fluviotile fill materials of recent origin.

Quaternary plateau basalts (QB) of Middle-Upper Pleistocene age (approximately 650 000 years plus or minus 30 000 years) and of generally alkaline olivine compositional range occupy a few broad areas in the south central part of the country (southeast of Kompong Chom).

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B. Intrusive rocks

1. Paleozoic High Alumina Granite

These granites of Early-Middle Paleozoic age are principally exposed in Cambodia in the Preas massif, some forty km north of Kampot, and in a number of related small stocks and plugs. The Preas granite is a sheared acidic biotite granite (16). The Bokham granite, which outcrops in the Se San valley and in many smaller stocks and batholiths in the northeast of Cambodia is dated as Indosinian (227 My) but may be of much greater age. Granites of this group are characteristically two-mica biotite granites with accessory cordierite and garnet. (34, 19).

2. Permo-Triassic Diorite-Granodiorite Association

These intrusives comprise a suite of rocks of intermediate compositions from the gabbro to granite range. The largest complex of this age in Cambodia lies along the northeast border with Viet Nam. The predominant rocks are equigranular biotite-hornblende granodiorites of pinkish colour (Phnom Tasson type of Tien (33, 34)).

This age group also includes a large number of sub-volcanic igneous stocks and small batholiths of generally biotite granodiorite composition, probably generated by the rifting movements of the Indosinian orogeny, dotted across the northern sectors of the country.

A muscovite granite of this suite from Bokham has been dated at 227 My, which is considered the age of its intrusion, placing it as coeval with the Indosinian orogeny's latter stages.

3. Jurassic – Early Cretaceous Diorite-Granodiorite Suite (Dinhquan Complex)

This association of rocks of generally intermediate to granitic compositional range, originally described as the Dinhquan Complex and so named for its wide exposure in the namesake area in the Dalat region of Viet Nam. Stocks, plugs and batholiths of this association are widespread throughout the region east of the Mekong River, and are common in the north central regions. A major intrusive complex of this age is centred near Rovieng (the Phnom Preck massif).

The suite exhibits a wide range of radiometric ages, ranging from 64 to 172 My, but most frequently falling in the 100 to 125 My range of the Lower Cretaceous (16, 19, 34).

4. Cretaceous – Paleogene Intrusive Suite

In Cambodia, intrusive rocks of this age are almost exclusively represented in the Kchol and Knong Ay massifs, in the west of the central region of the country. These are both comprised of monzonitic biotite granite with younger aplite and micro-granitic phases. The Knong Ay massif has been dated as late as Triassic-early Jurassic on the basis of field relationships (33, 34) although the later finegrained phases give much younger ages of less than 100 My, coincident with ages of late tage differentiated intrusions elsewhere in Indochina (34, p. 109). This would indicate that the complex acted as a resurgent centre of petrogenetic activity over a long period of time throughout the Mesozoic Era.

The designation is employed to indicate a small number of intrusives of generally gabbroic to granodioritic character of Cretaceous – Paleocene age which occur in north and southwestern Cambodia. These are of the same apparent age as the last group and represent more basic phases of the coeval granodioritic – granitic complexes. The largest group of these intrusives occurs in the region northeast of the Kchol massif, south-east of the town of Pursat. These have been referred to as Bamnak Type intrusions after the namesake geographical area.

The designations for intrusive rocks applied in this section are adapted from the classification employed by Tien et al (33, 34) and used by them on their "Geological Map of Kampuchea, Laos and Viet Nam" (33).

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C. Volcanic rocks

Limited information on the distribution of volcanics is available. The major known units are the Neogene and Quarternary plateau basalts and the rhyolitic-andesitic series of the Permian and the Triassic. These are discussed above under A(11), (12), (7) and (8) respectively.

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IV. TECTONIC HISTORY OF CAMBODIA

Cambodia occupies a small central part of a much larger tectonic unit which includes the Indochina peninsula, eastern Thailand, the Lao People's Democratic Republic, the southern Malay Peninsula, western Kalimantan and the intervening areas of the Gulf of Thailand and the Sunda Sea. This crustal block, known as Indosinia, was consolidated and raised by orogenic movements culminating in the late Triassic (the Indosinian orogeny). During the Jurassic, Cretaceous and Paleogene time it was mainly an emergent area, undergoing erosion and consequent continental sedimentation. It has been modified by tectonic events beginning in Neogene times which have produced the geographic configuration seen today. Figure 4 shows a sketch map of the main units and elements of the tectonic framework.

The central Indosinian core comprising the Kontum massif of central Viet Nam and northeastern Cambodia was consolidated before the Devonian, and is now seen as a metamorphic complex. A related but separated small sector of similar rocks exists in the Pailin area of western Cambodia. Areas of Variscan (Carboniferous) folding are seen to the north and south of this core, the southern fold belt lying across north and central Cambodia.

This period of tectonism and uplift created the Paleozoic terranes of the Rovieng – Stung Treng belt of northern Cambodia and also some similar areas in southwestern Cambodia, now represented by discontinuous outcrops of Devonian-Carboniferous age. In this way the Paleozoic Indosinian land mass was enlarged and consolidated.

This early landmass, comprising central and eastern Laos, northern, western, and northeastern Cambodia, central and southern Viet Nam and parts of Thailand, is referred to as Annamia (16). Marine sedimentation proceeded around this landmass, which remained essentially stable from the Middle Carboniferous onwards, with only marginal submergences by shelf and lagoonal seas. Through Carboniferous, Permian and Triassic time these sediments accumulated. In the late Permian, episodes of strong folding began which culminated in middle Triassic time. This period of uplift, dislocation and folding marked the Indosinian orogeny, which occurred along a broad linear mobile belt trending in a northwest – southeast direction across northwesten Lao People's Democratic Republic, central Thailand, and central-western Cambodia, and thence southward into the Gulf of Thailand. A sub-parallel belt of Indosinian folding reaches across northeastern Lao People's Democratic Republic and central Viet Nam. As a result of this orogenic episode, all formations up to and including Triassic are strongly folded about (the northwest-southeast (NW-SE) axes.

Tectonic reconstructions by Sangad Bunopas (27) and co-workers indicate that the Indosinian plate and the Shan-Thai plate collided and merged in mid-Triassic time, and the closure was complete by the Late Triassic. In this interpretation, the two micro-continents almost simultaneously collided and fused with the southern margin of the South China plate. The collision of the Shan-Thai plate and the Indosinian plate was instrumental in causing the general folding of continental margins, the volcanism and the widespread intrusions of granitic rocks associated with what is now seen as the Indosinian orogeny of Indochina. As well, in their view, the collision and consequent generation of a new linear subduction zone paralleling the west edge of the Shan-Thai block signalled the intrusion of the long belt of tin-bearing granite bodies of Triassic to early Jurassic age which extends from eastern Myanmar through southern Thailand, and Malaysia, to northwest Indonesia.

Following the final stages of the Indosinian orogeny (middle-late Triassic time), there was a general emergence of the land surface of the Indosinian subcontinent, which now extended from southern China as far south as western Kalimantan and to the region of Bangka and Belitung Islands in the Sunda Sea. Beginning in the Late Triassic (Norian stage), erosion of the great highlands created by the Indosinian orogeny laid down molasse-facies sediments over broad areas of the subcontinent. Volcanic activity was also locally extant. The locus of the greatest sedimentation was more or less in the centre of the Indosinian craton, where the vast fluvial valleys and deltas received continental/paralic clastic sedimentationthroughout the Jurassic and Cretaceous, culminating in episodes of evaporite precipitation in the Late Cretaceous Paleogene times, the result of a short period of marine incursions.

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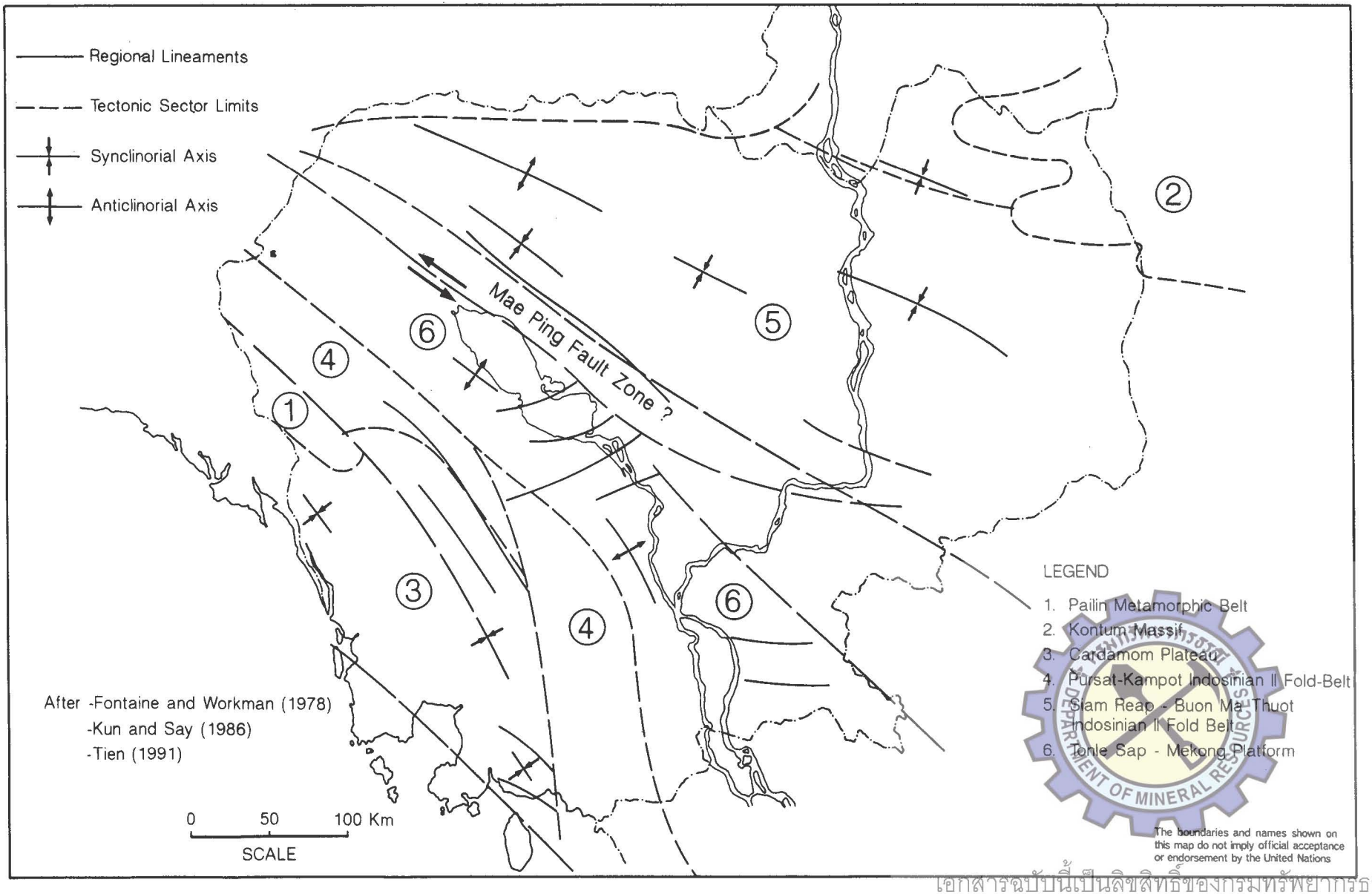

Figure 4. Principal tectonic elements and sectors of Cambodia

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Thus were created the extensive sandstones, conglomerates and siltstone deposits of the Khorat plateau, the Cardomom highlands and the related highlands of western Cambodia. The shift from marine to continental sedimentation is reflected also in the Bandon Formation of eastern Cambodia, of Lower to Middle Jurassic age, which terminates in its upper horizons in the continental "Terrain Rouge" sediments. It passes upwards into the continental Cato Formation (near Lomphat), which is correlative with the Khorat group and its equivalents in western Cambodia, but carrying considerable volcanic material. The Jurassic-Cretaceous continental sedimentation obviously at one time continuously covered a vast area of Thailand, Cambodia and southern Lao People's Democratic Republic with a great thickness of molasse-facies clastic sedimentary formations, covering many tens of thousands of square kilometres. This great mass of sediment is the result of the uplift and displacement, created by the Indosinian orogenic period and the long erosional interval which followed it .

It has been remarked (43) that during the Mesozoic the Indosinian terrane appears to have acted first in the Triassic as a foreland basin (during sedimentation) and later as a back-arc magmatic area in the Jurassic and Cretaceous, as evidenced by the larger number of granitoid intrusions in central and eastern Cambodia (see map) emplaced in this period. This is considered to be the main period of endogenic metallogeny in the country (see following section).

The final evolution of the Indosinian block was effected during the Cenozoic Era. Throughout this time the block, which had now achieved relative crustal stability, reacted as a single plate unit to the major tectonic forces at work in the larger surrounding region, namely sea-floor spreading to the east, the movement of the Shan-Thai plate northward and the development of the Shan Boundary Cretaceous-Tertiary orogenic system to the west and the Malay-Burma –Java geosynclinal basin to the southwest and south. The northern boundary lies against the NW-SE suture crossing north Viet Nam and southwestern China (the Red River suture zone).

Major translational and wrenching stresses appear to have diminished in the Paleogene. However, broadscale regional tectonic forces in Tertiary time produced major epeirogenic movements, transform faulting and plateau volcanism. These had several major effects, namely the uplift and warping of the central plate regions (including the whole Khorat basin complex), and the submergence of the southern part of the block forming the eastern part of the Gulf of Thailand and the Sunda Sea (27).

The Paleogene was the time of final thermal igneous activity on the plate (although minor geothermal hydrologic systems are still widely found) as seen in the form of many small quartzo-feldspathic intrusions of Paleogene age (50-70 Ma) throughout the peninsula (in Cambodia, the latter Knong Ay Kchol-intrusive suites most notably) and a number of small gabbroic-dioritic stocks and plugs of the same age, all probably a consanguineous sub-volcanic assemblage of now-eroded volcanic plateau basalts, their activity precipitated by tensional internal plate adjustments in early Cenozoic time.

Rifting activity began in the nascent Gulf of Thailand in Eocene time and continued through the Tertiary, beginning in the southern sector and extending northward with time, and opening up the gulf. This took the form of a widening series of intracratonic rift (pull-apart) basins with north-south orientation, resulting from east-west tensional forces. Thick sequences of mid- to late Tertiary lacustrine sediments have formed productive petroleum progenitor rocks, currently under exploration in the Gulf (32).

The Paleogene was also marked by broad long-amplitude folding and up-doming of the Mesozoic continental sequences, as seen now in the deformations of the Khorat and Cardomom highlands. As a result, large areas of these rocks were removed by erosion, and a broad area of doming about a NW-SE trending direction warped upward the pre-Quaternary rocks along the Tonle Sap – Lower Mekong axis. Probably related, the northeast-southwest (NE-SW) trending axes of swell raised cross-arches in the Pre-Tertiary basement of this central platform (See Fig. 2).

Tensional tectonic forces in the Late Cenozoic also gave rise to the formation of a number of major intermontane grabens in the Cardomom massif and Kravanh mountains, now filled with scores or even hundreds of metres of clastic fill of probable Pliocene-Pleistocene age. Smaller grabens are also seen in the Kratie and Snoul regions.

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Tensional tectonics also gave rise to the widespread eruption of plateau basalts starting in the Pliocene and continuing into Quaternary time (i.e. from approximately 5 Ma to 1.7 Ma, with most dates falling in the 1.7 to 2.6 Ma range). These basalts are important economically as source areas for gemstones and bauxite deposits (see later section). Meanwhile, the eastern and southern expanses of Indosinia gradually subsided to create the offshore shelf areas of the Gulf of Thailand and the South China Sea.

In the Quaternary the great lake basin of Tonle Sap and the central valley of Cambodia were formed by the slight subsidence of this broad area along northwest-southeast axes, leading to broad areas of thin to moderate Mid- to Late – Quaternary cover over this sector. It should also be noted that some recent workers have suggested that the Tonle Sap subsidence may be due to a major meteoritic impact event, postulated as having occurred about 700 000 years b.p., based on the ages of tektites in the Eastern Thailand-Cambodia region. This idea remains speculative at the time of writing.

In the northeast sectors, a second cycle of plateau basalts (age range 0.6-0.7 Ma) erupted along the broad east-west axes of the earlier Pliocene-Pleistocene volcanism, the younger lavas often superimposed on the older lava outcrop areas.

Since the Mid-Late Quaternary major tectonic movements have been absent, the only activity having been regional adjustments reflected in various terrace and beach levels observed between the present sea level datum and approximately 140 metres over datum.

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V. METALLOGENIC – TECTONIC EVOLUTION OF CAMBODIA

The formation of metallic and non-metallic deposits of potential economic value and of prospecting interest is commonly the result of the major orogenic cycles which have effected a given geologic terrane and more especially in the case of weathering or sedimentary deposits, of the erosional and sedimentary intervals between such orogenic cycles. In Cambodia, four such major cycles, or metallogenic epochs can be identified (43).

A. Pre-Indosinian Epoch(s) (Pre-Carboniferous)

Deposits of this epoch, for which evidence is very limited in Cambodia, will be limited to areas of the Kontum massif which extends into the north-easternmost corner of the country, of reported Proterozoic age, and to some other small local areas of exposed Pre-Indosinian folded basement.

Pre-Indosinian environments include:

1. Medium to low-grade metamorphic terraines including black schists, crystalline limestone, cherts, conglomerates and breccias, all attributed to the Devonian and Lower Carboniferous.

2. Quartzites, micaceous quartzites, graphitic and micaceous schists, metaconglomerates, attributed to the Cambro-Silurian.

3. Gneisses, schists and amphibolites of the Pailin metamorphic complex, which yield Triassic (Indosinian) dates but originated in Pre-Indosinian time.

4. Bo Kham Series granites, crystalline schists, and gneisses, dated as Triassic (220 Ma) but of much earlier origin. Similar granites and gneisses north of Kampot are attributed to the early Paleozoic. These granites and metamorphic rocks are hosts to small vein gold deposits and constitute the provenance of some local alluvial deposits of gold.

B. Indosinian Epoch (Carboniferous – Triassic)

The Indosinian orogeny occurred as a series of orogenic pulses in the Indochina peninsula, occurring in two sub-parallel mobile belts (see previous sections), not apparently everywhere active at the same time, but affecting different sectors at various periods over a long period from the mid-Carboniferous to the Carnian stage of the Triassic (upper Triassic), a period exceeding 100 million years.

During the early part of the Indosinian, the northern part of Cambodia seems to have been a subduction-related rift-zone, marked by carbonaceous sediments and andesitic volcanism. In the later part of the cycle, sedimentary rocks become generally more continental or paralic in character, reflecting uplift of mountainous terrains in this region.

Known deposits and occurrences associated with the period include limestones in western Cambodia; thin beds of bauxite (diasporic) are also intercalated in this sequence. Some small sedimentary iron deposits and a few copper occurrences northwest of Rovieng in the areas of andesitic outcrop are attributed to this orogenic cycle. The Indosinian at the present time shows little in terms of known mineralisation although the volcanic sequencies of the Permian and the Triassic should make this a most prospective Epoch.

C. Mesozoic Epoch (Jurassic – Neogene)

The Mesozoic metallogenic epoch is related to the emplacement of numerous granitoid intrusive bodies of widely varying size and age which were emplaced in eastern, central and western Cambodia over a long period, dating from 198 Ma to 79 Ma b.p., during which large sectors of Cambodia formed a backarc magmatic zone. These granites are shown on the 1978 map of Fontaine and Workman (Figure 2)

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and on the geological map accompanying this paper.

Mineralization, though widespread, seems to be concentrated in two areas (1) the Knong Ay Kchol district, about 100 km west and northwest of Phnom Penh and (2) the Rovieng district, about 200 km north of the capital (see Fig. 5). Mineralization activity is most frequently the result of thermal reactions and hydrothermal activity accompanying granitic and granodioritic intrusives penetrating Mesozoic sedimentary formations. In this way, numerous deposits of gold, tin, tungsten, molybdenum, copper, lead, fluorite and pyrite were formed. Unfortunately almost all so far identified have been too small to be of economic value and are merely of mineralogical or indicative interest.

Principal occurrences related to this epoch include the skarn magnetite iron deposits in contact aureoles near Rovieng. There are also quartz veinlets carrying sulfides, bismuthinides and tellurides, some with gold although these are all minor occurrences. Occurrences of tin, tungsten, lead, zinc, silver and barite have been found in the western part of the Knong Ay massif, but these are also reported to be subeconomic. Minor molybdenite and fluorite have been reported in quartz veins in the granites of Baphnom and Tan Lap on the eastern border.

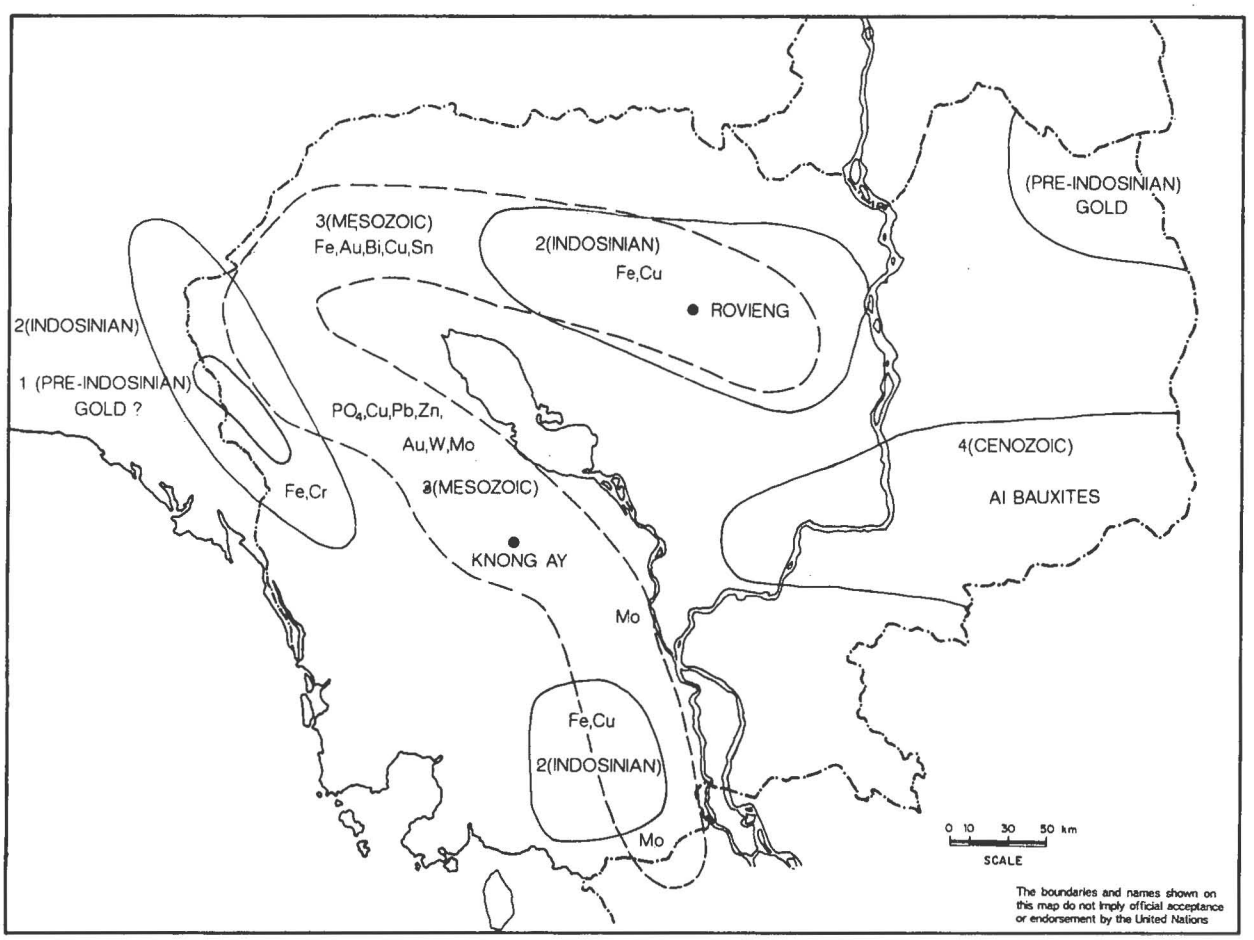

Figure 5. Metallogenic provinces of Cambodia

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D. Cenozoic Epoch

Mineral concentrations in the Tertiary and Quaternary rocks of Cambodia are principally related to weathering accumulations (placers), weathering products, and some low-value sedimentary deposits.

Placer accumulations are derived from pre-existing granitic rocks (gold placers at Bokham, cassiterite placers in the region of the Knong Ay massif) or from the weathering of the plateau basalts of Late Tertiary and Quaternary age (sapphires and rubies in the Pailin district, rubies of Samlot, zircons of Bokeo, and others) in which gem minerals occur as accessory minerals.

Weathering deposits of interest or of potential economic importance include the lateritic bauxite developed on the plateau basalts (Haut Chhlong), lateritic manganese crusts (Chhep district), kaolinite deposits in several areas, and phosphate accumulations derived from the weathering of Permian limestone (Battambang and Kampot provinces).

Sedimentary accumulations of interest include glass sands (Kampong Som Bay), peat deposits and numerous clay deposits of utility in the ceramic and construction industries, all of geologically recent origin.

In Figure 5 the approximate areas of the mineral provinces related to these epochs have been outlined, as indicated by mineral occurrence data and known geology. Areas are only approximate and indicative, and there are exceptions to these general guidelines.

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VI. MINERAL DEPOSITS OF CAMBODIA

In this section, the known mineral deposits and occurrences are located and described, as well as available source documentation allows. Locations are described by longitude and latitude position, and where possible, by relation to the nearest town or village. Deposits and occurrences are divided into four commodity grouping: (1) metallic minerals; (2) industrial minerals and construction materials; (3) gemstones and (4) fuel minerals.

The deposits and occurrences are listed alphabetically by metal or mineral commodity within each section. Locations are shown on the Mineral Resources Map of Cambodia at 1:1,500,000 scale (in pocket).

A. METALLIC MINERALS

1. Antimony

Sre Peang, Pursat province

The only occurrence on record of antimony in Cambodia is reported from a showing located 3 km SSW of the village of Sre Peang, at a point not far from Sre Peang chromite occurrences (see note on same). Location: approximately 103° 15 'East (E), 12° 2'North (N).

The mineralization is zinc-bearing stibnite which occurs in disseminated form in lumps of quartz vein material. The quartz veins apparently traverse an amphibole-biotite granite in the area. No analysis results are available. This prospect should warrant further systematic examination. (Ref.: 13, 19, 21, 29, 41).

2. Copper – Lead – Zinc

There are no large occurrences of actual or potential economic significance of these base metals yet reported in Cambodia. The following reports are given as indications to areas of possible future prospectivity.

1. Ban Chai district, Ratanakiri province

Copper mineral (plus galena) indications are reported in quartz veins in the southern periphery of a Paleozoic granitic massif, situated in the Ban Chai area in the Tonle San valley, Ratanakiri province. Location: 107° 15E, 14°O9'N (41, 19).

2. Sam Rong occurrences – Kompong Speu province

A vein-type lead-zinc mineralization is seen near Sam Rong, 110 km west north west (WNW) of Phnom Penh, in the western part of the province of Kompong Speu, 15 km north north west (NNW) the town of Choam (Cheom). Location: 103° 55'E, 11° 50N.

The occurrences are located in a group of hills situated southwest of the main mass of the Kchol massif, among which are the Phnoms Thmar Prak, Preah, Tang Vil, Ran Kol and Tabanchak.

The Kchol massif, of Jurassic-Cretaceous age, has intruded Triassic sandstones and graywackes in this area, and the deposits form lenses located in roof-zones of the intrusives, often also occurring as fillings in a post-Triassic fault network. Galena, sphalerite and pyrite are observed, with auxiliary chalcopyrite, pyrrhotite, arsenopyrite and marcasite.

The deposits have been worked by local miners. The Mission Miniere de la BRGM (21) investigated the showings; results were regarded as economically negative.

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There is an unsubstantiated report in ESCAP files of the discovery by French geologists of argentiferous galena in the area of the Kchol massif. Whether this report refers to the Sam Rong occurrence, or to some other location, cannot be confirmed.

3. O Chhung – Richomme Trail occurrences – Mondolkiri province

These occurrences are situated in western Mondolkiri province in a thinly settled sector of the country (106° 36'E, 12°42'N and 106°32'E, 12° 33'N).

In this area there are numerous outcrops of granodioritic and dioritic rocks which intrude a Triassic carbonate-sandstone sequence. Quartz-bearing skarn and hornfelsic rocks are weakly mineralized by sulfide minerals (galena and pyrite) (21).

4. Phum Pring area – Preah Vihear province

East of the village of Phum Pring, Preah Vihear province (approximately 7 km WNW of the town of Rovieng), remnants of hornfelses derived from sandstone sedimentary rocks are observed within the mass of the Phnom Deck granitic massif. The margins of the metasediments carry traces of pyrite, sphalerite, and galena. No apparent economic interest. Location: 105°04'E, 13° 24'N (21).

5. Knong Ay district, western Kompong Speu province

Chalcopyrite is reported in two locations in and near the Knong Ay granitic massif (Jurassic-Cretaceous) of central – western Cambodia. Location 104° 05'E, 11° 30'N.

At Daun Penh chalcopyrite, accompanied by blende and galena occurs in a vein system. To the west of Kdam Ngoeut, disseminated crystals and clots of chalcopyrite are observed in a diorite dyke. (19).

6. Lomphat area, Ratanakiri province

Sparse mineralizations of chalcopyrite and malachite are observed in the contact aureole of a small gabbro intrusive body which intrudes a calcareous sandstone sequence of Lower Jurassic (Lias) age. No economic interest.

The occurrence is located 7 km northeast of the town of Lomphat, along the O Cheng to Lomphat trail. Location: 106°58'E, 13°34'N. Ref.: 19, 21, ESCAP files.

7. Kroch Chhma, Kratie province

A group of chalcopyrite-bearing showings are reported in this area, 28 km west of the town of Kratie. The mineralization occurs in metamorphosed Triassic country rocks (sandstone, arkoses) in the vicinity of biotite granite stocks and intrusive dioritic sills and plugs. Chalcopyrite is also observed within the intrusive units. Of no economic interest, but perhaps indicative of copper prospectivity in the area. Location: 105° 46'E, 12°31'N.

8. Phnom Pel, Preah Vihear province

Surface showing located 33 km SSW of the town of Tbeng Meanchey, the provincial seat of government. Location: 104° 42'E, 13° 26'N.

Rhyolitic breccias of presumed Cretaceous age outcrop along the south margin of the Permian andesitic complex which outcrops in this area. Malachite and azurite have crystallized in vugs and fractures in the rhyolite breccias. No economic interest.

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9. Minor occurrences at Phnom Ke, Phnom Sekahom and Chamkar Keu

Three minor occurrences of related origins. Locations:

Phnom Ke: 105° lO'E, 13°20'N, Preah Vihear province.

Phnom Sekahom: 105° 22'E, 13° 44'N, Preah Vihear province.

Chamkan Keu: 105° 49'E, 13°32'N, in Stung Treng province (19 km South West (SW) of Phumi Thalibarivat).

The occurrences apparently all consist of minor showings of malachite in epidote-bearing hornfels and skarn rocks within and adjacent to intrusive granitic (sensu lato) rocks. Country rocks consist of metamorphosed Triassic schist-sandstone sequences with marble intercalations.

The first two are in the Rovieng-Chhep district, and the latter is nearer to Stung Treng.

References: 16, 19, 21, 39, 41 and ESCAP Data Files.

3. Bauxite

The known bauxites of Indochina occur at two points in the stratigraphic column: the Permian and the Quaternary. This holds true in Cambodia, where bauxite occurs in two settings, as intercalations in the Permian limestones of western Cambodia, notably in Battambang province, and as surficial weathering product of Pliocene – Pleistocene high plateau basalts in the Haut Chhlong district of southeastern Cambodia in Mondolkiri province. The low plateau basalts of the area north and west of Kompong Cham and in small areas in the Cardamom (Kravanh) Range (also Plio-Pleistocene) have also been examined, but to date are of no practical interest as sources of bauxite. Some bauxites are also developed as small local accumulations on basal marine sedimentary rocks of Devono – Carboniferous and Triassic ages but these are of only petrographic interest.

Investigations on the bauxites of Cambodia were begun by Saurin (30), and were followed by the reports of H. Cruys (12) and P. Carrel (6, 7). Since that time little systematic work has been done. This work was summarized by Workman (41) from whom the following paragraphs are adapted.

1. Battambang deposits

Thin beds of siliceous bauxite are interbedded within the Permian limestone found in the Battambang – Sisophon areas. The most intensively studied area is about 20 km west of the town of Battambang, at the hills known as Phnom Sampeou and Phnom Thmei (103° 02'E, 13° 04'N).

The greater part of north-western Cambodia is covered by the alluvium of the Tonle Sap basin. This alluvial covering is fairly thin in places, and the underlying bedrock is often exposed. One formation that crops out more than any other is the limestone group of Permian age which forms numerous isolated hills that rise abruptly from the plain, especially in the Battambang and Sisophon areas. The recognition of bauxite beds interstratified within the limestones in these hills is accredited to Saurin (30). The bauxite horizons do not in fact have the usual visible characteristics of bauxites but look rather like ordinary red marls or clays.

During the period of the early 1960s, the deposits of the district were the subject of much investigation, both in the field and in the laboratory. These studies fell into three categories: (i) chemical and assay data, (ii) mineralogy and (iii) field investigations.

(i) Chemical and assay data

Altogether 15 chemical analyses of bauxitic samples from the Battambang area are recorded. The first two, reported by Saurin and quoted by Cruys, are of siliceous bauxite with a very satisfactory alumina content:

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	1	2
Al_2O_3 (alumina).....	61%	56%
SiO_2 (silica).....	11	16
Fe_20_3 (ferric oxide).....	14	8
Ignition loss (water etc.).....	13	15

Four further samples were submitted to the Cruys mission by Saurin, and gave a somewhat wider range of results (Nos. 1-4 below). These were supplemented in the Cruys report by a further 8 analyses (Nos. 5-12).

Table 1. Chemical data on bauxites of the Battambang region

	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*
Al_2O_3	63	48	47	45	39	44	62	59	60	54	60	43
SiO_2	6	25	11	17	14	13	7	3	7	13	7	4
Fe_2O_3	14	7	26	21	20	23	15	23	15	17	17	43
Ignition loss.....	14	15	12	13	17	14	14	12	14	14	13	10

N.B. The analyses here are reduced to nearest whole numbers. All values quoted in per cent.

These materials can all be called bauxite, but cannot all be considered as good-quality aluminum ore, even excluding considerations other than the chemical analysis. In general, either the silica or the iron content, or both, is rather high. Low silica content (preferably less than 7 per cent) is an especially important criterion of good bauxite ore, as the presence of silica increases the consumption of chemicals used in the extraction process.

Not all the marl-like horizons in the limestones are necessarily bauxite. Sixteen analyses are quoted by Carrel (6, p.2) on samples from Phnom Krapeu, where also were obtained the 8 samples quoted by Cruys. Not one of Carrel's 16 samples proved to be bauxitic.

However, a later analysis of a sample from Phnom Kdong by Aluminium Suisse is quoted by Carrel in his 1968 report, and indicates a siliceous bauxite rather high in iron:–

Al_2O_3	51%
SiO_2	0%
Fe_2O_3	24 %
Ignition loss	12%

These figures are indicative of rather inferior bauxite ore, somewhat siliceous, although the ferric oxide content is just about within acceptable limits.

(ii) Mineralogy

Several samples were analysed microscopically by Cruys, and two particularly significant observations were made: firstly, there appeared to be an abundance of diaspore ($Al_2O_3.H_2O$) and secondly there was very little free quartz. The latter fact, taken in conjunction with the rather high silica content, suggests

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the presence of clay minerals such as kaolinite. This would mean that some of the alumina in the sample would not therefore be available as an economic source of alumina. The presence of diaspore is also detrimental, since this monohydrate of alumina is not amenable to the normal process of bauxite treatment (Bayer process) and is therefore not regarded as suitable material for the extraction of aluminium.

It appears that, so far, only one sample has been analysed by more precise non-optical methods, such as differential thermal analysis and X-ray diffraction. This was the same sample analysed chemically by Aluminium Suisse and its mineral content was calculated as follows:

Boehmite	44%	$Al_2O_3.H_2O$
Hematite	24 %	Fe_2O_3
Kaolinite	22%	$Al_2Si_2O_5.(OH)_4$
Gibbsite	5 %	$Al_2O_3 3H_20$
Anatase	2 %	TiO_2
Diaspore	2%	$Al_2O_3.H_20$

It should be noted that boehmite, although it has the same chemical composition as diaspore, is nevertheless amenable to the Bayer process while diaspore is not. (It could be that the diaspore identified by Cruys in microscope studies is also in fact boehmite). Even so, boehmite remains inferior to the trihydrate (gibbsite) as aluminium ore, because it still requires more difficult and more costly processing.

(iii) Field investigations

Field investigations were carried out in 1965 when, following the recommendations of the Cruys report, a programme of drilling and mapping was carried out by Carrel in the area southwest of Battambang. The field studies had as their principal object the size, geometry and possible sub-surface extension of the deposits.

Clearly, the hills of limestone which exist are merely the highest points of an ancient erosion surface, now mostly buried beneath the alluvium of the Tonle Sap basin. Carrel carried out a programme of 12 drill holes (each of 40 to SO metres) near certain limestone outcrops, and was able to prove the subsurface continuation of the limestone surface at no great depth in most of the holes. In addition, some of the holes intersected bauxitic horizons. However, the concealed limestone surface must be assumed to be karst-like, subject to sudden changes of relief, as are the above-ground limestone features of the area. This would make predictions of the depth of bedrock very hazardous. It is extremely improbable that bauxite occurring in these sub-alluvial limestone formations could be economically exploited even if the presence of high-quality ore were ascertained by drilling. (Nearly all bauxite mined today is produced by open-cast mining, although some underground mining is done in Europe).

It should be noted as well that in outcrop the bauxite marls are interstratified in the limestones in apparently horizontal beds. Field examination shows that in detail these beds are tectonically complex and may be disturbed, dissected and discontinuous in lateral extent (21).

Carrel gave a figure of 3 million tons of known reserves of bauxitic material available for opencast working. It should be noted that bauxite is very much a bulk-mining proposition, and in this context 3 million tons is not a large reserve. The quantity of bauxite available for open-cast working in the Battambang area is not likely to be greatly increased by the discovery of new outcrops, so that reserves must be reckoned as very minor on a world scale.

Conclusions (Battambang area)

At the present time, it can be stated that any development or exploitation of the bauxite of the Battambang area is made very doubtful by the following factors:

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1. The irregular quality of the ore,
2. The usual presence of too much silica, or ferric oxide, or both,
3. The apparent metamorphism of some of the ore to diaspore,
4. The apparent presence of other impurities, particularly kaolinite,
5. The small proven and probable reserves,
6. The high dips and structural complexity of the Permian strata, which rule out the possibility of inexpensive open-cast mining operations.

The above conclusions are Workman's. The French BRGM mission came to the same general conclusion (21).

2. Haut Chhlong Plateau deposits

Bauxitic laterites have been found in the Haut Chhlong plateau in the extreme eastern part of Cambodia (Mondolkiri province). The plateau forms an irregular geomorphological highland rising above the level of the central plain (about 200 m above mean sea level) to heights of 500 to 1,000 m, and covers approximately 4,500 square kilometres within Cambodian territory. This plateau, which extends into Viet Nam, is composed of a great thickness of basaltic lavas and volcanic ash which are probably mainly of Pleistocene age although the oldest flows may be Pliocene or Miocene. Similar high plateaus of basic rocks extend over much of central and southern Viet Nam (Plei Ku, Buon Me Thuot, Di Linh) and the southern Lao People's Democratic Republic (Bolovens).

In Cambodia only the Haut Chhlong plateau has been studied so far, firstly by the Cruys mission (12) and subsequently by Carrel (7).

Work to date has centred on the area between O Rang and Mondolkiri City (Senmonorom on some maps), at approximately 107° 12'E, 12° 23'N.

The favourability of these basalt plateaus for bauxite formation is dependent on the conditions of weathering under modern and recent climatic conditions.

The following table gives a summary of analytical results obtained by the Cruys mission (12):

Table 2. Chemical data on surficial materials, Haut Chhlong

	1	*2*	*3*	*4*	*5*
SiO_2	46-49	21-45	24-39	4-27	1-2
TiO_2	2	2-4	3-5	4-6	3
Al_2O_3	14-18	18-26	17	26-31	43-50
Fe_2O_3	12-14	14-41	19-36	24-36	17-24
Ignition loss	1-6	12-16	14-17	18-25	27-29

1. Basalt, slightly altered
2. Residual clays derived from basalts
3. Black, brown and grey soils
4. Red soils
5. Laterite nodules (range of 3 samples).

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The red soils (column 4), in spite of the wide range of silica content, are clearly ferro-aluminous. They commonly contain lumps of laterite, which in places consists of high grade bauxite (column 5). The three samples of laterite grouped in column 5 are very low in silica, and represent good-quality ferruginous bauxite ore.

Carrel (7) showed similar data indicating that basalt flows on the Haute Chhlong showed considerable residual enrichment of Al_2O_3 even while retaining their original textures and lithologic aspects:

Table 3. Chemistry of basalts, Haute Chhlong

all figures in per cent	*1*	*2*
SiO_2	18-25	6.2
TiO_2	5	5
Al_2O_3	25-30	33.7
Fe_20_3	25-30	33.1

1. Average composition of a basalt flow of Haute Chhlong (average of 10 analyses).
2. Average composition of highly altered basalt flow of Haute Chhlong (average of 8 analyses).

Analyses of beds of volcanic ash and agglomerate give even more evidence of laterite-type alteration, i.e., removal of silica and a proportional enrichment in alumina and ferric oxide (table 4).

Table 4. Chemistry of altered volcanoclastic rocks, Haute Chhlong

	1	*2*	*3*	*4*
SiO_2	12	10	11.0	14.7
TiO_2	5	5	–	–
Al_2O_3	31	36.3	34 .4	32 .4
Fe_2O_3	32	28	–	–

1. Average of 26 samples of altered volcanic ash, Haute Chhlong (quoted by Carrel).
2. Average of 14 samples of altered agglomerate (couche de lapilli), survey line 1, HauteChhlong (quoted by Carrel).
3. Average of 4 samples of altered agglomerate, survey line 2, Haute Chhlong (Carrel).
4. One sample of altered agglomerate, survey line 3, Haute Chhlong (Carrel).

Thus the data from the available analyses reveal that the weathering products of at least some of the volcanic rocks of Haut Chhlong are equivalent to siliceous bauxitic clays. Some individual analyses attain the requirements of bauxite ore.

Carrel draws attention to the possibility of more highly developed soil alteration in the valleys dissecting the high plateau, as compared to the soils of the plateau itself which were the subject of study of both the Cruys and Carrel missions. He observes that these more mature soils may have advanced further in the process of becoming alumina-rich (i.e., bauxitic). Carrel states "if in future it is desired to re-commence prospecting for bauxite on the high plateau, the method should.....consist of.....ascending to the plateau by one of the big river valleys". In any event, future prospection should certainly examine the entire exposed lava succession, and not merely the lava flows of the summit plateau.

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Conclusions regarding the Haute Chhlong plateau

The analytical results show an overall enrichment of alumina and ferric oxide and diminution of silica in the weathering products of the plateau basalts. Also, where laterite has been located, it has been shown to be bauxitic. A much more systematic programme of mapping, sampling and analysis is therefore indicated.

The discovery of bauxitic laterite on the high plateau of Haute Chhlong represents the first indication of Holocene (Quaternary) bauxite formation in Cambodia. This is significant because such deposits by definition rest on or near the present-day surface, and if they have any lateral extent are readily available for open-cast mining. In general, the amount of work achieved to date on the Haute Chhlong bauxites (firstly by the Cruys mission abd subsequently by Carrel) seems not to be by any means sufficient. The work was never followed up because of the insecure conditions prevailing in eastern Cambodia beginning in the late 1960s and continuing in subsequent years.

It should be noted that there are substantivated reports of bauxitic latosols of similar origin and quality noted on the Pleistocene basalt plateaus of the neighboring districts of Viet Nam.

J.P. Berrange in 1974 (cited in 16) reported Al_2O_3 contents of 40-50 per cent in geologically similar latosols on the Di Linh plateau in Viet Nam.

A 1992 ESCAP survey (2) of the bauxite potential of Indochina made the following observations:

"Russian and Vietnamese exploration work in the Dac Nong area (on the Viet Nam side of the Haute Chhlong plateau) started in 1976. Much of this work was used for the 1987 "feasibility study" prepared by the All Union Scientific Research and Design Institute for the Alumina, Magnesia and Electrode Industry (VAMI, 1987). After 1987 through to 1990 Vietnamese geologists of Geological Expedition No. 6 (Ho Chi Minh City) carried out additional work. So far, out of the total 850 km^2 of laterite plateau, approximately 500-600 km^2 has been explored.

On the basis of several hundred pits and drillholes a total possible bauxite reserve in excess of 5 billion metric tonnes was outlined. These reserves occur in 12 separate lateritized areas of the Dac Nong plateau.

The alumina content of the bauxite from the Dac Nong plateau as reported by the Council for Mutual Economic Assistance (COMECON) report generally ranges from 36-40 per cent total Al_2O_3. A significant improvement of this grade is obtained by wet-screening of the ore to remove smaller size fractions; this also removes part of the silica which for the bauxite from the Dac Nong plateau ranges from 5 to 10 per cent SiO_2.

The COMECON and Vietnamese geologists invariably used a simple wet-screening technique employing a 1 mm screen size. This produced a beneficiated product of considerably higher alumina content and much lower silica. For the whole of the Dac Nong plateau the average Al_2O_3 grade was improved from 38 per cent to 49 per cent in this manner, while the silica content dropped from 8.2 per cent SiO_2 to 2.3 per cent.

The beneficiated product is in all reports referred to as "concentrate" and most reserve figures quoted are in metric tonnes of "concentrate". Recovery figures after wet-screening usually ranged between 45 per cent and 50 per cent, i.e., about half the crude ore is washed out.

There are indications that if more sophisticated beneficiation is applied, much better results can be obtained. Test work conducted by a private laboratory on + 10 mesh material from this area achieved a concentrate with an alumina grade of 65 per cent and a silica content of less than 2 per cent with a 65 per cent recovery on bench-scale testing".

General conclusions and recommendations for bauxite prospecting in Cambodia.

1. The bauxites of the Battambang area have been adequately studied and do not present possibilities for exploitation. No further work is therefore recommended.

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2. Research into the aluminous soils of the Haute Chhlong plateau, however, has so far been of a preliminary and superficial nature. A more systematic study of this area is called for, with controlled sampling based on a preliminary study of the geomorphology of the plateau.

3. No other area of Cambodia is at present known to be of any interest regarding bauxite prospection.

The above were the conclusions of Workman and co-workers in 1972, and they remain valid and substantially the same at this time of writing.

References: 2, 6, 7, 12, 16, 19, 21, 30, 39, 41.

4. Chromium

Sre Peang, Pursat province

A chromite occurrence is reported to be situated 2 km South South West (SSW) of the village of Sre Peang on the Leach trail. Sre Peang is located 13 km east of the town of Phum Peam Prous in the Peam Prous depression, and 75 km West South West (WSW) of Pursat. Location: 103° 14'E, 12° 22'N.

In this area residual blocks and fragments of chromite are found over an area of some 10 square kilometres. The chromite is apparently the residue of the weathering of post – Jurassic gabbro outcrops of basic irruptive plugs and stocks which intrude the Devonian – Lower Carboniferous sandstones and schists that form the bedrock of the broad valley of the upper Pursat (Phoutisat) River in this district.

This broad valley forms a basin referred to as the Peam Prous depression. The depression is an erosional window in the Gres Superieures (Jurassic, Khorat group) in this area, floored by Devonian Carboniferous metasediments and on which some erosional klippen of Triassic sandy microbreccias stand in relief. This basement in this area has been intruded by numerous plugs and stocks of gabbroic and granodioritic parentage, probably of type gamma 4 (see geological map and explanation) of probable Late Triassic age (33, 34).

Given the probable origin of the chromite as a differentiate of the gabbroic phases of the primary magma which formed these relatively small intrusive masses, it is unlikely that a large well-defined source of the chromite will be located. The quality and extent of the residual chromite should be investigated, however, as it may form a suitable resource for artisanal mining activity. References: 13, 21,29, ESCAP data sheets.

5. Gold

Gold has been sought and won in Cambodia since historic times, as attested by a large number of old workings. Organized prospecting and mining for gold began in the late nineteenth century, presumably by French engineers and geologists working during the colonial regime.

With one exception (Bo Sup Trup), discoveries have all been small and of the scale suitable only for artisanal mining by local populations. There are however a large number of occurrences and therefore there is some potential for future discoveries and developments of commercial scale.

Gold prospects and alluvial workings are found in three main regions of the county: the northwest (Oddar Meanchey or Siem Reap province), north central areas (notably in Preah Vihear province), and in the northeast (Rattanakiri province). Other more isolated occurrences are also known.

Production figures have always been quite low and the certainty of the data is also limited. Production in 1963, the last year for which supposedly reliable figures are available, was 208 kg. From that time the industry was in apparent decline, as seen from reported production data:

1963	208 kg
1964	185 (est.)
1965	140 (est.)
1966	125 (est.)

Figures are from Workman (41). No data has been available since that time.

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1. Bo Sup Trup deposit – Oddar Mean Chey province

The Bo Sup Trup deposit is near Banteay Chhmar (103° 06'E; 14° 04'N), in the northwestern province of Oddar Meanchey about 15 km northeast of Thmar Puok (Phumi Thma Pol) and about 33 km southwest of Samrong.

Of the gold deposits thus far known in Cambodia, it is easily the most important, having been worked since at least 1877 by artisanal methods at the surface and from 1905 to 1911 by a company operating underground. In 1929, it was estimated that there was a reserve of 440,000 tons of ore carrying an average of 15 gm of gold per ton, i.e. 6.6 tons of gold metal.

Kun and Say (19) report the existence of 240,000 tons of ore carrying 15 grams per ton, with local sample tenors as high as 420 grams per ton.

The primary gold is in a network of quartz veins which are between 30 cm and 4 m thick, accompanying microgranite dykes intrusive into Devonian – Permian shales and limestones. Besides gold, the veins also carry traces of pyrite, sphalerite, galena, chalcopyrite and malachite. There is some silver. The primary ore control is apparently by a fault system which transects the eastern extremity of the Bo Sup Trup massif.

The vein-deposits, surface showings and tailings dumps were the object of a further investigation, including drilling, between 1961 and 1965 by geologists of the Bureau de Recherches Geologiques at Minieres de France (B.R.G.M.). Drilling totalled more than 1,000 m of core. Although some high values were obtained from individual core samples, the overall average grade of the principal group of veins was set at only 4.14 gm of gold and 3.74 gm of silver per ton rather low for an underground exploitation. Some veins gave much higher values than this (for example 94.42 gm of gold and 11.08 gm of silver per ton in one case), but invariably proved to be discontinuous.

A study of gold-bearing alluvium in the area showed an areal extent of some 20 sq km, averaging about 0.5 gm/cubic metre. Also, there is said to be an artificial lake that could be dredged.

While the B.R.G.M. report expresses doubt concerning any future economic viability for the deposits, the Bo Sup Trup district and its known primary and alluvial deposits show some potential for rehabilitation and possible further development. References: 11, 16, 19, 21, 41, 42, ESCAP data files.

2. Phnom Kambor (or Bar), Siem Reap (Oddar Meanchey) province

The occurrence is located approximately 28 km southeast of Samrong in Oddar Meanchey province at 103° 45'E, 14° 04'N. It is about 80 km NNW of Siem Reap town. The local name is Bar.

The gold occurs in quartz veins and shoots in Permian limestones. Permian limestone inselbergs and outliers repose on a folded anticlinal basement of Devono-Carboniferous schists and ironstones. (ESCAP data files). The occurrence is a consequence of the post-Triassic granitic intrusion activity observed elsewhere in this district.

The showing has been worked intermittently by local miners. There is also a small eluvial/alluvial placer deposit, a result of weathering and transport of adjacent associated gold-bearing lateritic deposits, which has also been worked by the local people.

3. Phnom Lung, Preah Vihear province

The deposit is reportedly 14 km west of the village of Phnom Deck, situated at 104° 57'E, 13° 22'N. The showing is situated at the eastern foot of the hill known as Phnom Lung, 6 km west of the Rovieng deposit described above.

The gold occurs in a network of quartz veins in aplite and granite of Indosinian age. The veins also contain traces of pyrrhotite, chalcopyrite, galena, sphalerite, covellite, chalcocite, mispickel, native bismuth and tellurobismuthite. Visible gold was reported by the B.R.G.M. Mission Miniere (1963), cited in 21.

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The country rocks are formed of marls and calcareous shales of Triassic-Jurassic age metamorphosed to hornfels and skarn facies. The Phnom Lung hill itself is formed of monzonitic granite, granodiorite and diorite, and is part of the Phnom Deck granitic complex of Jurassic-Cretaceous age.

The deposit has been worked by artisanal miners on a discontinuous basis in the years of the 1950s, 1960s and 1970s. Total output from Phnom Deck and Phnom Lung was reported as about 150 kg per year in 1962. In the period 1962-1964 the deposit was examined in some detail by the BRGM Mission Miniere; their report indicated a yield of about 3000 gm of gold from 96.7 tons of ore produced from pits and trenches, for a mean tenor of 31 gm per ton in this bulk sample.

The elluvial and alluvial placers derived from the primary deposit, which average about 20 gm of gold and 1.5 gm of silver per ton, were being actively exploited by artisanal methods at the time of Workman's report (41) though no production data are available. Phnom Lung was apparently the only active gold operation in the country at that time. References: 11, 16, 19, 21, 39, 41, 42, ESCAP data files.

4. Rom Dey vein deposit, Preah Vihear province

The vein deposit Rom Dey, Rovieng district, is located 8 km WNW of the village of Phnom Deck. Location approximately 104° 59'E, 13° 21'N.

Numerous quartz veins and veinlets, often auriferous, are noted in this district in which granites and granodiorite intrusions have metamorphosed the Triassic sedimentary rocks to hornfelses. The district has a long history of artisanal mining and an extensive area of placer deposition and activity.

The primary deposit is a gold-bearing quartz vein 3-90 cm wide containing in addition a little pyrite, chalcopyrite and mispickel. The vein cuts across a monzonite – granite intrusion.

Like the Bo Sup Trup prospect described above, Rom Dey was the subject of an investigation by drilling (over 500 m of core) during the period 1961-1965. The results were inconclusive. A number of very high values were obtained, with up to 191 gm of gold per ton in one case, but the wide variations and the small number of samples made a firm estimate of reserves impossible. It was calculated however that there were 12,639 tons of ore, and that the average grade might be as low as 7.16 gm/ton or as high as 20.4 gm/ton, meaning reserves of between 90 and 258 kg of gold. Altogether, the results seemed to rule out any systematic exploitation of the deposit underground, but further exploratory work was recommended. References: 19, 21, 41, 42, ESCAP data files.

5. Phnom Deck alluvial deposit – Preah Vihear province

This deposit was in fairly continuous production for a period after 1956 into the 1960s. The deposit is a lateritic placer of presumed eluvial – colluvial (as well as alluvial) origin located near the village of Phnom Deck in the Rovieng district. Location 105° 01'E, 13° 19'N.

The origin of the deposit is unclear, but according to the BRGM geologists it has an evident relation to the quartz vein phases of the Phnom Deck granitic massif (21).

This placer, or residual deposit, has yielded gold of quite coarse grain size of 0.1 to 3.5 mm . Some rare nuggets weighing up to 150 gm have also been found (41). Workman reported in 1972 that the alluvials were said to be exhausted, and that the primary source area for the gold had not been located. The coarse grain size may indicate, first, that the gold has not travelled far, and secondly that there may be reserves of finer gold which escaped collection . File notes (ESCAP) report that the gold is "linked to a network of small gold-bearing quartz veins localized in post-Triassic biotite calc-alkaline granites". No data is available on past output or gold content.

6. Gold prospects in Rattanakiri province

In the valley of the Tonle Sap and its tributaries, there are numerous trace showings of alluvial gold which have been worked by simple panning methods from time to time. An organized prospecting

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mission was launched by a private company as long ago as 1894, but revealed no gold deposits of immediate commercial interest, and was eventually abandoned for lack of capital. Since then, there does not appear to have been any further systematic prospecting in this area.

At Bo Kham (107° 21 'E; 13° 51 'N), there are auriferous gravels and also gold-bearing quartz veins related to diorite intrusions. There is reported to be some panning activity, more or less intermittent, in this area (16, 19, 21, 41). In northeastern districts, about 20 km northwest of Voeun Sai (Rattanakiri province) there are reported to be small gold placers derived from hornblende-bearing granites and from quartz-diorites rich in pyrite. Location: 106° 45'E; 14° 05'N (16).

There are also a number of other localities where auriferous veins occur, all of which are, in the area of crystalline schists and granitic intrusions of possible Pre-Cambrian age in the extreme northeastern corner of the country.

7. Phnom Thmar Meas, Battambang province

Located near the Thai frontier in westernmost Cambodia, approximately 90 km west-northwest of Battambang (102° 21'E, 13°18'N). The showing is in a remote area, accessible only by seasonal trails from Pailin or from adjacent Thailand towns.

The occurrence consists of gold-bearing quartz veins in faulted and deformed Permian limestones. In this region of Cambodia and Thailand, prominent northwest-southeast-trending ridges of Permian limestones have been pushed up by abrupt tectonic shortening, resulting in faulting, detachment and uplift of horsts and fault-blocks, accompanied by strong brittle and plastic deformation of the limestones. This diostrophism gave rise to local veins and stockworks of quartz, and minor intrusions of rhyolite and microgranite.

Quartz veins are locally weakly mineralized by native gold and gold-bearing pyrite, and there is gold reported in adjacent eluvial/alluvial materials. Both have been worked in the past on a small-scale basis. There is a reference to investigations of this location in 1933, but the occurrence is apparently of limited economic interest. (Location also reported as 102° 21'E, 13° 13'N, but this is in (or almost in) Thailand)

8. Pailin District, Battambang province

There are reports (ESCAP files)of placer workings in the Pailin district of western Battambang province. The location and size of the workings is poorly known.

9. Phnom Chi – Sud occurrence, Kompong Thom province

Deposit is located at Phnom Chi – Sud (south) in eastern Kompong Thom province (location approximately 105° 39'E, 12° 45'N) near the border of Kratie province.

This showing is apparently consisted of fracture-filling quartz veins in a sheared Triassic hornfelsic country rock (probably Triassic). This system is related to intrusions of granitic rocks (monzonites, diorites) in the vicinity. Gold is weakly dispersed in the veins.

There is also evidence of former alluvial workings in the area. The prospect was considered to be of no economic interest by the BRGM mission (1970) but may be of indicative interest.

10. Memung occurrence, Mondolkiri province

Reported location is 106° 55'E, 12° 35'N, near Keo Seima. The primary deposit is a gold-bearing quartz vein of 0.6 m wide, also mineralized with arsenopyrite, pyrite, chalcopyrite and galena. The vein(s) traverse a Triassic-age shale sandstone series. Microscopic studies show that the gold particles vary in size from 0.05 mm to 5 mm and are associated with arsenopyrite. The occurrence was studied by the Department of Geology and Mines in 1991. The deposit is reported to be worked on a small scale by local miners.

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11. Romchek occurrence, Kompong Cham province

Reported location is situated 8 km north of the village of Memot, at 106°11 'E, 11° 54'N, at Chom Ta Mao. The occurrence is described as "lateritic placer". Gold occurs in colluvial material above the eroded primary vein system in a soil-zone some 3-4 m thick, within which values up to 11 gm per ton were found, with an average yield of 2.72 gm per ton. Adjacent placer material reported 7 gm per cubic metre. The underlying vein system has apparently not been tested. Gold grains are reportedly quite coarse, in the size range of 0.1-3 mm. This occurrence was studied by the Department of Geology and Mines in 1992.

12. Phnom Lok, Kampot province

An occurrence is reported at Phnom Lok, 13 km south-southwest of Kompong Trach in southern Kampot province, on the sea coast near the border of Viet Nam. The description would place the occurrence at 104° 25'E, 10°27'N. The occurrence is found on an isolated hill surrounded by Quaternary rocks. Here sandstones of probable Devono-Carboniferous age are traversed by three sets of quartz-filled systems striking at 40, 90 and 160 degrees, with dips of 50 to 60 degrees. The gold is accompanied by chalcopyrite, pyrite and molybdenite. No data is available on the extent or grade of the mineralization.

References: 11, 12, 16, 19, 21, 39, 41, 42, ESCAP data files.

6. Iron Ore

Most iron ore deposits which have been prospected or worked in Cambodia are apparently of the contact metasomatic type ("skarn" type), most often near the margins of granodiorite intrusions, and usually of post-Triassic ages (see Geological and Mineral Map).

The principal iron ore prospect known in Cambodia is the Phnom Deck deposit, located 15 km southwest of the town of Rovieng.

Iron ore occurrences appear to cluster in two main areas – the above-mentioned Rovieng district and in the Stung Treng district; a lesser group is seen between Takeo and Kampot.

Chinese geologists in the 1960-1962 period (10) examined 20 showings in the northern (Kompong Thom and Preah Vihear) provinces, three in Stung Treng and one in Rattanakiri provinces. Most were found to be very small. Apart from *in situ* occurrences of clearly metasomatic origin, a number of occurrences of residual type were observed, i.e., surficial areas marked by blocks of massive haematite and iron-rich lateritic debris.

Total iron reserves in the Phnom Deck area are estimated at 5-6 Mt (41) and other deposits may add 2-3 Mt to this figure. Iron deposits have not been systematically exploited because of political uncertainties and lack of suitable transport and infrastructure. A small steel mill was reportedly under construction during the 1970's, but work was later indefinitely postponed. (Mining Annual Review, 1980).

Described deposits:

1. Phnom Deck, Preah Vihear province

This is the principal iron ore prospect known in Cambodia at present. It is situated in the northcentre of the country, at 105° 03'E, 13°14'N, 15 km southwest of the town of Rovieng and 74 km north of Kompong Thom.

The superficial deposits which are amenable to open-cast mining have been known for more than a thousand years, and the ore has been excavated by manual methods at various times in history, for example between the ninth and thirteenth centuries. During the modern era various attempts have been made to evaluate and develop the Phnom Deck prospect, by French and Japanese concerns during the period 1881-1945 and by the Government of Khmer Republic after independence. None of these projects reached the development stage, mainly because of a lack of transport routes (and insufficient reserves to

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justify their construction) combined with inadequate local sources of fuel and power. The most recent evaluation was undertaken in about 1960 with assistance from the People's Republic of China.

The primary deposits at Phnom Deck are contact-metasomatic replacements in metamorphosed Triassic sandstones and marls near the margin of a vast post-Triassic granodiorite batholith having a diameter of some 30 kilometres. Deposits formed by supergene enrichment of the ferriferous contact rocks contain some high-grade masses of magnetite-hematite with up to 64-68 per cent iron The overall grade is, however, rather low: estimates put the bulk of the ore at only 51-56 per cent iron. There is said to be generally 14-27 per cent silica and 3 per cent alumina in the ores. Phosphorus and sulphur are however very low, each less than 0.03 per cent. The total tonnage was tentatively put at 5 to 6 million tons (10).

2. Phnom Deck 2, Preah Vihear province

This is apparently a related zone of the same deposit, reportedly located at 105°03'E, 13°19'N. This reported location is approximately 10 km NW of the above site. Described by the BRGM mission (21).

At least four occurrences were described by the Chinese geologists in the area surrounding the Phnom Deck massif: Phnom Deck, Phnom Thmak, Katkeo and Popun. All are apparently comprised of magnetite and hematite in a quartzose gangue associated with the contact zones of the granodiorite irruptive rocks (10, 11).

Analytical ranges as reported by Kun and Say (19) indicate – Fe content 37.52 – 54.6%, SiO_2-17.28 – 44.6%, S – 0.013 – 0.040% and P – 0.024 – 0.030%

3. Rovieng district – Phnom Kbol, Preah Vihear, province

Iron ore occurs on the Phnom Kbol area, Rovieng district, north of the Stung Sen (river). Approximate location: 105°14'E, 13°35'N.

The country rocks are reported as grey-brown Lower Permian schists, sandstones, white-gray limestones and various calcareous and sandy schists of apparent Triassic age. The local structure is dominated by a NE-SW faulting which has brought Permian and Triassic rocks into contact.

A granodioritic intrusive suite consisting of granite, granodiorite, quartz diorite porphyry, tonalites and related rocks has invaded the above country rocks.

Magnetite mineralization occurs as a result of skarn reaction near the intrusive contacts of the granitic rocks. Little information is available on the quality or extent of the deposits.

4. Rovieng district – Authmor Sreal, Preah Vihear province

Occurrences of magnetite are reported at this location, 6 km west of Rovieng. Quartz diorite prophyry and rhyolites are noted in the vicinity. The ore minerals of interest are magnetite, specularite and massive haematite.

Analysis of three iron-rich samples are given as follows:

Sample No.	*Ore-type*	*Total Fe %*	*SiO_2 %*
HF 21	Hematite Ore	37.52	44.60
HF 22	Specularite Ore	55.62	17.28
HF 23	Magnetite Ore	54.60	19.77

The deposit is of apparent skarn origin. Approximate location: 105° 03'E, 13°23'N.

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5. Rovieng District – Koh Keo deposit, Preah Vihear province

The Koh Keo (Koh Keo island) magnetite occurrence is situated about 16.5 km north north east (NNE) of Rovieng, situated on a track accessible in good weather conditions. Location: 105°11'E, 13° 28'N.

An excavation at the site shows magnetite, skarn rock and epidotized quartz-bearing metasediments. A 3 m thickness of magnetite striking N60E is observed, rich in the central core and becoming leaner toward the margins where it becomes a granular magnetite dispersed in a skarn rock. One analyzed sample reports a total iron value of 53.71 per cent.

6. Rovieng district – Phnom Prolean, Preah Vihear province

This magnetite occurrence lies about 8.5 km southwest of Rovieng, about 0.7 km from the mountain of this name. Blocks of magnetite of approximately lm thickness are reported in a larger area of fragments of magnetite, hematite, limonite, and ferruginous rock fragments. The geologic setting is not reported but this magnetite is probably of skarn origin. Location: 105° 03'E, 13°19'N.

7. Chhep, Preah Vihear province

Chhep 1. Two showings. One between Sniet and the Chhep-Rovieng road, the other along the trail from Chang-ha to Chrach. As well, hematite blocks near Phnom Chirait on the trail from Chhep to Chrach (21). Reported location: 105° 23'E, 13° 33'N. Reportedly in the Devonian-Carboniferous sandstones and grit rocks (ESCAP data files).

Chhep 2. Surface showings located 5 km north-northwest of the town of Chhep in the province of Preah Vihear; the mineralization occurs in the form of lumps of massive hematite over an area of about 3 km by 2 km on the surface of a much-folded subvertical schisto-sandstone series of Devonian to Lower Carboniferous age. No sampling has been carried out before 1969. Location: 105° 22'E, 13° 47'N.

8. Chrang, Stung Treng province

Surface debris of iron ore is reported 3 km east of Chrang in an area of weathered mica schists. Location: 105° 49'E, 13° 43'N.

9. Sam Ang, Stung Treng province

In the same district, 6 km south of Sam Ang on the Chrang – Thalibarivat road there is a bed of magnetite in a series of Paleozoic quartzites and cherts. A possible affinity with banded ironstone type of deposit has been suggested (41). Location: 105°53'E, 13°38'N.

10. An Long Chey, Stung Treng province

Surface showings located 38 km west-northwest of the town of Stung Treng in the province of the same name; the mineralization occurs in the form of lumps of massive hematite for a length of 3 km on the surface of a much-folded subvertical schisto-sandstone series of Devonian to Lower Carboniferous age. No sampling has been carried out before 1969. Location: 105°40'E, 13° 38'N.

11. Stung Treng District, Stung Treng province

Surface showings located about 10 km southwest of the town of Stung Treng in the province of the same name; the mineralization occurs in the form of lumps of massive hematite for a length of about 4 km on the surface of a concordant and folded subvertical series of sandstones, schists and phthanites of Devonian to Lower Carboniferous age. No sampling has been carried out before 1969. Location: 105° 54'E. 13° 31'N.

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12. Phnom Khbal Stung, Siem Reap province

Located on the upper Stung Chi Kreng River, approximately 55 km east north east (ENE) of the town of Siem Reap. Location: 104°19'E, 13°35'N.

The prospect consists of strongly folded magnetite-bearing Cambro-Silurian quartzites which form inselbergs mostly surrounded by rhyolites and andesites of apparently younger age (41). Probably of little economic importance.

13. Phnom Rumdek – Siem Reap (Oddar Meanchey)

This occurrence is reportedly located 20 km east south east (ESE) of Sam Rong in Siem Reap (Oddar Mean Chey province) in northwesternmost Cambodia. The ore is in the form of massive eluvial hematite derived from the weathering of iron-rich shales of Devono-Carboniferous age (41). Crusts and blocks of hematite are dispersed over a large area on the slopes around the hill. Of very limited economic interest (21). Approximate location: 103° 50'E, 14° 05'N.

14. Phnom Pours, Siem Reap (Oddar Meanchey) province

This deposit is reported to be northeast of Siem Reap, where iron is observed as magnetite and limonite in a series of quartzites and tuffs of probable Devonian-Carboniferous age in the Phnom Pours range of hills. The best mineralized zone is reported to be in the banded quartzites at Phnom Chamkouy, where the iron-bearing quartzites (strike EW, dip vertical) achieve an apparent thickness of 20-30 m. (ESCAP data files).

Location not precisely known. If as described, this prospect is one of some potential, given its structure and sedimentary origin.

15. Phnom Prean deposits, Battambang province

Phnom Prean is located in the Sam Loth District, about 54 km southwest of Battambang.

The deposit is set in Lower Permian grey to gray-white brecciated limestone, schists and calcareous breccias, occurring in two areas of the mountain, referred to as Phnom Poil and Phnom Prean. Two large fault blocks show dips of about 40° to NW and SW respectively.

At Phnom Prean outcrops of a speckled granodiorite porphyry are noted. Around the intrusive, the limestones are marblized, with grossular garnets present, and with skarn-derived magnetite in numerous beds, especially on the western flanks and western top of the hill. The deposit is described as a discontinuous series of pods extending over a distance exceeding 100 m, and the iron is reported to have a thickness of about I m. The deposits have been worked irregularly in several places on the hill.

The principal iron mineral is a dark gray magnetite, occurring in relatively rich zones and also in lean zones with abundant silica. Specularite and hematite are also noted. Three analyses show the following data:

Sample No.	*Total Fe* %	*Soluble Fe* %	*SiO_2* %	*S* %	*P* %
HF 64	30.58	30.34	54.72	0.03	0.08
HF 65	40.95	40.95	39.11	0.04	0.13
HF 66	67.84	67.81	11.14	0.01	0.05

The deposit(s) show some potential for greater development. Approximate Location: 102°52'E, 12°46'N.

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16. Kampot Group of Iron showings – Kampot province

This group is located near the village of Tani, 35 km SSW of the provincial center of Takeo, south of the town of Ang Co Chay. The mineralization consists of a large number of veinlets and fracturefillings of magnetite and hematite in metasediments (sandstones, phyllites, schists) of Devono Carboniferous age. This mineralization occurs throughout the Tapok Tang Dong upland in this area. A nearby granite intrusion (the Tan Lap massif) may be the origin of the mineralization.

The weathering of the surface of these rocks in the area has concidentally produced a thick ferruginous laterite crust of great durability, which is locally mined and used as building blocks. Of doubtful economic interest as a source of iron ore (41, 21, 19). Location: 104°40'E, 10°44'N.

References for iron deposits 10, 11, 16, 19, 21, 38, 39, 41, 42, ESCAP data files.

7. Manganese

Chhep 1, Preah Vihear province

This occurrence is reported as lying 6 km west of the town of Chhep, southwest of the village of Phum Troun. Location: 105° 21'E, 13° 46'N.

The prospect consists of manganiferous laterite containing a layer of psilomelane and pyrolusite usually less than lm, but locally up to 1.8m thick, admixed with iron minerals. Reserves are estimated to be 60 000 tons and grades are 11 to 26 per cent Mn, rising locally to possibly 40 per cent Mn.

The lateritic bed is the product of weathering of a rhyolite substrata, weakly mineralized with primary manganese minerals. The rhyolites overlie or are intercalated within a sequence of continental sandstones and grits of presumed Triassic age.

Chhep 2, Preah Vihear province

A similar deposit is reported to occur at a point 7 km ENE of Chhep, along the road from Chhep to Phum to Chang Ha. Location: 105° 28'E, 13° 47'N.

The size, thickness, quality and setting are the same as described for the Chhep 1 occurrence above.

Chhep 3, Preah Vihear province

A possible similar showing is reported to occur at a locality 9 km west of Chhep but no details are available.

Reference: 16, 21, ESCAP data files.

8. Molybdenum

Occurrences of molybdenite are generally small and rare in Cambodia. The usual mode of occurrence of molybdenum minerals is as late stage crystallates in vein systems in and near granitic (Sensu Lato) masses.

1. Phnom Basset, Kandal province

Located about 20 km northwest of Phnom Penh at 104° 45'E, 11° 41'N. Two small inselbergs of monzonitic granite of Lower Jurassic age, fine-grained and locally porphyritic in texture, rise from the central plain. The monzonite carries molybdenite in quartz veins and also weakly disseminated in the ground mass.

The molybdenite occurs in disseminated crystals and clusters in the granite, and as coatings and platelets in the quartz-filled fracture-filling vein systems. Pyrite, chalcopyrite and fluorite are reported companion minerals.

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The veins which strike generally at 100° and dip steeply (80°) to the southwest are usually 5-10 m long, 2 to 10 cm in width and have a density of about one vein per metre across strike, with 5 to 8 per metre occurring locally. These veins conform to the major jointing direction. The molybdenite is concentrated along vein walls. The total mineralized zone is about 80 m in width and extends discontinuously to some 800 m in an east-west direction. A sample of quartz vein material assayed by the Chinese mission (1975) carried 0.14 per cent Mo and a sample of nearby biotite granite reported 0.010 per cent Mo.

Recently the deposit was re-examined by geologists of the Phnom Penh Institute of Technology, who determined that the monzonite showed a concentrically zoned character indicative of an intrusive dome, a favourable environment for molybdenum mineralization. Their studies indicated an overall grade of 0.036 per cent Mo and an available reserve of approximately 10 to 20 million tons (assuming a 50 m depth.)

The showing has been known for a long time and has been examined by various groups. It appears that the generally low level of mineralization, its dispersed character and the so far limited size of the zone preclude any prospect of economic interest for the occurrence.

References: 11, 18, 21, A. Ejov (personal communication), ESCAP data files.

2. Phnom Den, Takeo province

Near to the Vietnamese border in this area there occur a number of small hills formed by a very coarse biotite calc-alkaline granite. The easternmost of these (Phnom Den) bears showings of white quartz veins mineralized with molybdenite. The site is about 45 km south of the town of Takeo. Location: 104° 53'E, 10° 36'N.

The intrusive granite stocks and plugs are of Upper Cretaceous (79.3 My) age and are of a petrologic type not known elsewhere in Cambodia; they appear to be similar to and are probably genetically related to the Tri Ton granites of adjacent Viet Nam. The country rocks and setting are unknown, as the hills rise from the alluvial plains.

The molybdenite occurs as coatings and seams in a network of quartz veins, often long but discontinuous in character, accompanied by lesser chalcopyrite and pyrite. The BRGM geologists considered the prospect to merit further examination and suggested that more detailed prospecting should cover Phnom Den, with extension to the adjacent Phnom Bayang if results were indicative. There are to date no known reports of sampling or assay data. It may be noted that there is a reported past production of a few tens of tons of molybdenite concentrates from related granitic rocks at Tri Ton, 20 to 30 km to the southeast in Viet Nam.

3. Phnom Thong, Preah Vihear province

Surface showings are reported at the location 35 km south of the town of Tbeng Meanchey at 105° 00'E, 13° 20'N. The showings are located in the Phnom Deck district, approximately 12 km WSW of the town of Rovieng. Molybdenite mineralization occurs in disseminated form in a network of small discontinuous quartz veins within a monzonite granite of probable Jurassic-Cretaceous age which intrudes Lower Jurassic (Lias) sandstone breccias. No sampling or assay data are available. The occurrence was reported by the Chinese geological mission of 1960-1962; it appears to be only of mineralogical interest.

References: 10, 18, 21, 41, ESCAP data files.

9. Silver

There appear to be no native sources of exclusively silver ores in Cambodia, notwithstanding the well-known production of chased and embossed silver handicraft work, one of the specialties of local craftsmanship.

Silver occurrences have been reported in association with the Bo Sup Trup gold deposit, the Phnom Kambor (or Bar) gold occurrence, and also with the Sam Rong group of sulfide mineral occurrences in

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Kompong Speu province, where argentiferous galena is reported at Phnom Thmar Prak. More recently, press reports (1991) have indicated the discovery of silver occurrences by the Department of Geological Survey and Mines in the areas of the Kchol massif (western Kompong Chhnang province) and near Siem Reap (Odder Meanchey province).

References: 21, Mining Magazine (London, Aug. 1991).

10. Tin

1. Knong Ay occurrences, Kompong Speu province

Cassiterite was discovered in an area on the eastern side of the Cardamom mountains, in southwest Cambodia in 1964. Subsequently intensive prospecting was carried out by the Bureau de Recherches Geologiques et Minieres of France for the Government of Cambodia in 1964-66, and a final report was issued in 1967 (1). The conclusion was that no exploitable concentrations of cassiterite had been discovered either in the parent rocks or in the alluvium, but that this first discovery of tin mineralization in Khmer Republic might point the way to more extensive deposits being found in other geologically favourable areas.

The tin mineralization occurs at a locality known as Knong Ay (104° 07'E, 11° 33'N), in the province of Kompong Speu, 95 km west of Phnom Penh. The primary mineralization is of hydrothermal type and occurs in a network of veins and dykes of microgranite, aplite, greisen and quartz. These veins and dykes cut across the marginal zone of the granite massif of Knong Ay, itself a southward extension of the larger monzonite granite massif of Kchol. Some of the veins and dykes also cut the enclosing Triassic and Devonian-Carboniferous sedimentary rocks, which are altered locally by contact metamorphism. The intrusions and the mineralization are considered post-Triassic and probably Lower Jurassic, and have been dated at 188 Ma b.p.

Cassiterite occurs disseminated in the veins of aplite and above all of greisen, in tiny grains ranging from 0.12 to 0.5 mm diameter. Some of the quartz veins contain a little galena, sphalerite, chalcopyrite, barite and fluorite. There are also occurrences of scheelite and wolframite associated with the cassiterite. The greisens themselves have a reported mean Sn content of 0.02 per cent Sn. (ESCAP file-note).

Downslope from outcrops of the cassiterite-bearing veins there are discontinuous elluvial and alluvial deposits of cassiterite. The alluvial cassiterite was investigated and found to be of limited areal extent and extremely fine-grained: 95 per cent of the cassiterite grains less than 0.5 mm in diameter and 62 per cent less than 0.12 mm, hence indicating the inexploitability of the deposits. Elluvial and primary occurrences equally were found to be inexploitable, partly because the greisen and aplite veins themselves have been comparatively little affected by weathering, making the recovery of the cassiterite uneconomic.

In spite of the apparent inexploitability of the Knong Ay deposit, the discovery of tin-bearing granite in Cambodia for the first time was significant, and the Bureau de Recherches Geologiques et Minieres recommended a careful examination of other granite intrusions in the country. This was especially encouraged in the western parts of the country by discoveries of cassiterite in adjacent parts of southeast Thailand at about the same time.

The occurrence of tin was also investigated in numerous other valley deposits in western Cambodia by the BRGM group, notably in the valleys west of Daun Penh and Prachuol, which also receive drainage from the Knong Ay massif. Notwithstanding some encouraging local values, no economic concentrations were found. Given the deterioration in the economics of tin since that time, much more favourable indications would be required at this time to be of any potential economic interest. References: 1, 21, 41.

11. Tungsten

In Cambodia, tungsten mineralization is rare and of very limited scale.

The only occurrence so far examined has been in the Knong Ay district (104° 06'E, 11° 35'N), where traces of wolframite accompany the cassiterite (refer to section on tin). The mineral occurrence is of no economic interest. References: 4, 21.

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B. INDUSTRIAL MINERALS AND CONSTRUCTION MATERIALS

12. Clay minerals

Clay mineral occurrences and showings are very numerous and occur in almost all parts of the country. They are derived either from the recent clay deposits of the central alluvial plains (the Tonle Sap plain and the Mekong plain) or from the deep weathering and alteration of previously existing igneous rocks (usually granitic or rhyolitic species).

The clay resources can be generally classified into three groups: those suitable as refractory materials (brick, tile, furnace linings etc.), those of a quality adequate for ceramic manufacture (kaolinitic clays) and those whose quality, composition and location makes them useful for portland cement manufacture.

Investigations by BRGM (8, 9) and by geologists of the Chinese mission of 1960-1962 (10) examined a large number of occurrences, which form the primary sources for much of the data on the principal occurrences described in this section. It should be noted however that these descriptions are far from complete because there are hundreds of small pottery works all over the country using local clays.No country-wide survey of clay resources, quality and potential has even been undertaken although this is an important primary material for local manufacturing.

According to Bakker (1993, ESCAP, personal communication) the aluminous clays of the Haute Chhlong, long considered for their potential as a source of bauxite, do not (wherever observed) contain sufficiently pure horizons of grit-free clay to offer possibilities for quality clays for ceramic or refractory applications.

(a) Refractory and ceramic clays

1. Phnom Krom, Siem Reap province

This deposit is located on an inselberg at the village of this name, 6 km south of the town of Siem Reap on the road to the Tonle Sap Lake, and near its shore. Location: 103° 49'E, 13° 17'N.

The clays are derived from the decomposition of tuffaceous horizons in the rhyolitic rocks of presumed Triassic age. Beds dipping generally to the south at dips of 30° offer relatively easy access, although the altered tuffs show numerous intercalations of rhyolitic or sandy layers. Reference: 21.

2. An Dong Srey, province of Kampong Chhnang

This area is located a few kilometres southwest of the town of Kompong Chhnang, one of the principal centres of the ceramic industry in Cambodia.

A few kilometres south of the town, on the east side of the road to Srok Ach Romeas, 2 km south of An Don Srey (Prey Khmer), is a deposit of residual clay derived from an underlying granite. Location: 104° 39'E, 12° 09'N.

Here the kaolinic clay occurs as fillings in irregular pockets, with a capacity of the order to one cubic metre, formed as a result of the deep weathering of a Lower Jurassic biotite granite rich in potassic feldspar. No analysis results are available (9).

The deposit was worked to supply clay to a ceramics factory at Kompong Chhang, whose main product is cups (half million were made each year) for catching tapped rubber. At the place where the clay is obtained, it is slightly spotted with iron oxide stains. This gives the fired product a yellow colour reducing its value as chinaware.

3. Phnom Traok, Kampong Chhnang province

Phnom Traok is a small promontory situated approximately 5 km southwest of Kampong Chhang, along the road to Phum Phsar Romeas (Srok Ach Romeas). Location: 104° 39'E, 12° 13'N.

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Here at a point 300 m west of the road lies a depoosit of white compact kaolinitic clay containing very little iron. At another location, a similar distance on the east side, near the base of a granitic slope there is another occurrence of similar white clay. The clays, of good ceramic quality, have been worked for a long time on the artisanal scale. The kaolinic clay occurs in a more or less continuous weathered seam of undetermined thickness formed as a result of the deep weathering of a Lower Jurassic biotite granite rich in potassic feldspars. No analysis results are available. Reference: 8, 9 39, ESCAP data files.

4. Pre Kak, province of Kompong Cham

Located 30 km northeast of Kompong Cham on the west shore of the Mekong River, a kaolinitic clay deposit is located near the village of Prek Kak, also given as Phum Prek Kan. Location: 105° 32'N, 12° 15'N.

A small deposit has been sporadically worked at this site. The resource is potentially large, however. The clay deposits are part of a thick sequence of ancient alluvium which covers several square kilometres, occurring at several levels in ancient terraces of the river. The principal bed is some 20m thick, and consists of white feldspathic clay with some fraction of quartz grains and rock fragments.

The deposits have been related by some authors to the deep weathering of Cretaceous rhyolites (41, 42), or to subjacent granitic massifs of Mesozoic age (21). These provenances remain conjectural. No analysis data are available.

In 1967 an investigation was made (8) to see whether sufficient quantities and quality of clay were available for making pottery and chinaware. This appeared to be the case from the on-site survey, but further mineralogical and chemical tests were to be the basis of a decision whether or not to open a quarry and screening plant for preparing the clay for shipping. The results of these studies are not available for this report (ESCAP data files).

5. Trasey district, province of Pursat

See reference to Section 6.15, Ornamental and Craft Stones, to the description of the pagodite depoosit at Trasey. Location: 103° 45'E, 12°00'N.

The pagodite is in fact a kaolinitic clay variety of compact habit, formed as a result of the deep weathering of Lower Cretaceous rhyo-dacitic volcanic rocks. Measured reserves are estimated at 200,000 tons (21).

In view of the apparent lack of high-quality kaolin deposits in Cambodia, an investigation was undertaken in 1967 by BRGM geologists to learn whether the pagodite deposit at Trasey, which has been worked since ancient times as a source of ornamental and craft material, could serve as a ceramic raw material.

The survey found that in certain sectors of the ceramics industry, the pagodite could serve as china clay. An admixture of a certain alluvial white clay would provide the plasticity which the pagodite lacks. The pagodite would ensure an adequate alumina content in the mixed clay.

Trasey is a remote hamlet about 110 km south of Pursat, and the deposits are grouped 3 to 4 km west of the village. The present isolation creates a major problem in exploiting the pagodite.

The most accessible of the deposits was investigated in detail and found to have reserves adequate for both sculpture and ceramic use for several decades. The distribution of pagodite suitable for statuary and for ceramics was adequately determined. The other deposits in the area, being less accessible, were not examined in detail (9, 39).

DEPARTMENT OF MINERAL RESOURCES

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(b) Clay for cement manufacture

1. Phnom Sar (Phun Trapeang Sala), Kampot province

The clay pit of Phnom Sar is situated southeast of the main Phnom Penh-Kampot road (national route 3) at km 141, approximately 10 km northeast of Kampot. Location: 104° 16'E, 10° 39'N.

The deposit is 7.5 km from the cement works at Chakrei Teng, which it supplies with clay extracted from the argillaceous beds of the Kampot delta alluvial plains.

The clay deposit was surveyed and tested by a mission from China in the early sixties and a concession of 12 hectares was granted to the cement company, comprising a calculated 813,000 tons of clay suitable for its purposes. The bed of clay (of recent age) has a regular thickness of 5 m. Up to the end of 1967 about 100,000 tons had been removed. Because of the limited size of the available reserves, a UNDP-assisted investigation was carried out in 1967, and was able to extend the known area of the clay deposit to 30 hectares. Also it was found that there was more than one suitable clay horizon in depth in some of the drill-holes, and that in places the thickness of utilizable clay might exceed 10 m. A partial calculation over seven blocks, each of 40,000 sq m, gave proved reserves of 2.1 million cubic metres of clay. Assuming an overall specific gravity of 2.6, this would be about 5.5 million tons, thus there are adequate reserves to supply the cement works for many years.

References: 8, 9, 10, 21, 39, 41, 42, ESCAP data files.

13. Construction materials and dimension stones

Raw materials for architectural and construction stone are not lacking in Cambodia. Dolerites, sandstones, basalts, granites, limestones and laterites are present in various parts of the country. Quarries have been opened in areas of outcrop situated favourably with respect to transportation and markets, in or near urban and industrialized zones usually, and near road or rail access.

The Triassic sandstones constitute the best building stone resource in the country. Quarry works are found in the Srang hills, to the southeast and south of Kompong Speu, as well as in numerous isolated hills between Phnom Penh and Takeo.

Basalts are quarried in the areas around the borders of the volcanic massifs of Kompong Cham, Snoul and similarly located towns.

Granites are currently exploited as road material and construction aggregate at Phnom Basset, 20 km northwest of Phnom Penh, and sporadically in the region around Kompong Chhang. It may be noted that none of the large granitic massifs (Knong Ay, Kchol, Phnom Tbeng, Phnom Ker,for example) are being exploited, doubtless due to their distance from principal centers.

Limestones are extensively used as road material in the western part of the country, most notably around Sisophon, Battambang and Kampot, where Permian limestones are abundant.

Rhyolites, probably of the Jurassic-Cretaceous volcanics, are found and quarried around the flanks of several hills to the north of Phnom Penh, along the road toward Skon. Some silicified horizons form an extremely tough and durable rock.

The younger sandstones ('gres superieurs') of Jurassic age, which outcrop widely is western Cambodia, are also extensively used for road material and construction, notwithstanding their often poor degree of induration. Some horizons near the base of the series are more completely cemented by iron oxides and form a more durable material and are extracted for flagstone and facings. The fine-grained graywacke beds in these formations are especially desired and are often seen used in temple construction throughout Cambodia.

Quartzites of the Cambro-Silurian are also locally used in the region of Takeo and Kep for road material, railroad ballast and sometimes for construction.

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Laterite blocks are quarried and used as building stone material. Laterite is often used around temples for construction of walls, pavements, and foundation stones.

Marbles are available in the Chvang district northwest of Stung Treng, but exploitation has been very limited.

There is apparently no organized building stone or dimension stone industry in Cambodia. Although the industry is highly developed in Thailand and an abundant variety of stone raw materials is available (as seen in this brief summary), a local industry based on these resources has not yet developed. Demand for ornamental and dimension stone will no doubt grow in Cambodia and neighbouring countries with increased levels of economic development. References: 21, 33.

14. Fluorite

A small deposit of fluorite is reported at Ba Phnom, in Prey Veng province, approximately 60 km southeast of Phnom Penh. Location: 105° 22'E, 11° 14'N.

At this location a number of small quartz veins which traverse outcrops of leucocratic alkaline granites carry weak fluorite mineralization.

The site was re-examined in 1992 in a diploma study by a student of the Phnom Penh Technical Institute. He reported that fluorite veins up to 4 cm in width can occur as fracture fillings in alaskite granites of presumed Jurassic age. Green, violet and colorless varieties were observed. References: 21, 23, ESCAP data files.

15. Graphite

There is a report of beds of graphite in the Permian limestones of Phnom Chac Khley, near Tuk Meas in southern Cambodia. Location: approximately 104° 30'E, 10° 40'N. Also in brecciated limestones at Phum Kaoh Sla, in the upper valley of the Kampot river, also in southern Cambodia. Location: 104° 15'E, 10° 54'N. References: 41, p. 106.

16. Gypsum and Anhydrite

No information exists of any significant gypsum-anhydrite deposits in Cambodia, but it is expected geologically that such deposits may well occur in Mesozoic red-bed formations or the Quaternary alluvial deposits covering much of the country.

No rock salt nor potash occurrences were known in the country. The Mesozoic red-bed formations covering large areas in eastern and southwestern Cambodia may quite possibly contain saline horizons, but formations corresponding to the major salt-bearing formations of northeast Thailand and Laos are believed not to exist in the country and no prospects for comparable thick and widespread salt beds are expected in Cambodia. Reference: 38.

17. Heavy mineral sands

Beach sands with high proportions of heavy minerals (mainly ilmenite) occur at intervals along the east coast of Viet Nam and at a few points along its south coast, and have been prospected. Interest has also been expressed in heavy mineral sands on the southeast coast of Thailand in the area of Trat. Ilmenite and zircon are generally the dominant minerals where investigated, followed by rutile and monazite. The character of heavy mineral suites depends on the terranes from which the major rivers derive their sedimentary load. The Cambodian coastline, from Koh Kong southward to Kompong Som and Kampot, has very clean sands with few heavy minerals because rivers in that sector drain the Cardomom Highlands, which is a terrane of rather clean sandstone (sensu lato) rocks.

East of Kampot to the Vietnamese border, streams drain frorn an upland of more variegated metasedimentary, metavolcanic and granitic rocks. Hence there may be some potential along this 60 kilometre reach of coast for heavy minerals of commercial value.

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18. Limestone and Dolomite

(a) Limestones

Limestone here refers to the commercial rather than the sedimentological usage of the term. Such limestones, suitable for the production of lime for construction purposes and cement manufacture, are generally classified as rocks containing at least 80 per cent calcium and magnesium oxide which, when calcined, give a product that slakes upon the addition of water. Such rock is an important raw material for the construction and chemical industries. To be commercially useful, the rock must not only have a reasonably high and consistent purity, but must also occur in beds of sufficient horizontal and vertical extent to make for practical quarrying, and in locations near the available transportation system and potential markets.

In Cambodia the most common suitable unit for limestone resource use is the Permian limestone series of the western provinces. These outcrop as inselbergs and ridges in the provinces of Battambang, Pursat, Kampong Speu and Kampot. As well, there are a few areas where limestones of the Carboniferous age may meet the necessary standards for utilization (for example, in Stung Treng province).

Two districts which offer potential raw materials for limestone extraction are the Battambang-Sisophon area and the Kampot area.

1. Battambang – Sisophon district

This district can be described as two areas, which can be located thus:

Battambang:	103° 00'E, 13° 00'N (approx.)
Sisophon:	102° 50'E, 13° 45'N (approx.)

Numerous outcrops and inselbergs of Permian limestones of which many are situated near available transportation links. Some of these hills show near-horizontal bedding, while others are severely faulted or folded. Many have already been quarried for limestone or road material, and in some locations the limestone is of chemical grade purity (generally, greater than 95 per cent Ca CO_3).

Large reserves are available. It remains to classify the deposits on the basis of their chemical quality, available tonnages, dimensions and proximity to transport and markets.

2. Kampot district

In the Kampot district, the principal source of supply has been at Bos Dambang, situated 22 km northeast of Kampot city. Locations: 104° 16'E, 10° 39'N and 104° 21'E, 10° 44'N.

Here lie a number of outcrop areas of the Permian limestones, less imposing than the outcrops of the same units in the Battambang or Tuk Meas areas, but well situated with respect to transportation and markets, and of good chemical quality. The available limestone beds have a thickness of about 125 m (41, 42) and reserves are estimated at 90 million tons (19).

The Cambodian cement industry has concentrated exclusively on the Kampot limestones, which are very conveniently situated on the main Phnom Penh – Kompong Som transport arteries (39). The first cement plant was inaugurated at Chakrei Ting, 14 km northeast of Kampot, in 1964. This is at Km 131 on the road from Phnom Penh. Production in 1967 was 58,000 tonnes against national consumption of about 170,000 tonnes, the remainder being imported. An expansion of the Chakrei Ting plant, started in 1967, was to increase capacity to 110,000 tonnes in 1969. A second cement factory was planned to have a design capacity of some 300,000 tonnes per annum.

The Chakrei Ting plant was sited following an investigation by a team from China, which confirmed a deposit of cement-grade limestone in the vicinity (minimum reserves 9 million tonnes) as well as a clay deposit 8 km away (reserves 813,000 tonnes) suitable as a cement ingredient. The iron and gypsum necessary for cement manufacture were imported, as no suitable indigenous raw materials were found near the plant site.

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A second investigation was launched in 1966/67 to evaluate additional limestone and clay raw materials in the Kampot district in view of the expansion of the industry. The target was an additional 50 million tonnes of cement-grade limestones and 10 million tonnes of clay. The mission was able to show reserves to 90 million tonnes of cement-grade limestone on the southeast side of Phnom Laang, a hill 9 km east of the Chakrei Ting factory. As part of the systematic prospection, 69 samples of limestone were analysed and the results showed that no impurity was present in impermissible quantities. In general, the limestone is high in calcium and silica; iron, alumina and sulphate are all low.

Production ceased in 1970, reportedly because of security problems (42, 39).

Other large reserves in the same stratigraphic horizon of the limestones may be found in the hills to the east of Bos Dambang, notably at Phnom Loang and in the Tuk Meas area. Location: 104° 32'E, 10° 40'N.

3. Stung Treng district

There are a number of areas in the north and north-central parts of the country which offer outcrops of limestones, most often in the Devonian-Carboniferous terranes. Large areas of these units exist to the east, north and northeast of Stung Treng city in the province of the same name.

This is a region in which future cement works may be justified, in view of the great distance from the available cement plants in the southwest of Cambodia (Kampot) and in the case that future large amounts of concrete are required for the construction of power or transportation installations on the Mekong River system.

Geologists of the Chinese mission in 1975 examined limestones in twelve locations in Cambodia, mostly Permian, and analyzed 17 samples. Some of the locations yielded material of very high purity. Since the sites are named but not located accurately in their report the data is unfortunately of limited use (11).

In conclusion, Workman has noted (41) that much of the lime production of eastern Cambodia (and of central Viet Nam) is believed to be from impure limestones found mainly in the Pre-Cambrian and Triassic-Liassic formations; this is because the output is used essentially to satisfy local domestic and agricultural needs for lime and therefore is obtained from the raw materials that happen to be available. It follows that the siting of village limestone-burning operations for lime is not necessarily a guide to the (sighting) occurrence of high-calcium or cement-grade limestone suitable for industrial use.

(b) Dolomite

Chvang, Thalabarivat district, Stung Treng province

In this district there are numerous hills and outcrops of dolomites, saccharoidal in texture and locally metamorphosed to marbles (see Section 6-23, Ornamental and Craft Stones). The outcrops are near the village of Chvang, 43 km by road northwest of Stung Treng town, along the road from Stung Treng to Chhep. Location: 105° 46'E, 13° 44'N.

The small massifs are probably of Devonian-Carboniferous age, part of the large terrane of folded sandstones, shales and grits which extend east and west of the Mekong River in this sector of Cambodia.

The dolomites are well recrystallized, massive in structure, white to pink and often pass laterally into limestones or zones of calcareous or dolomitic marbles. The purity is quite high, although few figures are available. The one available composite analysis (ESCAP files) indicates a molecular ratio Ca:Mg of 58:42, a low SiO_2 content (0.3 per cent) and no other significant impurities. Small quantities (a few tons per year) have been extracted for use in the glass industry at Phnom Penh.

References: 11, 19, 39, 41, 42, ESCAP data files.

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19. Phosphate minerals

There are two phosphate provinces in Cambodia, the Battambang area in the western regions and the Kampot area in the south. The phosphate in each case is in the form of supergene replacement and fissure-filling deposits in weathering Permian limestones. The limestones occur as numerous steep-sided erosional outliners or inselbergs which arise abruptly from the plains (locally called "phnoms"). Evidently any of the numerous phnoms of southern or western Cambodia may contain pockets of phosphorite rock analogous to that seen near Battambang and in the Tuk Meas district of Kampot province, although these two locations are the only areas where the deposits have been significantly worked. Equally, there are almost certainly concealed deposits in or on the subsurface parts of the Permian calcareous rocks of western Cambodia.

The deposits are marine in origin and residual in occurrence. They occur as vuggy fracture-fillings, cavity-linings, cave debris and as coarse irregular clinker-like concretions in karstic openings in the vadose zones of the weathering limestones. Some nodules and cavity linings show a laminated or varved structure indicating their origin by solution deposition. The material is often a variegated yellow-brown-gray color, poorly crystalline and carries impurities of sand and clay particles. Chert pebbles and fragments are common.

The deposits apparently originated as thin beds, bands or stratigraphic laminae usually in the upper Permian (Upper-Lower Permian to Upper Permian, Kungurian to Tatarian stages) fusulinid limestones. At the weathering interface between soils and limestone, or at the exposed surface on the phnoms, the phosphate is attacked by acid waters and released to reprecipitate in fractures and cavities or as encrustations. This material eventually breaks away and accumulates on the limestone surface or as part of the soil debris in the areas of limestone outcrops and sub-crops. The mineralogy is not well known but in similar deposits (for example, Tennessee, U.S.A.) the primary mineral and also the reprecipitate is a mixed-crystal calcium carbonate-fluoro apatite (francolite) (37). Where determined, the mineral in the Cambodian deposits is most commonly a calcium carbonate-apatite, with lesser collophane in some samples.

Impurities of iron and manganese oxides are common, giving a red, brown, black or purple cast to the material.

There is some limited potential for bedded phosphate (laminar or nodular) in marine shales of the Lower Devonian which occur in the Chhep-Stung Treng belt and in other areas. Beds up to lm thick are noted in the Siluro-Devonian (Pak Chom Formation?) of the Chiang Mai and Phrae districts of Thailand (26). These may find stratigraphic analogues in the Stung Treng belt which is Devonian-Carboniferous in age.

1. Battambang province

Several occurrences of secondary phosphate accumulations are known in this province, notably near Sisophon, south of Mongkol Borei and west of Battambang town. These were examined by a Chinese geological mission in 1975. The following paragraphs and location maps are adapted from their report (11).

These geologists noted several types of phosphate depositional modes in this area, namely:

- Botryoidal (grape-like) phosphorite. Apatite francolite-collophane in radiate crystals. Gray-brown-white masses.
- Crustified or sponge-like phosphorite. Bioclastic, crustified, cellular textures. Often discolored by limonite, hematite.
- Earthy phosphorite. Brown-red, finely crystalline to massive, often including bioclastic material. Limonitic.
- Conglomeratic phosphorite. Gravels of limestone, phosphorite and quartz, poorly cemented by a clay, carbonate and phosphatic matrix. Light gray to yellow-green color.

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– Sandy-clay conglomeratic phosphorite. Gray-brown color, clastic texture of ill-sorted fragments cemented by a phosphatic and clay matrix.

The mineralogy was examined in a number of samples and can be summarized as follows:

	Francolite	*Collophane*	*Clay*	*Limonite*	*Other**
Botryoidal	93-95	3-5	1-3	0.2-2	1-5
Crustified	66-95	5-20	1-7	1-5	1-10
Earthy	60-70	–	3.25	5-25	1-2
Conglomeratic	60-65	–	20	3	1-8

* Includes calcite, quartz, chlorite, oxides of manganese. All values in per cent.

None of the deposits are large. The report indicates maximum dimensions of 30 m to 180 m but the form and morphology of the deposits are not described. According to Kun and Say (19) the majority of the materials is localized in caves and in karstic depressions.

a. Sisophon area

Three deposits are known in this area, at **Phnom Bak** (two deposits) and at **Phnom Chung Chaing** both of which lie one to two km northwest of the town of Sisophon (Location app. 102°57'E, 13°35'N). At Phnom Bak 1 (the eastern of the two), an earthy phosphorite mass 60 m long and 2 m thick forms a north-south dyke-like cavity-filling. At Phnom Bak 2,250 m to the west, outcrops of crustified and conglomeratic phosphorite are aligned in a north-south direction for a length of some 30 m. At Phnom Chung Chaing, a ferruginous earthy phosphorite mass similar to Phnom Bak 1 is seen on the south slope of the hill (see figure 6). Analytical data are shown in table 5.

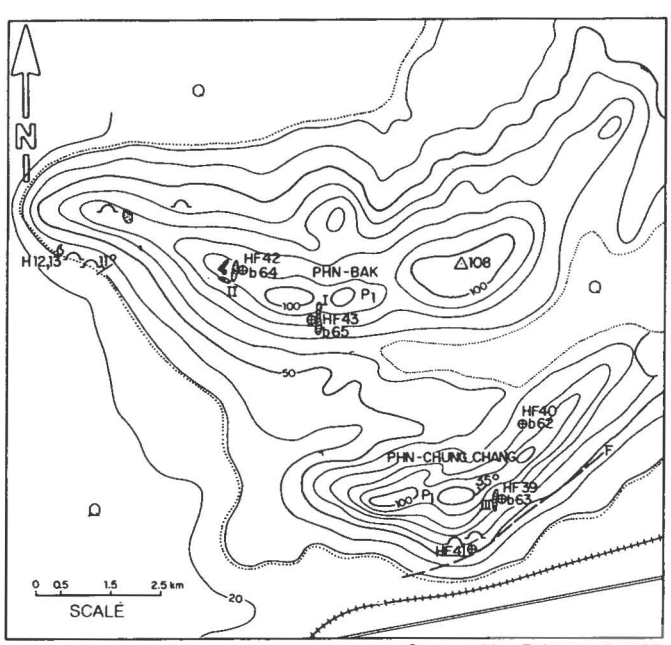

Figure 6. Geologic sketch of phosphate deposits at Phnoms Bak and Chung Chaing, Sisophon area.

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Table 5. Chemistry of phosphate deposits, Sisophon area

Deposit	*Ore Type*	*Components*							*Solubility*	
		P_2O_5	*Soluble* P_2O_5		*Insoluble residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
			Acid	*Neutral*					*Acid*	*Neutral*
Ph. Bak1	clayey	23.34	7.94	2.59	23.48	10.22	34.40	2.38	34.01	11.09
Ph. Bak2	encrusted	33.76	15.78	6.04	8.72	6.46	41.83	1.84	46.74	17.89
Ph. Chung Chaing	earthy	22.75	8.20	3.35	–	5.65	42.44	12.00	36.04	14.72

The northwest dipping limestones are transected and moderately deformed by near-vertical faults. Steeplydipping north-south fracture fissures are the main host structures for the residual phosphorite accumulations.

b. Mongkol Borei area

Phnom Ban Teay Neang. This hill is situated about 2 km south of Mongkol Borei adjacent to the railroad (see Figure 7). Location 103°01 'E, 13°30'N.

The setting is similar to that at Sisophon (see above). Lower Permian limestones strike east-west and dip northward at 18°. On the north slope, north-south tensional fractures dominate. On the east and south slope, two sets of fractures at 100° and 60° are observed. Those fracture sets are the loci of accumulation of phosphatic debris, several bodies of which are seen on the flanks of the hill.

The south deposit is about 35 m long and 1-2 m wide, and consists of crustified phosphate with some manganiferous phosphorite lenses. The north deposit, 35 m long and 5-10 m in width, is a conglomeratic phosphorite mass occupying a large tensional fracture. Chemical data are reported as follows:

Table 6. Chemistry of phosphate deposits, Phnom Ban Teay Neang

Deposit	*Ore Type*	*Components*							*Solubility*	
		P_2O_5	*Soluble*		*Insoluble residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
			Acid	*Neutral*					*Acid*	*Neutral*
South	Encrusted	37.28	17.28	7.27	0.67	1.02	52.22	2.86	46.35	19.50
South	Mangani-ferous	32.92	14.92	5.81	7.22	2.95	47.20	2.47	46.32	17.65
East	Conglo-meratic	8.72	3.19	1.89	29.50	7.63	31.53	17.25	36.58	21.67

Phnoms Thom and Prasat: These two hills are situated about 10 km south of Mongkol Borei, located respectively on the west and east sides of the road and railroad corridor (see figure 8). (Location: 103°01'E, 13°28'N).

Here the succession of Permian limestones strikes in a general north-south direction with moderate to strong (10°-50°) eastward dips. The deposits are small, no more than 60m in length and about 2m or less in greatest width. The available chemical data are as follows (the first three samples are from the south end of Phnom Thom, and the remaining sample is from Phnom Prasat):

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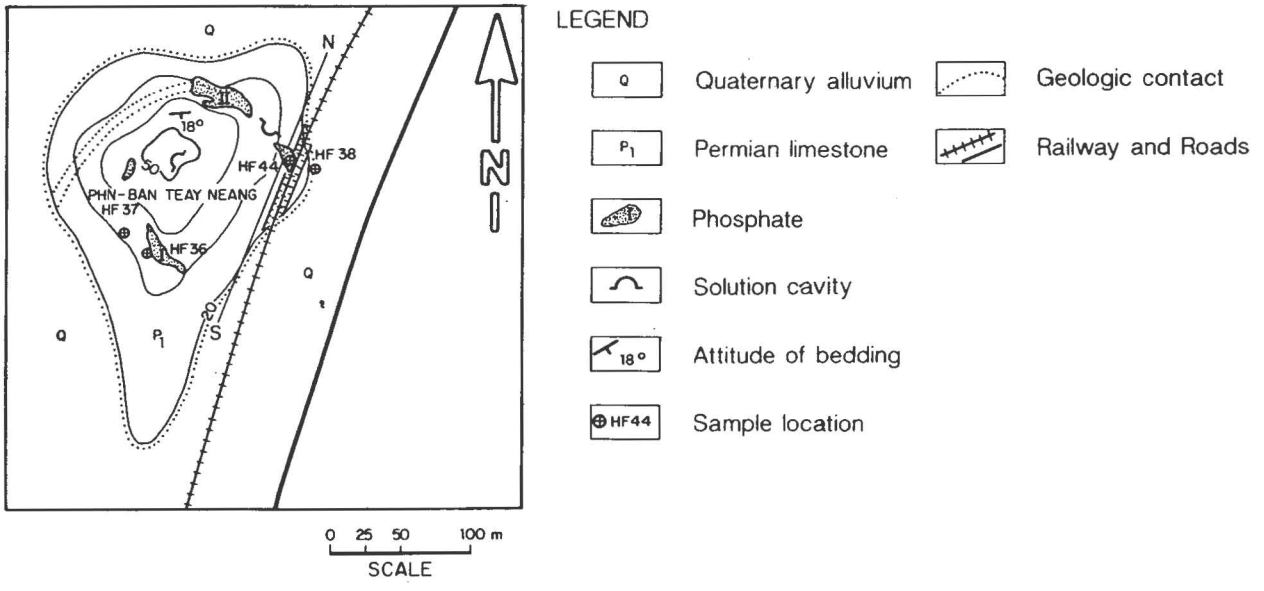
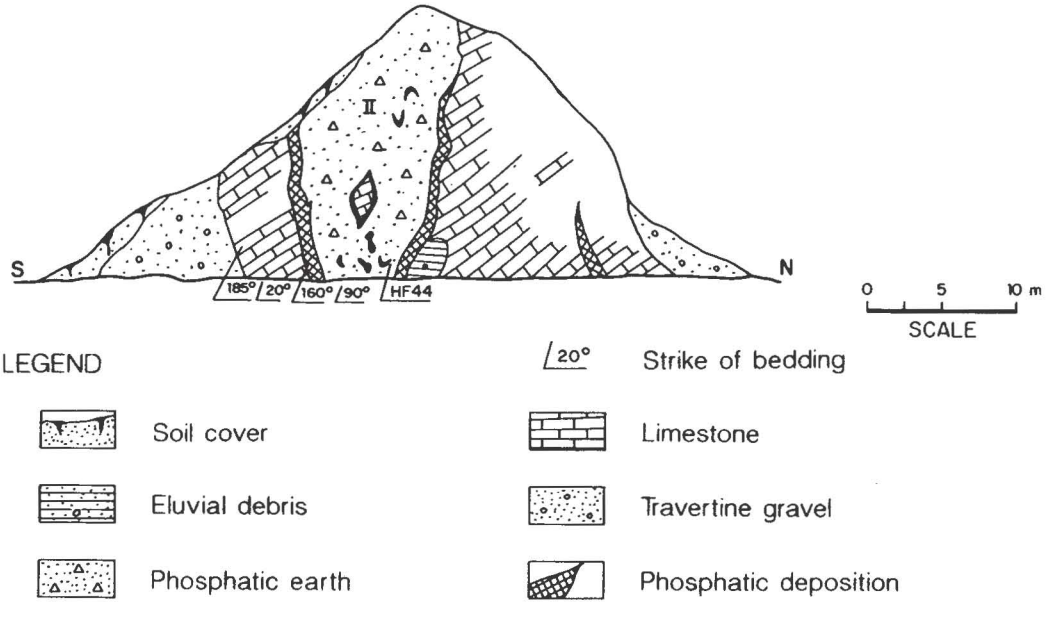


Figure 7A. Geological sketch of phosphate deposit at Phnom Ban Teay Neang, Mongkol Borai area.



Source : After Reference II, 1975.

Figure 7B. Geological S-N cross-section through east flank of the Phnom Ban Teay Neang phosphate deposit

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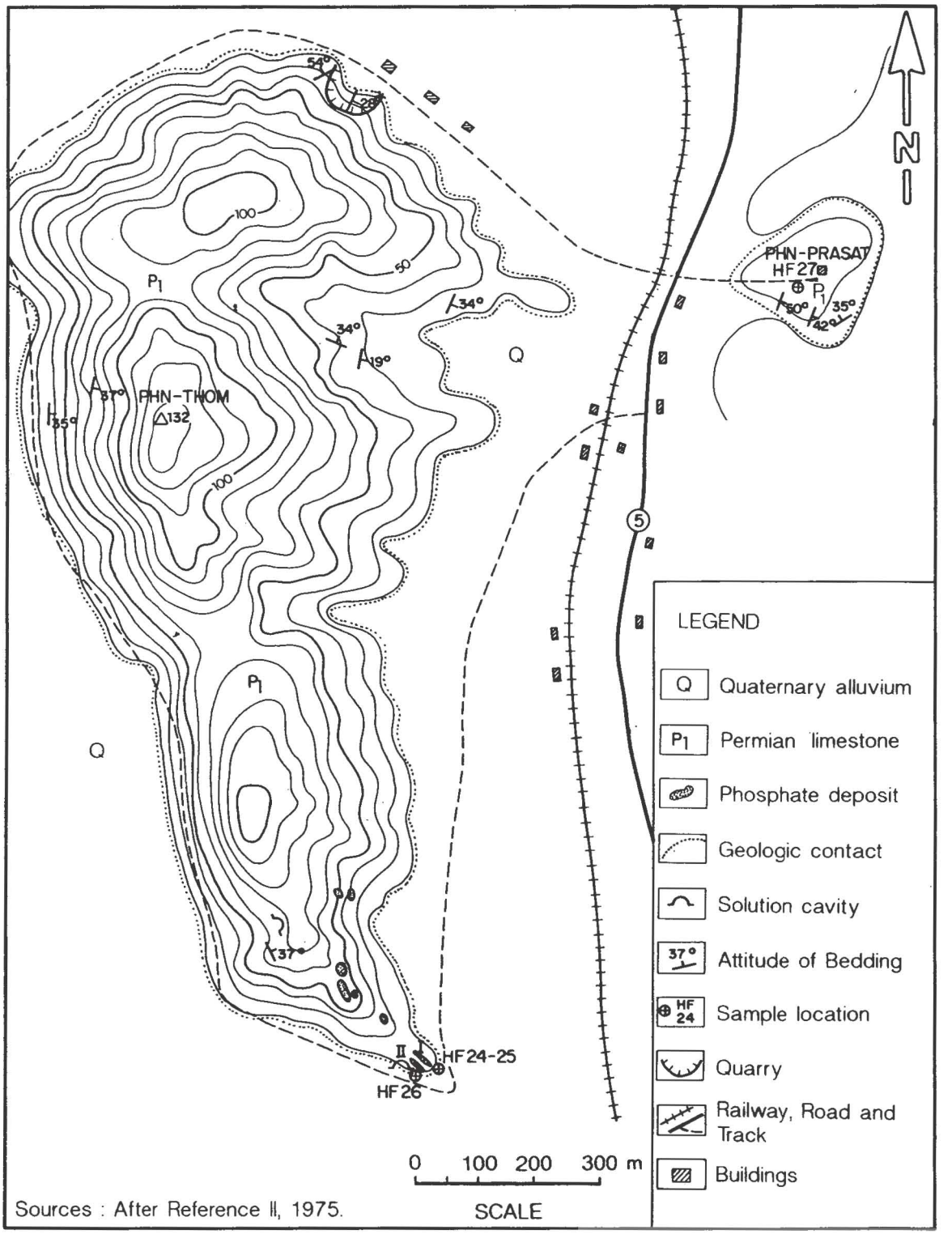

Figure 8. Geologic Sketch of Phosphate deposit at Phnoms Thom and Prasat, Mongkol Borei area.

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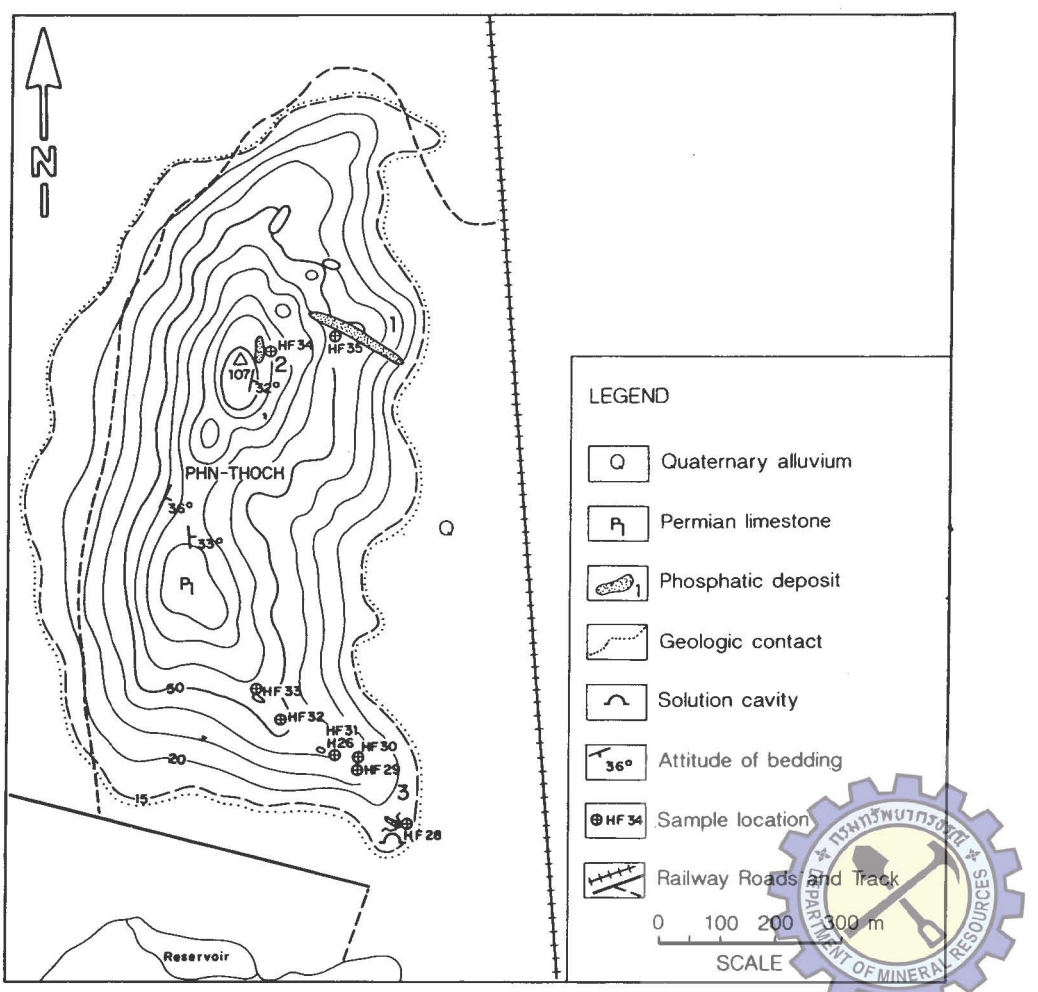
Table 7. Chemistry of phosphate deposits, Phnoms Thom and Prasat

Ore Type	*Components*							*Solubility*	
	P_2O_5	*Soluble* P_2O_5		*Insoluble residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
		Acid	*Neutral*					*Acid*	*Neutral*
conglomeratic	19.82	11.61	5.10					58.57	25.73
botryoidal	30.13	19.23	8.19	19.03	3.35	40.34	2.35	63.82	27.18
encrusted	31.22	18.84	7.92	17.12	1.56	43.28	2.81	60.35	25.36

Phnom Thoch: This hill is located immediately west of the railroad about 14 km south of Mongkol Borei and about 3 km south of Phnom Thom, to which it is similar in lithology and structure (see above and figure 9). Location: 103°01'E, 13°24'N. Seven phosphatic deposits are known, all on the east face, all occupying verticallyoriented solution-modified fracture zones in the limestones. The largest of these (1 on figure) is 180 m long and about 4 m wide, and comprises crustified phosphate intermixed with earthy phosphorite in the lower part, and conglomeratic material in its upper part. Deposit 2 consists of crustified and conglomeratic phosphate and is about 50 m long and to 2 m in width. Deposit 3 is smaller, approximately 17 m long and 1-2 m wide and consists of porous conglomeratic phosphate. Chemical data is reported in table 8.



Source : After Reference II, 1975.

Figure 9. Geologic Sketch of Phosphate deposit at Phnom Thoch, Mongkol Borei area.

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Table 8. Chemistry of phosphate deposits, Phnom Thoch

Deposit	*Ore Type*	*Components*							*Solubility*	
		P_2O_5	*Soluble* P_2O_5		*Insoluble residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
			Acid	*Neutral*					*Acid*	*Neutral*
1	Botryoidal	37.85	23.60	10.97	0.92	1.32	51.88	3.20	62.35	28.98
2	Conglo-meratic	24.85	11.14	4.60	31.21	4.45	32.39	1.46	44.86	18.51
3	Sponge-like	33.26	22.86	10.46	12.16	2.06	44.04	2.11	68.73	31.44

c. Battambang area

Phnom Sampeou: This hill is located approximately 15 km southwest of the town of Battambang, along the road leading to Pailin. Location: 103° 02'E, 13° 04'N. It is one of a group of hills at this location (a similar group (Phnom Thmei et al) lies about 5 to 7 km to the northwest) all of which bear phosphate accumulations (Phnoms Krapeu, Khong, Kdaong, Banteuy, Banon etc.) but Phnom Sampeou is the most important of these (figure 10). It is located at 103° 06'E, 13° 01'N.

The deposits on the northeast part of the hill have been exploited in the past. The phosphate occurs as phosphate-cemented limestone breccias in solution zones and in irregular pockets in the Lower Permian limestones of these inselbergs. The general strike of the rocks is north-south, with shallow dips; NW-SE striking steeply-dipping debris-filled solution cavities and karstic depressions, probably developed along fracture zones, traverse the limestones. The largest of these is reported to be 40m long and 1-2m wide. Chemical data is reported in, tables 9 and 10.

Table 9. Chemistry of phosphate deposits, Phnom Sampeou

Sample No.	*Ore Type*	*Components*							*Solubility*	
		P_2O_5	*Soluble* P_2O_5		*Acid residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
			Acid	*Neutral*					*Acid*	*Neutral*
HF 50	botryoidal	37.41		6.74	2.00	2.20	50.46	2.46		18.01
HF 52	encrusted	32.59		5.08	5.47	4.12	46.49	4.23		15.59
HF 51	sandy-conglome-ratic	26.74		4.82	19.95	15.33	27.77	1.23		18.02
HF 56	green-yellow phosphorite	23.17	4.76	2.05		37.78		1.00	20.54	8.84

Soils in the area also carry considerable phosphate, and can themselves be utilized for direct-application phosphate fertilizer, as seen in samples from the vicinities of the following phnoms (table 10).

d. Exploitation of the Battambang province deposits

The phosphate deposits have been known and intermittently worked for many decades. There are references to production in the 1920, and according to an annual report of the Mining Bureau of Indochina of 1927-1928, the three hills Sampeou, Khong and Krapeu held a possible reserve of 0.4 to 1 million tons assaying 20 to 30 per cent P_2O_5. In 1930 a French group undertook commercial production but after a short period the operation ceased due to financial difficulties (37).

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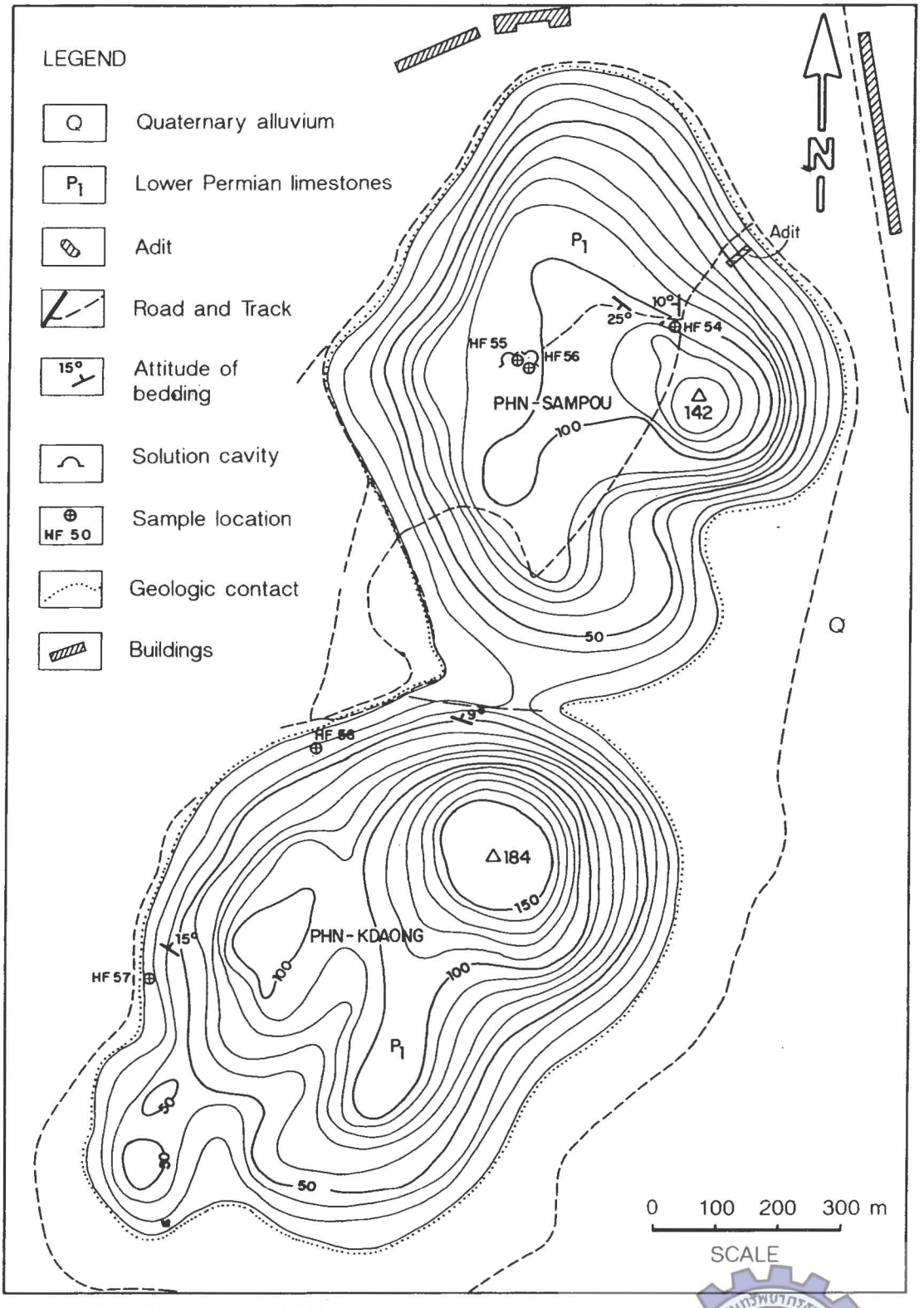

Source : After Reference ll, 1975.

Figure 10. Geologic Sketch of Phosphate deposit at Phnoms Sampeou and Kdaong, Battambang area.

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Table 10. Chemistry of phosphate soils, Phnom Sampeou area

Location	*Sample No.*	P_2O_5 (%)	*Soluble* P_2O_5		*Humic acid*	*Solubility*	
			Acid	*Neutral*		*Acid*	*Neutral*
Phnom Banteuy	HF 48	3.11	2.90	2.84	0.65	93	91
Phnom Sampou	HF 55	4.08	3.76	3.07	0.80	92	75
Phnom Krapeu	HF 61	6.81	6.26	5.54	0	92	81
Average		4.66	4.31	3.82		92	82

In 1960 the Societé Financière du Cambodge undertook a study of the deposits, and arrived at a much smaller reserve figure of 60,000 tons probable and an additional 300 000 tons of possible ore. (Reference: 41, p.87).

Interest in production revived in the 1960s. Quarrying methods were employed to mine the deposits at Phnom Sampou and at Phnom Thom. The ore was transported to Battambang for fine-grinding and then transported by barge along the Mekong River to Saigon. The phosphate rock so produced had a P_2O_5 content of about 20 per cent (41). Chemical data were reported as follows (11) on these samples from the Battambang plant:

Table 11. Chemical data on phosphate product, Battambang plant

Sample No.	*Grain size (mm.)*	P_2O_5 %	*Soluble* P_2O_5 %		*Solubility*		*Average* P_2O_5 %
			Acid	*Neutral*	*Acid*	*Neutral*	
HF45-1	>0.18	22.13	11.27	1.72	50.92	7.77	
HF45-2	0.18-0.149	21.07	10.76	2.07	51.07	9.82	21.68
HF45-3	<0.149	19.85	10.34	2.71	52.09	13.65	

In 1975-1978 the Phnom Thom plant was enlarged and rehabilitated through local funds and the assistance of some non-governmental aid organizations. According to one report (39), it has continued to operate since that time at a rate of 4000 tons per year, but the plant at that time required further improvement. According to Kun and Say (19), both the Phnom Sampeou and Phnom Thom deposits were being operated in 1978.

There is some potential for increased production in this area, probably using a combination of artisanal mining, organized (commercial-scale) mining and haulage, and improved centralized plant processing operations (14).

2. Kampot province

The general setting of the phosphate deposits of Kampot province is like that in Battambang province. Most phosphate occurs as a residual product of weathering on and near isolated inselbergs of Permian limestones, formed by leaching and redeposition of the phosphate as earthy nodular masses. See figure 11.

The deposits are 40 m to 80 m in length in general. The shapes of deposits is controlled by fractures and karstic fissures, mainly oriented in a NW-SE direction following primary jointing and faulting directions in the limestones. The same textural and morphological variations are seen as are noted in the Battambang district, with the addition of two other types:

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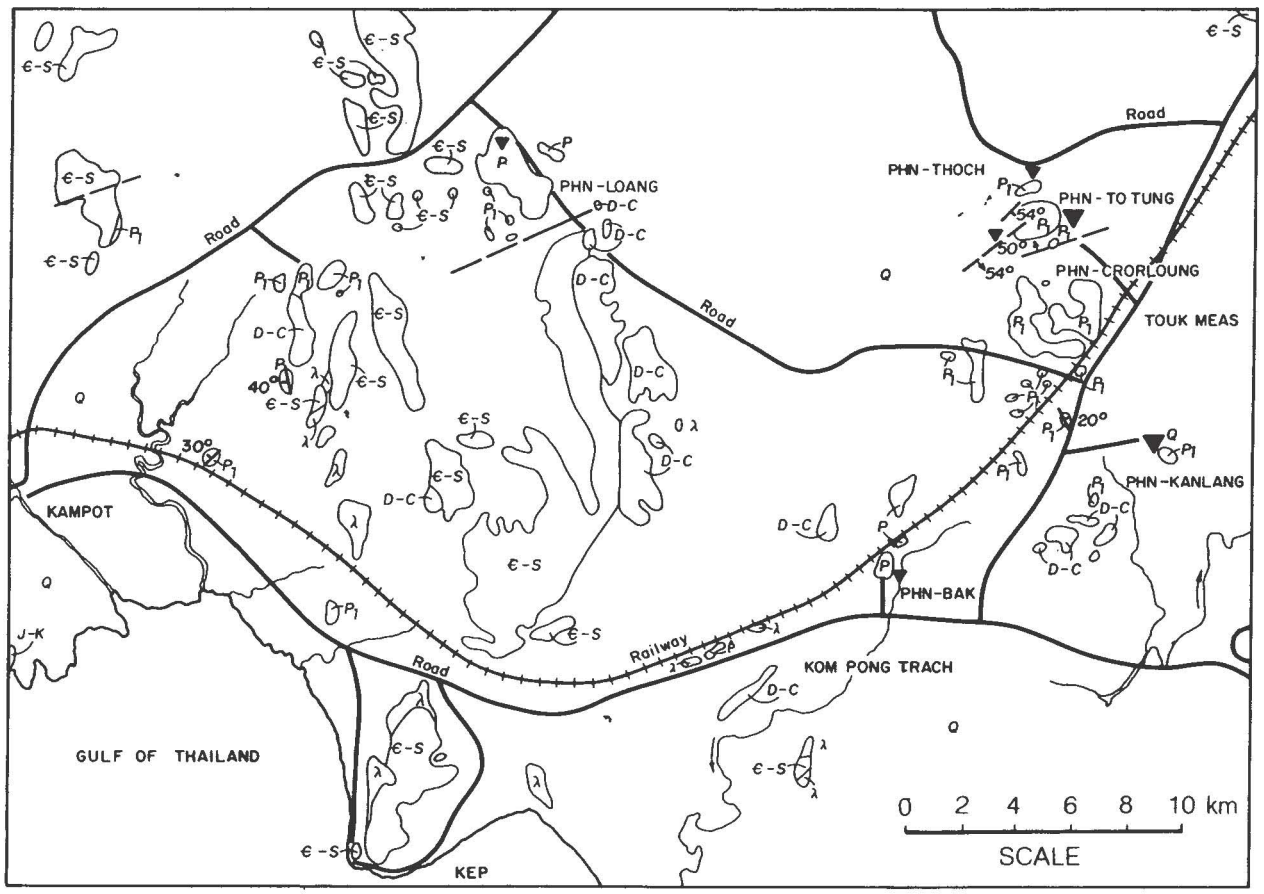

The boundaries and names shown on this map do not imply official acceptance or endorsement by the United Nations

Source : After Reference II, 1975.

LEGEND

Q	Quaternary	β	Basalts
J - K	Jurassic-Cretaceous grits		Geologic contact
P_1	Lower Permian limestone		Assumed fault
P	Permian undifferentiated	54°	Attitude of bedding
D-C	Devono-Carboniferous, Seds.	▼	Phosphate deposit
ϵ - S	Cambro-Silurian schists	▼	Phosphate occurrence
λ	Rhyolite		

Figure 11. Geologic map of the Kampot province phosphate district

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Brecciated phosphorites: Fragmented phosphate material cemented by carbonate, or siliceous fragments cemented by phosphate material. Gray-yellow-brown colors, cryptocrystalline.

Silty phosphorite: Silty fine texture, perhaps cross-bedded, gray-white-yellow-green colors.

The phosphate deposits of Kampot province are mainly located in the Tuk Meas district, although other occurrences are known, notably at Phnom Loang, 22 km WNW of Tuk Meas, and at Phnom Bak, about 2 km north of Kampong Trach.

The mineralogy of the phosphates in this district is shown in the following table.

Table 12. Mineralogy of phosphate deposits – Kampot province

Ore type / *Composition*	*Colloidal encrusted phosphorite*	*Brecciated phosphorite*	*Sponge-like phosphorite*	*Silty phosphorite*	*Sandy-conglomeratic phosphorite*	*Manganiferous phosphorite*
Carbon apatite	75-93	>52-70	60	62-85	<25	55.60
Collophane						
					<15	
Calcite	1-5	0.20	1	trace-1	40-45	
Quartz & chalcedony	0.5-15	5-32	3-5	5.35	<25	0-10
Manganese oxide	0.5-20	0.2-1	3-5	0-5		25-38
Iron oxide		0-1	10		<25	0-5
Illite			20			
Montmorilonite	0-5	0-7		0.2-5	<15	2-20
Mica	0-trace			0-1		0-3
Chlorite				0-2		

Data from Chinese Geological Mission report, 1975. All data in per cent.

The dominant mineral is carbonatic apatite or francolite. Collophane is less important.

a. Phnom Totung, Tuk Meas district

The Phnom Totung deposit is located about 4 km NW of Tuk Meas at 104° 32'E, 10° 41'N. The deposit occupies a fracture (fault?) system traversing the hill striking N 40° W and dipping near-vertically. There is also a large alluvial/eluvial fun of phosphate-bearing talus and debris. (See figure 12).

ESCAP conducted a detailed examination of the deposit in 1992 (25) and noted three material types in the fissure system:

1. Concretionary phosphate rock, consisting of very well indurated phosphate concretions in a matrix of clay, iron oxides, manganese oxides and phosphatic material. This is the most important phosphatic lithology;

2. Phosphate rock breccia, consisting of angular clasts of black phosphate material in a matrix of lighter colored silica and clay material. Their angularity suggests they originated as karstic collapse debris;

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3. Limestone block breccia cemented by lime-phosphatic material, dark brown-red in colour.

The ESCAP mission noted a large deposit of phosphate material in the outwash fan derived from the fissure deposit. This outwash fan covers 23,000 m^2 and is over 210 m long, over 150 m wide and on average 3 to 4 m thick. There are two-lithologies: a lower Yellow Zone (2-3 m), presumably a residual deposit on the extension of the fracture zone, and an upper Red Zone which is a talus debris deposit of gravel, cobbles and boulders (average 1 m thick) derived from weathering of the concretionary material in the fissure.

A reserve of 56,000 tons was calculated for the fissure zone. The outwash fan was estimated to hold a minimum of 22,000 tons and a possible maximum of 49,000 tons. Moreover, there is additional potential reserve along strike in the fissure zone and across strike in the limestone block facies of the deposit, whose full width could not be observed.

Chemical composition of the Phnom Totung phosphatic materials is shown in table 13.

Table 13. Chemical analysis of phosphate materials, Phnom Totung area, Kampot province

Ore Type	*Components*							*Solubility (%)*	
	P_2O_5	*Soluble* P_2O_5		*Insoluble residue*	Al_2O_3 + Fe_2O_3	*CaO*	CO_2		
		Acid	*Neutral*					*Acidd*	*Neutral*
brecciated phosphorite	9.26	6.11	1.74	46.52	8.6	21.03	8.34	65.98	18.79
brecciated phosphorite	23.12	12.00	2.95	19.71	3.09	38.46	8.68	51.90	12.75
sponge-like phosphorite	24.10	12.57	6.03	23.77	5.18	25.68	2.23	52.16	25.02
siliceous brecciated phosphorite	20.53	14.03	6.06	41.87	2.05	27.76	2.83	68.34	29.52
silty phosphorite	22.96	4.15	2.09	22.98	22.77	15.45	1.50	18.07	9.10
sandy-conglomeratic phosphorite	8.14	1.42	1.15	14.26	7.79	39.59	26.48	14.13	

The Chinese mission (11) also noted that the soils in the vicinity of the hill were enriched in phosphate, carrying up to 8.02 per cent P_2O_5.

Phosphate occurrences are also reported on and around nearby hills to the south (Phnoms Chau Tong, Kean Khlong, Kan Sat). These cannot be explored, however, because of the presence of unexploded military ordinance. They indicate the possibility of future additional reserves.

The figure from the Chinese mission report also indicates a similar deposit on the southwest side of Phnom Totung (see figure 12).

The main Phnom Totung deposit is considered to have originated as a karstic filling deposit which accumulated in a linear solution-created void which coincided with the axial keel of a syncline in the Permian limestones. As the limestone dissolved away along the axial joint system, widening the void, phosphatic material accumulated upon progressive dissolution of the upper limestones above, depositing clay, phosphate nodules from the rock, collapsed limestone blocks and re-deposited phosphatic fragments in this axial zone. Secondary weathering and mass washing caused the phosphatic material to form the alluvial fan adjacent to the hill. The deposit on the southwest side of the hill might be the outcrop of the phosphoritic limestone unit on the southwest limb of this syncline (rather than a similar fracture zone), although the little structural data available leave this open to question.

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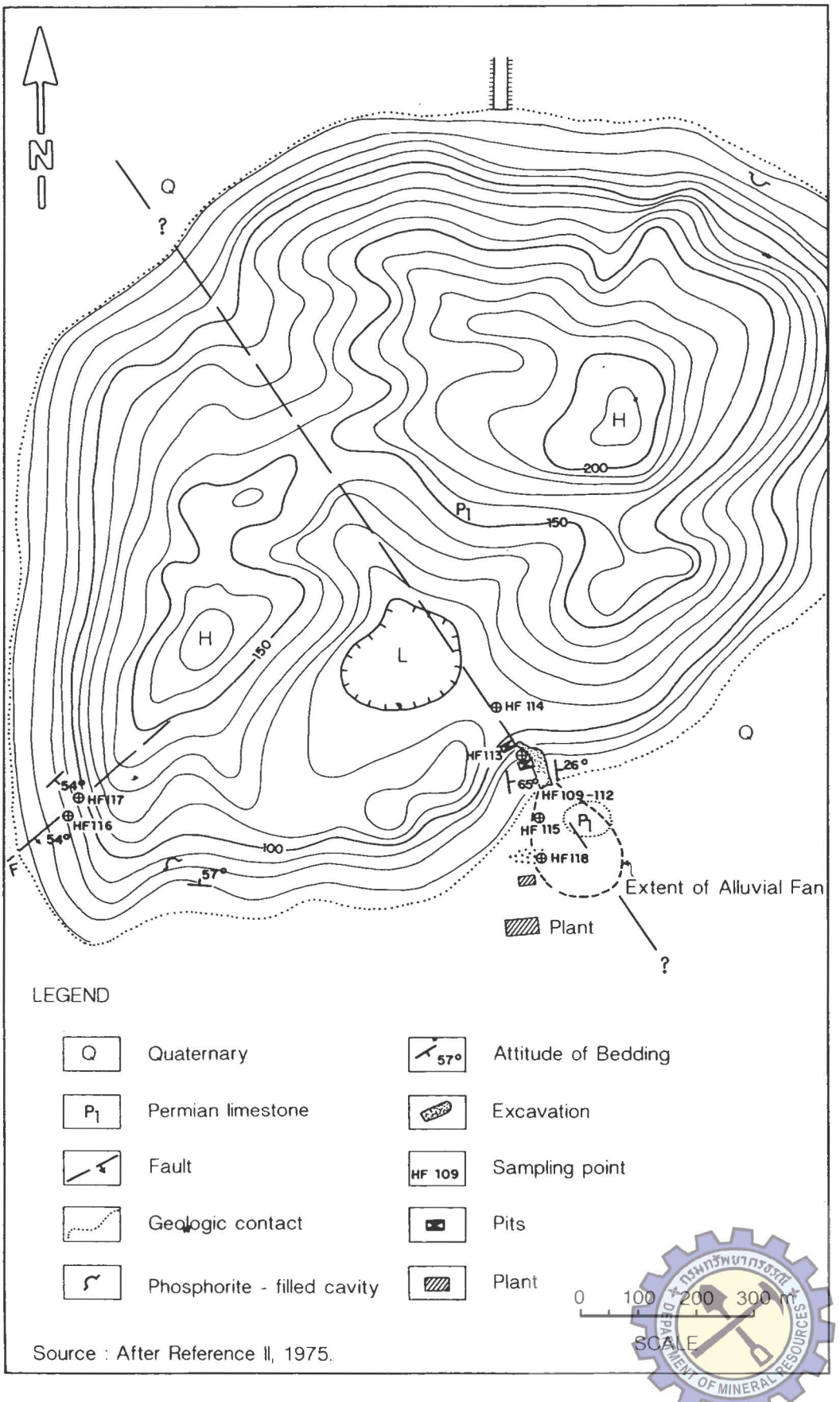

Figure 12. Geologic Sketch of Phosphate Deposit at Phnom Totung, Tuk Meas District.

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b. Phnom Kanlang, Tuk Meas district

Phnom Kanlang is a prominant hill 6 km south of the village of Tuk Meas. Location 104°34'E, 10°38'N. It is reached by a small local road which leads eastward off the Tuk Meas – Kampong Trach road at a point 6 km southwest of the former.

There are three phosphorite exposures and the geologic settings are very similar to those at Phnom Totung. See figure 13.

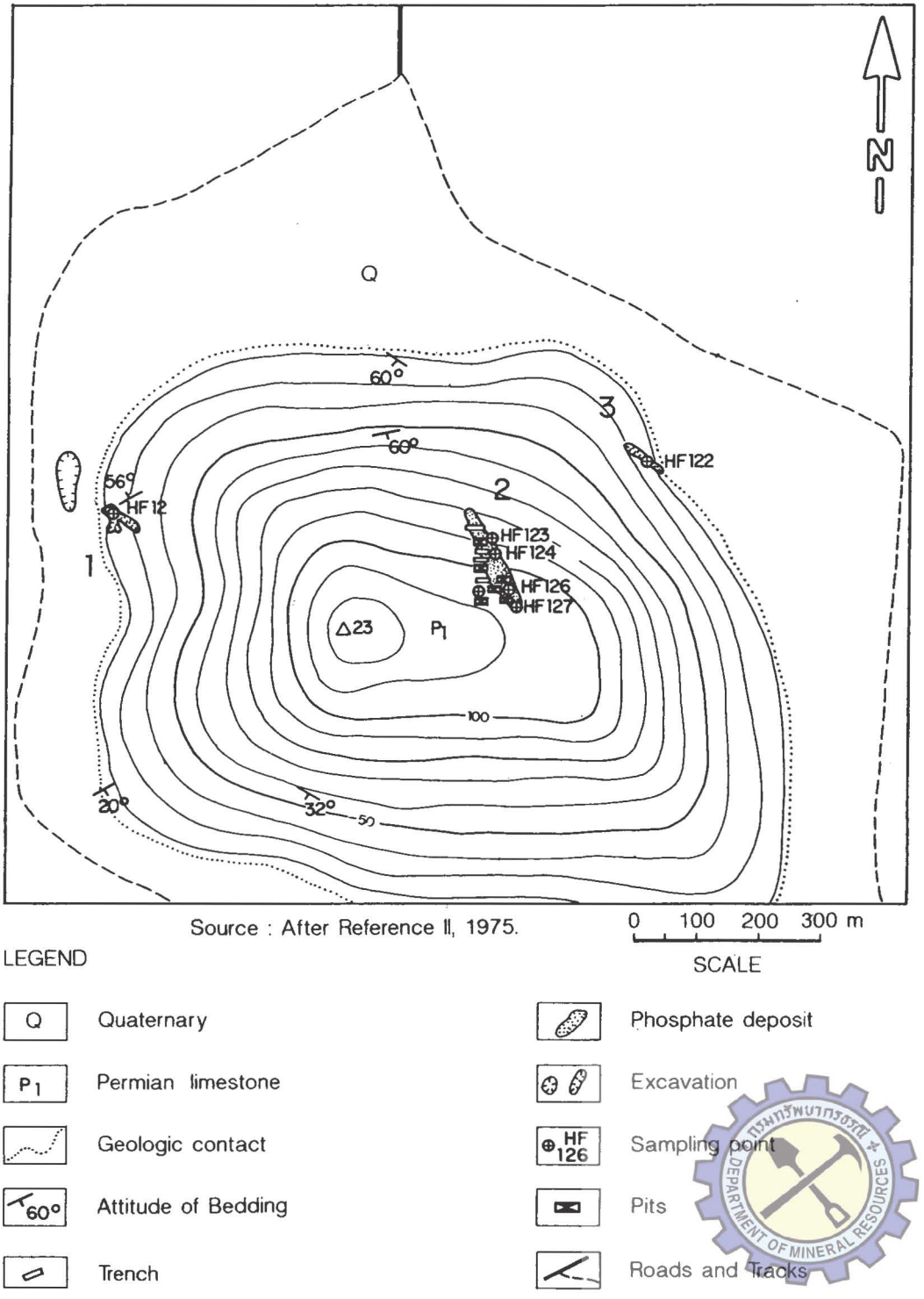

Figure 13. Geological Sketch of Phosphate Deposit at Phnom Kan Lang, Tuk Meas district.

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Deposit 1 occurs at the west slope of the hill at the joint of two fissures of directions NW and NS respectively. It is 40m long and 1-2m wide and consists of colloidal-crustified, banded phosphorites with some manganiferous phosphorite. Deposit 2 occurs in the middle part of the north slope of the hill, around 60m from the bottom. It is 60-70 m long and 10m wide, striking 160°, and is dominated by brecciated phosphorite and subordinate silty phosphorite. The ores are always mixed with siliceous and carbonate fragments. Deposit 3 strikes in the 130° direction on the east slope of the hill, extending 40m long and 4-8 wide, and consists of colloidal, crustified, pisolitic and microcrystalline phosphorites.

3. Exploitation of the Tuk Meas phosphate deposits

The deposits at Tuk Meas were mined between 1919 and 1930, with recorded output in 1930 of 1100 tons. The operation became uneconomic and production ceased. In early 1960, interest renewed when the Societe Financiere du Cambodge re-examined the deposits. This investigation indicated a 350,000 ton reserve of material with P_2O_5 content between 11 and 26 per cent, an average solubility factor of 57 per cent, and fertilizer qualities superior to imported ground phosphate if ground to a 100-150 micron screen-size. The reserve in 1968 was measured at 320,000 available tons. This is some five times larger than the estimate of the main reserve at Phnom Totung and probably includes other nearby deposits and perhaps some nearby phosphatic soils.

A small plant with a rated capacity of 12,000 tons per year of fertilizer was opened in July 1966, and in 1968 reached a production level of 11,330 tons. Material was crushed and blended to yield a product rated at 20 per cent P_2O_5 content, suited to direct application on acidic rice soils. This plant was set up by SOKPHOS, a largely state-owned company, and the product was distributed by OROC, the state fertilizer monopoly. The plant was destroyed twice during the 1970s, first by bombing and later by looters. The plant was rebuilt on another site in the early 1980s and by 1988 was producing some 5,000 tons per year.

The operation of this plant was inhibited by two factors; first, the lack of suitable methods and equipment to exploit the relatively rich (35 per cent) upper level deposits needed to maintain product grade quality, and second, the lack of adequate storage and transport equipment to effect efficient product distribution. As a result, the plant ceased operation in 1991 owing to lack of demand for the product (14).

To address the first problem, a 2-man mission composed of a geologist and a mining engineer was organized by ESCAP in 1992 to (1) assess the reserve and to locate 50-100,000 tons of material to support a 5 to 10 year mine life at an expanded rate of operations and (2) produce recommendations for an improved mining and production plan to process such a volume. As indicated earlier a reserve of 88 to 105,000 tons of available material in the primary fissure deposit and its outwash fan was indicated. Recommendations were made (40) for mine engineering and processing improvements.

At the time of writing, arrangements are being awaited for funding to implement these recommended actions (14).

4. Cheam Phosphate-Marble prospect

At Phnom Cheam (103° 54'E, 11° 41'N) a number of small outcrops of marble, derived from contact metamorphism of the Permian limestones by the granitic intrusions of the Kchol massif, are found south of the former village of Cheam (now called Roleak Kang Choeung). Phosphate accumulations are seen in the karstic depressions and cavities.

The deposit has been described (21, 42) as an accumulation of phosphatic-cemented oolin breccia in irregular pockets in the marbled limestones. No analytical data are available and no estimates of available tonnages have been recorded.

See the reference to Cheam under marbles, Section 6.23.3.

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5. Phosphate demand and utilization

The demand for phosphate fertilizer has been difficult to measure in recent years, owing to unsettled conditions and the availability in some years of heavily subsidized fertilizer imports. Similarly, the production of local phosphate fertilizer has varied widely.

Fertilizer experts at ESCAP and local authorities suggest that local demand for locally produced phosphate for direct application purposes (such as has been produced at Tuk Meas and Battambang) is probably in the area of 6000 to 8000 tons per year, and that the figure would increase in future years given continued economic stability and a growing economy (14).

This quantity appears to be readily achievable, since maximum production at Tuk Meas itself reached 9000 tons in 1987 (14). The demand for the product is however dependent on improved and constant quality (say 20 per cent P_2O_5 and 6 per cent available phosphate) and improved handling characteristics (the material produced now is too finely powdered for convenient application).

Such future requirements will require a programme of action which includes the following elements:

1. The characteristics (geology, mineability and phosphate assays) of the various small deposits should be identified for their future potential exploitation;

2. Quality control, by planned extraction and materials blending, should be introduced;

3. Product characteristics (granulometry, solubility) must be standardized and improved through better processing.

20. Silica sands and glass raw materials

1. Coastal sand deposits

Quartz sands suitable for glass manufacture are excavated at Tuk Sap (103° 41'E, 10° 35'N) for the local glass container industry. This is a beach sand deposit located around the headland from the village of Tuk Sap, 20 km ESE of Kompong Som to O Tres (103° 37'E, 10° 32'N), 15 km SSE of that town. Sands in this area have a very high purity and whiteness, and their proximity to Highway No. 4 has improved their utility. The sand which has an average grade of 99.3 per cent $Si0_2$ is shipped to the state glassworks near Phnom Penh.

Another deposit, with similar purity and quality has been described at Phnom O Chrou, east of Tuk Sap. Location: 103° 48'E, 10° 33'N.

Very pure beach sands also are found at several locations on the east side of Kompong Som Bay (See figure 14). Analyses of the chemistry and granulometry of sand samples from a variety of locations along this shore are given in tables 14 and 15.

Data are from the report of the Chinese Geological Mission of 1975 (11). Areas on this coastline which have been tested include:

Phum Sralau (103° 42'E, 11° 05'N),

Phum Nesath (103° 40'E, 11° 00'N),

Prek Krapeu (103° 42'E, 10° 56'N),

Phum Stoeng Thma (103° 43'E, 10° 53'N), and

Phum Thma Rong (103° 41'E, 10° 46'N).

Beach sands of these qualities are found at many points around the western coast from Koh Kong in the north to Kompong Som and along the south coast from Kompong Som to Kampot. These sands are the product of the weathering of the sandstones and grits of the Elephant Range and the Cardamom highlands to the north and east.

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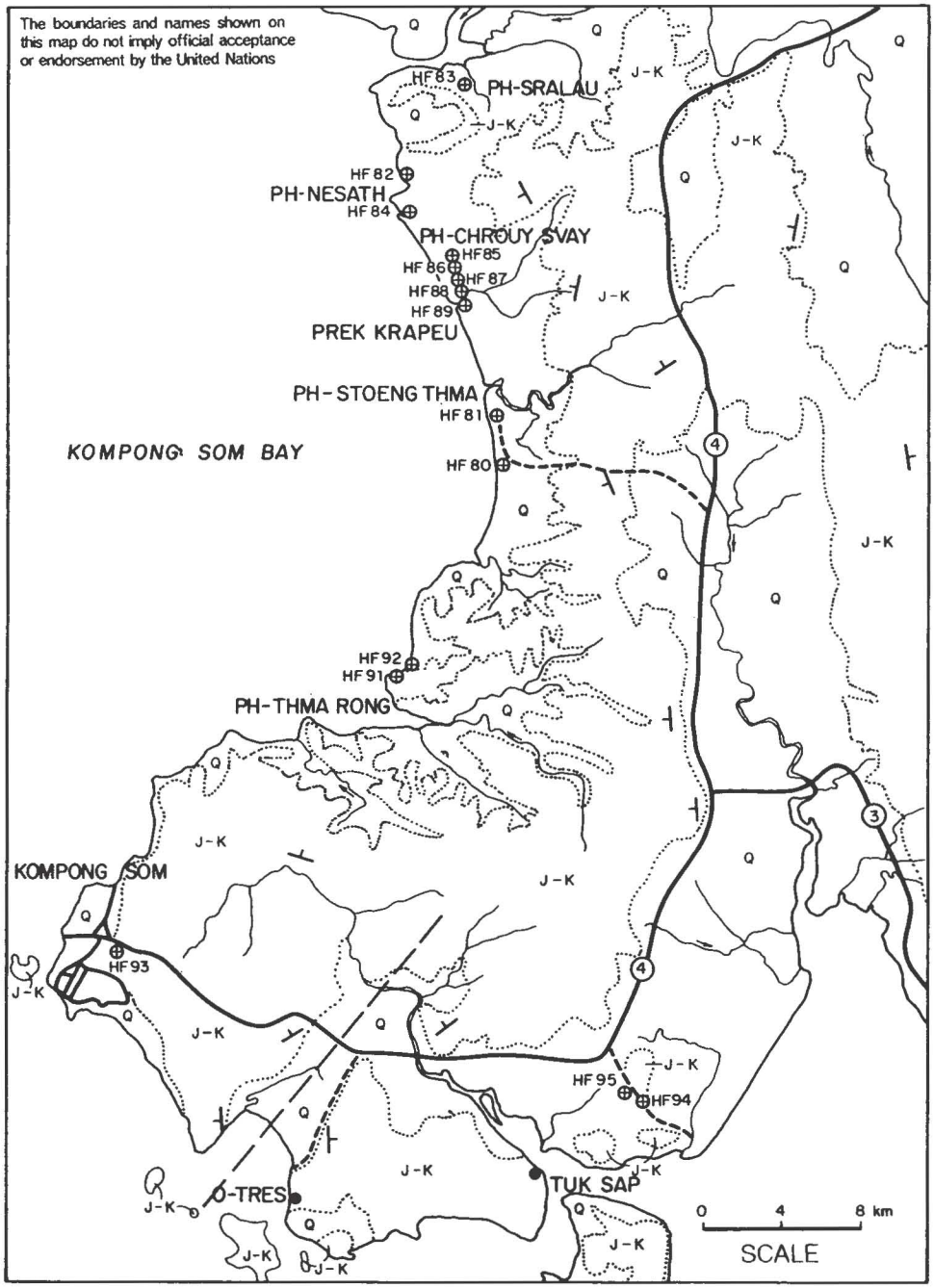

Figure 14. Geological map of the Kompong Som Bay area.

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Table 14. Chemical analysis of Kompong Som district glass sands

Sample location	*Sample no.*	*Chemical composition (%)*				
		SiO_2	Fe_2O_3	Al_2O_3	Cr_2O_3	TiO_2
1. ph. Stoeng Thma	H F80	99.64	0.017	0.08	0.0003	0.024
2. ph. Stoeng Thma	H F81	99.55	0.018	0.10	0.0007	0.043
3. ph. Nesath	H F82	99.66	0.021	0.09	0.0005	0.035
4. ph. Sralau	H F83	99.05	0.044	0.39	0.0010	0.093
5. ph. Chrouy Suay	H F84	99.62	0.018	0.09	0.0005	0.043
6. Prek Krapeu	H F85	99.52	0.016	0.17	0.0006	0.050
7. Prek Krapeu	H F86	99.20	0.017	0.09	0.0008	0.055
8. Prek Krapeu	H F87	99.59	0.014	0.14	0.0004	0.046
9. Prek Krapeu	H F88	99.62	0.012	0.11	0.0004	0.033
10. Prek Krapeu	H F89	99.34	0.024	0.08	0.0006	0.031
11. Prek Krapeu	H F90	99.65	0.012	0.09	0.0006	0.035
12. ph. Thma Rong	H F92	99.59	0.019	0.14	0.0002	0.013

Table 15. Granulometry of Kompong Som district glass sands

Sample Location	*Grain size (mm)*	*1-0.50*	*0.5-0.25*	*0.25-0.10*	*0.1-0.05*	*<0.05*
1. ph. Nesath	R3		29.50	69.50	0.50	
2. ph. Chrouy svay	R4		2.00	72.00	25.00	1.00
3. Prek Krapeu	R5	2.80	31.50	59.00	5.00	2.00
4. Prek Krapeu	R6	2.60	10.90	78.40	7.50	0.50
5. ph. Thma Rong	R8		82.50	82.50	1.20	

These uplands are formed by the Jurassic-Cretaceous "grès superieurs", fine to medium-grained sandstones of considerable purity (96-98 per cent SiO_2) which are poorly cemented. Once eroded, they are further cleaned by fluvial transport and the action of long-shore drift currents in the ocean, and are thus accumulated as very pure beach sands. Prospecting and testing of silica sand resources in this area was continued as recently as 1991. The resources are practically limitless, within the constraints of chemical, mineralogical and granulometric quality requirements. Current extraction rates of glass sand raw materials are not known.

The glass industry in Cambodia

Glass was produced at four locations in the 1960s including the state-owned company Verrierie d'Etat, at Sung Meanchey. The production was interrupted during the 1970s but one plant resumed operation in 1979, and another in 1986. Glass products are in quickly rising demand in Cambodia at the present time. Total production capacity is about 40 tons per day, almost all of which is used for locally-required containers and other consumer goods. Much of this raw material is supplied from the Tuk Sap area. References: 11, 39, ESCAP data files.

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2. In-situ Silica deposits: Tram Khnar

An occurrence of high-silica jasper is reported at Tram Khnar, just east of the village of that name. Tram Khnar is 30 km SSW of Phnom Penh on the Route Nationale No. 3. Location: 104° 45'E, 11° 17'N.

Here a sequence of cryptocrystalline cherts and jaspers (phtanites) of Devonian-Carboniferous age outcrop in a series of five small hills lying along an anticlinal ridge oriented with N60W strike. The meta-sedimentary rocks were recrystallized by the intrusion of the nearby Tonle Bati granites, creating a cryptocrystalline highly pure quartzite or jasper from the pre-existing sandstones, of such purity that the material has been taken for vein quartz (which is also present at this site). The material represents a potential silica raw material but is probably not economic in the face of the availability of abundant silica sands in southwest Cambodia. The rock has been excavated for use as railway ballast. References: 21, 42.

21. Zirconia

1. Bokeo district

The Bokeo district is described more fully under the section on gemstones, as the district is primarily known as a source of gem-quality zircons, accompanied by lesser corundum and garnet stones. Location: 107° 10'E, 13° 40'N.

The district is also a potential source of zircon for industrial use. Under the present system of artisanal mining, much zircon of small size or low (non-gem) quality is discarded. Zircon has industrial uses as an abrasive and as a refractory agent. It is also a source of zirconium and hafnium oxides, compounds with a number of industrial/chemical applications. Bokeo zircons, while relatively pure, are low in hafnium content.

Most of the world's zircon is produced from beach sands, in which the mean size of zircon grains is about 0.1 mm. The larger size and relative mineral purity of alluvial zircons, such as could be produced from deposits such as Bokeo, can command a higher price, particularly for the abrasives industry markets.

Exploitation of zirconium reserves would require regularizing and rationalizing mining in order to improve the efficiencies and economics of mining. Lacombe (20) estimated the declared production under current methods to be about 1.2 to 1.5 tonnes per year. In his study, an area of 31 hectares of alluvial deposits was estimated to have a recoverable reserve of 25,600 kg of zircon, made up of 6100 kg of gem and near-gem quality stone and 19,500 kg of industrial grade stone. It was calculated that under the prevailing system of working some 65 to 85 per cent of the zircon is irretrievably lost in rejected ground and in tailings. Nearly all the recovered zircon was sold by the kilogramme and of that only some 2.5 per cent was cut and sold as individual stones, indicating that loss and waste levels on the resource are very high, well over 99 per cent. References: 20, 21, 39, 41.

2. Other zircon districts

Although Bokeo is the best known zircon-producing district, other gem-producing areas in Cambodia (Pailin, Somlot, Chamnop etc.) also produce abundant zircon as a by-product of gem operations. Much of this material could find potential use as chemical- and industrial- grade zircon.

C. PRECIOUS AND ORNAMENTAL STONES

22. Gemstones

In Cambodia there are two principal associations of gemstones and a further group of miscellaneous stones:

1. The sapphire-ruby-zircon-spinel association, in Neogene-Quaternary basalts;
2. The quartz-amethyst-opal association, in rhyolites of Paleozoic and Mesozoic ages;

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3. Miscellaneous gem minerals-garnets, spinels etc.

1. Association 1.

The minerals of the first association are worked exclusively in eluvial and alluvial deposits derived from adjacent outcrops of Quaternary and Late Neogene basalts (basanites) in which the gemstones occur as rare phenocrysts or xenocrysts. Basalts of this age are widespread in the southern Indochina peninsula (see figure 15). The largest areas are in southern Viet Nam; however, no sapphires are known in that country, and there is only one known occurrence of zircon (at Xuan Loc). The Cambodia and eastern Thailand volcanics on the other hand have produced large quantities of gems. In Cambodia the most important areas have been Pailin in the west of Cambodia, and Bokeo in the northeast (Ratanakiri).

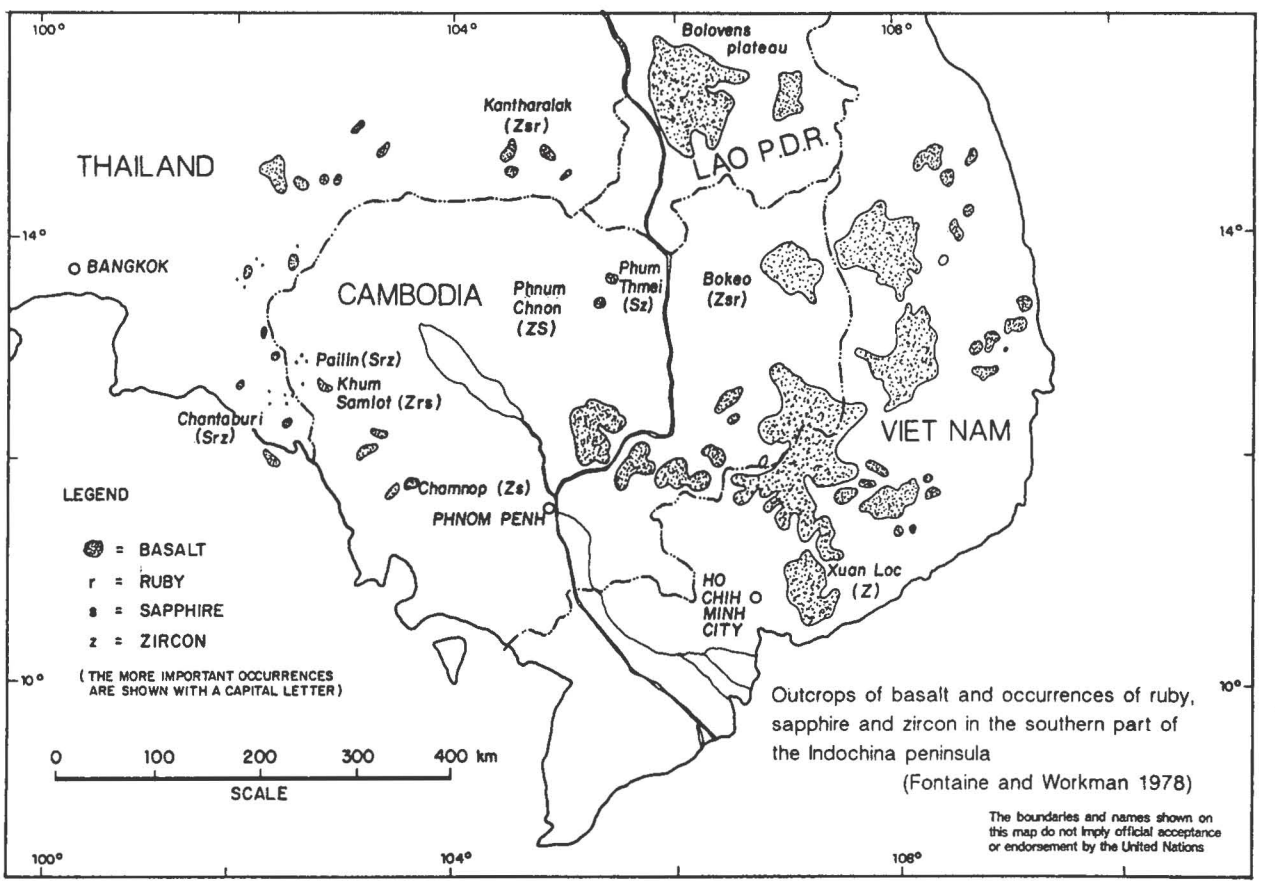

Figure 15. Gemstone Occurrences in Southern Indochina.

Gemstones of the first association also occur at Samlot, Borlang (in Battambang province), Phum Chnoun, Phum Thmei (both in the Rovieng district of Preah Vihear province), and Chamnop (Pursat province). Other smaller local occurrences are also known.

There is an observed zonation in the occurrence of these gem minerals, notwithstanding their common mode of origin. The relative commercial importance of the gem minerals at the main centres varies from west to east approximately as follows:

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	Sapphire	*Ruby*	*Zircon*
Chantaburi west, (Thailand)	1	3	3
Ban Bo Na Wang, (Thailand)	3	1	2
Ban Thung, (Thailand)	3	1	3
Bo Rai (Thai border)	3	1	3
Pailin west, (Cambodia)	3	1	3
Pailin town, (Cambodia)	1	4	3
Samlot, (Cambodia)	4	1	3
Chamnop, (Cambodia)	2	3	1
Chnuon, (Cambodia)	2	4	1
Thmei, (Cambodia)	1	4	1
Bokeo, (Cambodia)	4	4	1

(1 - dominant, 2 - common, 3 - some, 4 - rare)

In very general terms, zircon assumes more importance, and ruby is less important, as one proceeds from west to east. Sapphire's importance is more mixed but generally decreases in abundance eastward (there is very little at Bokeo relative to the zircon). Pongsak (24) has made detailed observations in color variations in the neighbouring fields in Thailand but such data are not yet available on the same scale in Cambodia.

The origin of the corundum and zircon phenocrysts (or xenocrysts) in the basalts is a matter of petrologic speculation. Many areas of basalt in Indochina are barren of these minerals, and the colour distribution of the corundums (blue sapphire, red ruby, plus green, yellow varieties) and the variation in abundances of these megacrysts of corundum, zircon, spinel, garnet and other large crysts in the groundmass of the basalts in various districts indicates great variability in their parent magmas or in the rocks with which they have mixed or reacted prior to their extrusion.

Berrange and Jobbins (3) in their examination of the gem-bearing basalts of the Pailin district came to the following general conclusion on this aspect:

> "We consider that the sapphire, ruby, zircon, spinel, magnetite, ilmenite, garnet, augite, olivine, and maybe the phlogopite and feldspar, are mineral xenocrysts that originally formed at a great depth below the surface by metamorphism and/or metasomatism on the contact of a differentiated igneous pluton with the largely amphibolitic (metavolcanic) Pailin Crystalline Complex. At a much later period in geological time the rocks of this contact zone were intruded and assimilated by basic magmas which incorporated and carried the corundums and other megacrysts up to the surface. The xenocrysts were not assimilated on account of their refractory monomineralic nature. The presence of enclaves of granite, greenstone, quartzite and peridotite (this last recorded by Lacombe (20) from the Ratanakiri gem-field) observed in the basanite, the apparently hybrid nature of the lavas, and the very marked corrosion (reminiscent of that on meteorites) seen on the surface of the rough gemstones and the pyroxene xenocrysts supports this hypothesis. A similar mode of origin has been proposed by Lacombe to account for the genesis of the zircons of Ratanakiri in particular and the gems of SE Asia generally".

In this interpretaion the alkali basalts are the vehicle by which the crysts are transported from their place of metamorphic or magmatic origin.

In the Pailin district (and other areas) it has been possible to locate the vent area in the basalt outcrop zones. Berrange and Jobbins noted that stream sediments and residual soils derived from rocks near the vent areas carry higher concentrations of gem materials than other parts of the basalt piles.

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The gem minerals are widely but sparsely distributed in the basalt parent rock and locating them in situ is infrequent. It is not practical or economic to extract them directly from the source rock. It is only when natural concentration, by eluvial or alluvial means, has taken place that mining becomes economical. Weathering of the source rock, so that it is susceptible to mechanical excavation, or its erosion and transport by stream action so that the gems are available for washing or dredging, are necessary pre-concentration agents.

There is no diagnostic characteristic of fertile versus infertile basalt fields. Some large areas of basalts have been found completely barren whereas some eruption centres of very small dimension (even too small to appear on national maps) have been very productive (e.g. Pailin).

Figures on the economic volume, industrial employment and productivity of the gem-mining industry are almost totally unknown as the industry is almost completely unregulated. Such production data as exists is anecdotal and based on estimates of worker activity, productivity and potential product value. Gem-mining activity is reported as far back as the late sixteenth century (21, p.5, p.36) and was observed and described in the Pailin area over 100 years ago, the deposits having been reportedly discovered and opened up by Burmese traders in 1874 (3). According to Workman (16), the pre – World War II production amounted to 2000 to 5000 carats per annum at Pailin. As recently as in 1974 the British geological mission which completely examined the Pailin deposits and operations (3) was unable to make any estimate of net productivity of the industry in the area, although they estimated that the industry probably supported between 100,000 and 200,000 people in that region.

In more recent times production of several hundred thousand, or possibly over one million carats per annum have been estimated by ESCAP geologists for the total annual western Cambodian production of rough gems (ESCAP, internal documents). This has been made possible by mechanized pitting and sorting operations, which were forbidden until the 1980s (1).

It is clear that regulation and control of the industry following the lines suggested by Berrange and Jobbins would allow the capture of considerable resource income by the Cambodian government. This would also lead to incentives for more efficient mining of the raw gem material, more sophisticated prospecting methods for future reserves, and better evaluation and marketing of raw and finished gems.

1. Pailin area, Battambang province

This gem district lies on the margin of the Triassic Tadeth massif, an uplifted area, where three small irruptive centres have laid down areas of basaltic flows each of the order of 2 to 3 square kilometres in area. These small intrusives appear to lie on or near a major NW-SE fault system which separates the Tadeth group (on the southwest) from rocks of the Pailin crystalline complex of presumed Pre-Cambrian age rocks. Location: 102° 36'E, 12° 50'N (Pailin town). See figure 16.

The volcanic rocks are laid down on the Tadeth group graywackes and sands and on the crystalline schists and amphibolites of the Pailin crystalline series, on an interface of Older Alluvium developed on the older rocks, which places the age of the basalts near the Lower Quaternary-Upper Pliocene boundary, and this is confirmed by fission-track ages on zircons which centre (with considerable variation) on 2.14 ± 0.20 Ma b.p. (3).

Minerals exploited at Pailin proper (Phnom Yat)are mainly sapphires, secondarily rubies, plus accessory zircons, spinels, and garnets. At the two other deposits nearby, Phnom Ko Ngoap (6 km WNW of town) and Phnom O Tang (7 km NW of the town), the rubies are more common and zircon is rare. This again indicates that the proportions of xenocrysts, and the colors of the corundum minerals, are dependent on the particular petrologic history and mineralogy of the magma pulse which produced each eruptive centre.

The deposits are eluvial, colluvial and alluvial. In the alluvial deposits, a thin basal gravel layer, covered by several metres of barren silt, is the productive zone. The deposits are worked by a variety of surficial mining techniques (trenching, cleaning, gravel-working, etc.) which, according to the authors'

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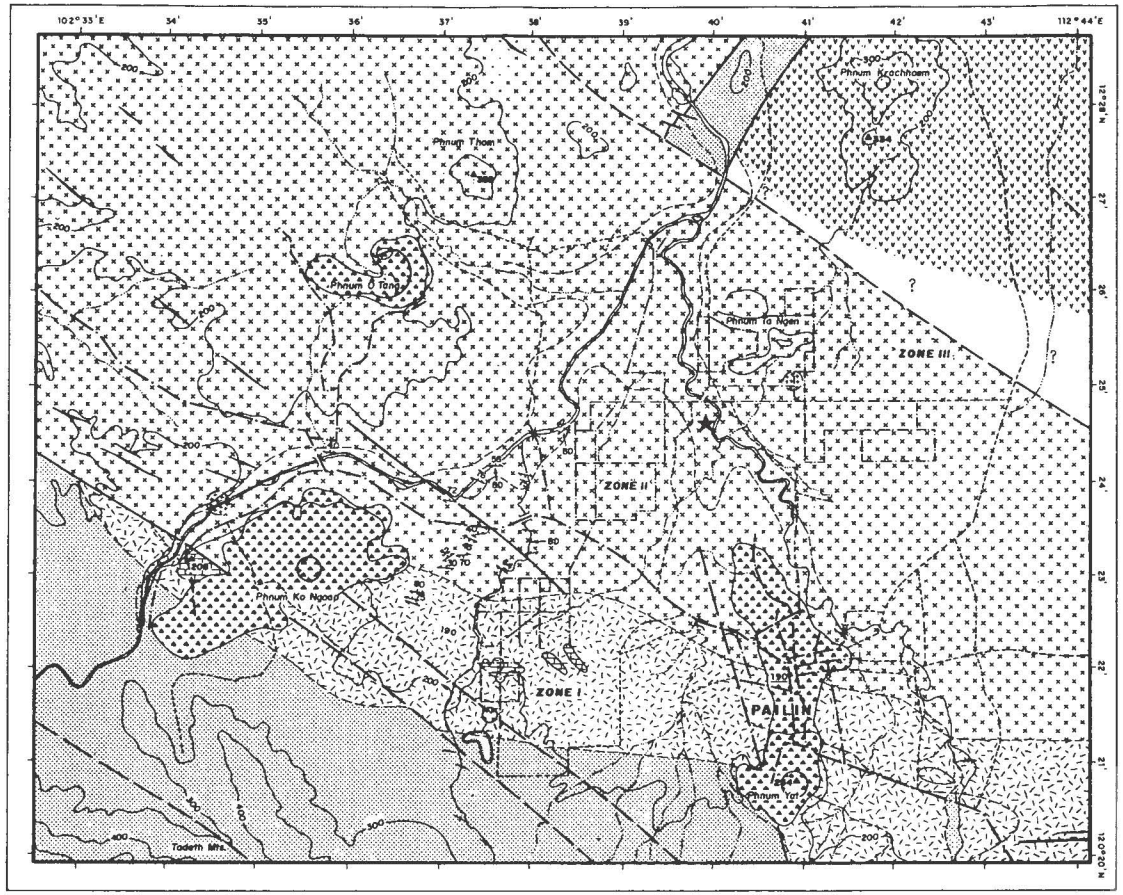

Source : J.P. Berrange. Overseas Division, Institute of Geological Sciences, London, 1974.

OTHER SYMBOLS

- Lithological contact. Defined, approxmate.
- Margin of Recent Alluvium Approximate.
- Fault zone. Defined, approximate.
- Lineament From air photographs
- Bedding
- Secondary foliation
- Remnant of volcanic pipe, vent or crater
- Iron pyrite & Arsenopyrite mineralisation
- Road : National Route No. 10 loose surface
- Road : loose surface
- Track: Suitable for tractor or cart
- Contour lines
- Spot elevation
- River
- Creek
- Bridge
- Dam : Earthen

LEGEND

Quaternary	Recent Alluvium. Mainly reworked Ancient Alluvium.	
Pliocene	Basanite Flows. Aqalcime basanite hosting megacrysts of sapphire, ruby, zircon, spinel, magnetite, ilmenite, garnet, olivine, clinopyroxene, phlogopite, feldspar.	
Triassic	Tadeth Group Greywacke sandstone with minor intercalations of black shale, siltstone, breccia & conglomerate.	
Permian	Ba Tang So Member. Crystalline limestone.	
Devono – Carboniferous	O Smoet Formation. Jasper, chert, phanite, acid volcanics, quartzite, schist, phyllite, siltstone & shale.	
Devono – Carboniferous	Phnom Krachhoem Formation Acid volcanics & subvolcanic granite.	
	Gabbro Plutons	
Precambrian	Pailin Crystalline Complex. Granodiorite, diorite, amphibolite gneiss & schist, greenstone, quartzite, synplutonic basic dykes (BD).	

Figure 16. Geological Map of the Pailin Gemfield.

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observations in 1976, were "highly inefficient and wasteful" (3). They proposed simple improved methods of extraction and concentration which have not yet been implemented. Nevertheless, Pailin was and remains today the major centre of gem production in Cambodia.

2. Samlot, Battambang province

The Samlot field is located in the administrative district of Sangkum Meanchey, about 22 km SSE of Pailin town, near Kbal Stung. Location: 102° 41'E, 12° 40'N.

In the midst of the Tadeth massif in the valley at the headwaters of the Sangke River, several small eruptive centres are distributed between Samlot village and the Thai border some 30 km to the west. Towards the south a tributary of the Sangke, the O Slap Chmar also has gem-bearing river flats and gravel bars, although no obvious eruptive centre has been located here (21).

According to Sok Leng (31), the lavas are thicker than at Pailin and clearly show flow directions northward and southward. The basalts are olivine limburgites.

The industry in the Samlot field is centred at the village of Bolanh (102° 38'E, 12° 40'N) and the field has been exploited for centuries. The minerals of interest are almost exclusively ruby, although a few sapphires and zircons and very rare spinels (ferropicotite) are also found, occurring in the alluvial terraces and in the alluvium of the present river course. Sok Leng (31) estimated that production was 80-85 per cent rubies. There is no data on production volume or value (Workman, 42).

It is of interest to note that the large basalt flow outcrops of Samlot-Tassanh, to the southeast of Bolanh, have been found to be barren and non-productive of gems.

3. Chamnop, Koh Kong province

This location is in the midst of the Cardomom highlands, 70 km ENE of the town of Koh Kongville and 9 km north of the village of Chamnop (= Chum Nop, Chamna).

In this area the country rocks are the sandstones and grits of the "Grès Superieures" (Jurassic-Cretaceous). Younger basalts (Pliocene-Quaternary) have been extruded via regional faults in this area, forming valley flows. Near Chamnop one such flow occupies a long NE-SW oriented valley. At the head of this valley (the eruptive centre of the flow) eluvial and alluvial material have yielded sapphires and zircons of good quality. Other flows in the district have reportedly produced gem-quality stones but the extent, location and potential of the deposits in this district is thus far poorly known. According to Workman, zircon dominates over sapphires in this district. (16, p.591).

4. Phnom Chnuon, Preah Vihear province

At this location, near the village of Phnom Chnuon, 17 km northeast of the town of Rovieng, a small area of basalt flows lies on the local Triassic country rocks composed of sandstone, conglomerate and less common schists. Location: 105° 15'E, 13° 26'N.

The gem minerals (zircons with lesser sapphire) (16, 31) are formed in alluvial deposits in and near the Stung Sen stream, which are derived from the nearby weathered basalts. The deposits are worked sporadically by local people using simple panning and washing methods. The source area is reportedly very small (21) so that the future potential of the deposits is very limited. An interesting feature of the basalts at Phnom Chnuon is that they contain phenocrysts and nodules of the sodium feldspar albite.

5. Phum Thmei, Preah Vihear province

At Phum Thmei, which lies 35 km NNE of Rovieng, there are two separate occurrences of gem minerals, one located two km west of the village, and the other three km to the southeast. This village is shown on some maps as Phum Kongao. Locations: 105° 15'E, 13° 39'N and 105° 17'E, 13° 38'N.

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The area is marked by a large terrane of apparently Permian andesitic flows lying above a Devonian-Carboniferous basement of folded meta-sediments. On this terrane a number of small cones and sheets of Neogene – Quaternary olivine basalts (basanites) have been extruded, the source of the gem minerals. These are mainly sapphires with secondary zircons (16, 21, 41, 42, ESCAP files), which have weathered to produce eluvial and alluvial concentrations. These have been worked in a sporadic fashion by local villagers.

6. Bokeo, Ratanakiri province

The gem deposits of Bokeo (district of Andaung Pich) lie about 30 km northeast of the provincial capital of Lomphat. The centre of the deposit area lies about 5 km southeast of the village of Bokeo and some 3 km to the south of national route 19. Location: 107° 10'E, 13° 40'N.

The area is a vast basalt plateau covering some 1,500 km^2. The basalt is the parent rock of the gem-stones, which are obtained primarily from scattered but extensive elluvial and alluvial deposits. The most widely accepted theory of origin of the gems is the same as for those of other basalt-related deposits in the Indochina peninsula (20).

By far the most common gem at Bokeo is zircon, which often occurs as large crystals of high quality. Sapphire and ruby are comparatively very rare. There are some occurrences of gem-quality garnet and spinel (picotite). Exploitation is exclusively manual, using pans, sieves and hand-operated jigs.

There are several gem-producing districts: Bo Loi, Bo Noac, Bo Tum etc. In a detailed prospection of the deposit at Bo Hei, in the Bokeo district, Lacombe (20) estimated that an area of 31 hectares carried a potential reserve of 1.5 million carats of recoverable zircon, made up of 6,100 kg of gemstones and 19,500 kg of industrial-grade zircon. It was calculated that, under the present system of working, 65 to 85 per cent of the zircon was irretrievably lost in rejected ground and tailings. Nearly all the recovered zircon was sold by the kilogramme and only 2.5 per cent of the production was cut and sold as individual stones. The official production estimate was about 1,200 to 1,500 kg of zircon per annum (1966) without any indication of value (39).

The deposits, in common with other similar eluvial/alluvial gem deposits, have always been worked by artisanal methods by local people, in a sporadic fashion. It is understood that there is no current production (1993).

Apart from their utilization as a gemstone materia, the zircon deposits of Bokeo have furnished zircon concentrates for use as industrial raw materials (see section VI.21). The large available tonnages of zircon-bearing soil and alluvium in this area, and their ease of extraction by mechanical methods indicate a potentially valuable mineral resource, if production were to be systematically managed by appropriate industrial methods.

The zircon product, being of high purity and large crystal size compared to beach-sand-derived zircon, would command a premium price on world markets for chemical and abrasive applications. The potential has however never been evaluated and the production has always been irregular.

2. Association 2. Quartz – amethyst – opal (hydrothermal) association.

This association of silica-mineral semi-precious gemstones is usually affiliated with cavity and fracture-filling deposits in rhyolitic lavas and, to a lesser degree, in granitic rocks. Production is small and is largely consumed in the handicraft and small-scale jewellery industries.

Crystal quartz is widespread in such terranes and is found most often in the deluvial-proluvial accumulation zones at the foot of slopes in rhyolitic terranes (22).

Identified districts include the following:

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1. Khone district, province of Stung Treng

Just at the point where the Mekong River crosses the frontier from Lao People's Democratic Republic and enters Cambodia, not far from the Khone Falls and Phrapeng Rapids near the village of Preah Angkoal, quartz and amethyst have been found and exploited. Location: 105° 57'E, 13° 56'N.

The amethyst and hyaline quartz occur as linings in vugs and geodes in scattered outcrops of dacites and rhyolites of presumed Cretaceous age. According to Workman (42), a small deposit has been sporadically worked by local miners who would dig out the geodes and crystals. A BRGM report refers to an effort to commercialize the deposit in the past, but gives no details (21). The interest at that time was presumed to be in the potential of the quartz for electronic applications.

2. Phnom Chi 2, Kompong Thom province

Phnom Chi, a prominent hill which rises some 1700 m above the surrounding plain, lies near the eastern border of Kompong Thom province, and 65 km NNW of Kratie. The hill comprises a granodiorite massif of Late Jurassic – Early Cretaceous age. The rocks have been locally strongly faulted and sheared. Location: 105° 59'E, 12° 54'N.

The southwest slopes of the complex are traversed by quartz-filled veins in which the axial zones often contain amethyst. Veins are sufficiently frequent that the occurrence has been referred to as a "vein field" (42, ESCAP data files). The amethyst is pale in color and of medium quality in general, but some good quality material has been also found. Hyaline quartz is also reported from this area. This occurrence was the subject of detailed prospecting in 1970 by a Cambodian Government Agency (ITSAKS) but the purpose or results of the study are not known. References: 21, 42, ESCAP data files.

3. Miscellaneous gem minerals

Garnets of gem and semi-precious quality are reported from various parts of the country, most often as accessory minerals in sapphire-ruby workings. Spinels (picotites) of gem quality, and black-grey sapphires (used to produce so-called black star sapphires) are found in the same manner.

There is a report by Vietnamese workers of pegmatitic gem minerals (beryl, topaz, garnets) occurring at Vansai and Songpong in Cambodia (22), but this cannot be confirmed.

The same authors refer to occurrences of opal, formed by post-erruptive accumulation of silica in joints and cavities in basaltic rocks. There is however no defined area of discovery or exploitation in Cambodia.

23. Ornamental and craft stones

Cambodia does not have a wide variety of ornamental and craft stones. The principal stones of this class include pagodite, jet, marbles, amethyst and minor chert and jasper occurrences.

1. Pagodite

Pagodite is a soft, rather soapy stone; it has a variable chemical composition close to that of muscovite (white mica). It occurs exclusively as an alteration product of other alumino-silicate minerals such as feldspar, nepheline etc. In texture it is invariably amorphous and massive.

Pagodite occurs in a wide variety of colours. It is easily carved and is thus used for making ornaments. The name comes from its use by the Chinese in making miniature pagodas and other religious images.

Certain materials which are hard enough to be carved are called "pagodite", even though their chemical composition may be that of a clay rather than a mica. Such deposits may also have applications in the fields of ceramics and refractories. This is true of the pagodite deposits of Cambodia and Thailand, which are described briefly below and referred to again in the section on clays and ceramic materials (Section 6.12).

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The desirable characteristics of the material are: (1) it is sufficiently soft to be carved, (2) sufficiently hard to be durable after crafting and (3) it can be polished to an attractive lustre.

In Cambodia, pagodite is exploited principally at **Trasey** (approx. 103° 45'E; 12° 00'N) in the province of Pursat, about 110 km south of Pursat itself (41, p. 132). Pagodite has been carved into statues and ornaments at Trasey for an unknown period of years. The pagodite, which is found in green, violet, beige, pink, grey and white colours, comes from the decomposition (alteration) of rhyolite lavas and tuffs. It is sometimes incorrectly known as "Pursat marble", and even the name "pagodite" is strictly speaking a misnomer. A chemical analysis of the Trasey pagodite, given by Carrel (8), shows that the sample, at least, does not equate with the standard definition of this mineral but is very close to pure kaolinite in that it contains only traces of potash and other alkalis.

	SiO_2	Al_2O_3	K_2O	H_2O
Pure muscovite.....	45.3	38.4	11.8	4.5
Trasey pagodite.....	44.2	39.4	0.4	4.2
Pure kaolinite.....	46.5	39.6	–	13.9

The material at Trasey is apparently a product of leaching and alteration of the primary rhyolites and dacites, followed by some re-silicification, which may be irregular in degree and extent.

The pagodite is located within a large rhyolitic mass in the vicinity of the Kchol granitic massif. The rhyolites are apparently of the age of the upper "gres superieures", probably Lower Cretaceous. The pagodite owes its origin to intense weathering of the rhyolites and dacites along major regional fault zones (striking N15W in general) which border the western edge of the massif in this region.

Reserves of the material are very large in relation to the quantities mined. Exploitation is sporadic. The material is mined and transported as crude blocks to workshops in Pursat and other towns where it is cut into smaller blocks, polished and sculpted usually into small art objects and adomments.

Efforts have been made to find a wider range of application for the materials, for example as architectural trim and facing stone. This has apparently not been possible owing to the general heterogenety of texture, colour and mechanical strength of the material (21).

Investigations have also been made of the suitability of the mineral as a raw material for the ceramic industry, substituting for kaolinitic clays. This is discussed in the section on Clay Minerals (section VI.12). The main mineral component of the pagodite is kaolinite with a high regularity of internal structure, according to Tran Xuan Toan (35) (Also Carrel P, 1968 Etude de gisement du pagodite de Tasey. PNUD report, Phnom Penh).

Pagodite is also found near Kompong Chnang at a location four kilometres southwest of the town in an area mapped as underlain by Carboniferous-Permian volcanics. According to Toan (35), the mineralogical characteristics are similar to the rock at Trasey. Location: 104° 33'E, 12° 13'N.

2. Jet

This black ornamental material is a form of silicified wood. It is found in a number of localities in Cambodia; the setting is usually in lignitic beds in Jurassic sandstones in the lower sections of the Gres Superieures series. There are two principal areas of occurrence:

1. Kompong Sralau district, Preah Vihear province

This occurrence is in the extreme north of the country, near the Lao People's Democratic Republic border. There are two neighbouring sites: Don Ton (105°17'E; 14° 02'N) and Anlong Chau (105° 13'E, 13° 57'N), respectively 45 km NE and 33 km ENE of Tbeng Mean Chey, the capital of the province.

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The jet occurs in a horizontal bed of partially silicified fossil wood, in the basal section of the series of Jurassic sandstones that make up the frontier hill range of Dang Rek.

The mineral in situ has the aspect of a lignitic bed, with some zones more compact and silicified than others.

There has in the past been sporadic excavation of the more compact lenses of jet, for local use as a jewel or ornamental stone. It is said that between 1930 and 1940 there was an average annual production of between 10 and 15 tons of jet (41, p. 131).

2. Kompon Smach, Koh Kong province

This occurrence lies about 10 km north of the town of Koh Kong near the west coast of Cambodia. Location: 102° 58'E, 11° 41 'N.

The geological setting is very similar to that of the Kampong Sralau occurrences described above, in the same basal section of the "Gres Superieures" sequence.

Jet was formerly produced at several small workings to the north of the town. Records indicate an annual production of 400 to 1000 kg per year during the period 1915 to 1925 (42).

3. Khone, Stung Treng province

Old reports mention a small occurrence of jet, of similar geologic setting, located near Khone, where a few tons of jet were extracted during the years 1929 to 1935 (ESCAP data files); also mentioned in the BRGM mission report of 1970 (21) (informally handwritten in report). Location: not known precisely.

3. Marbles

Marbles are abundant among the metamorphosed limestone and marls of the Paleozoic sequences in southern, western and northern Cambodia. In only a few places have they been developed as an architectural and ornamental stone source. The Perman limestones offer considerable potential, although their surfaces, where exposed, are deeply weathered and incised by karst-style erosion. In a few locations they have been exploited as a source of raw materials for ornamental and handicraft applications.

1. **Chvang 2 site.** This area is located some 20 km north of the town of Stung Treng. Location: 105° 45' to 55' E, 13° 43' to 48'N, northwest of Tharaburivat town, along the road to Chvang village. The village itself is located at 105° 47'E, 12° 43'30"N.

The occurrence consists of a number of outcrops of folded, metamorphosed dolomitic limestones of probable Devono-Carboniferous age, although some references in ESCAP files suggest that these rocks are Permian in age. Locally the metamorphism has produced a coarse-grained marble within a more extensive field of saccharoidal recrystallized limestones. Reports mention the marbles as occurring in six hills outcropping over a total length of about 12 km (21, 42, ESCAP data files). The dolomitic material has been used in small amounts as a source of raw material for the glass industry. Apart from this utilization, little examination has been made of the extent or quality of the marble rock as a potential building material or industrial mineral resource.

2. **Stung Kleach.** This showing is located 53 km WNW of the town of Kompong Speu in the province of the same name. The setting is on the eastern margin of the Cardomom Highlands at the contact of the Knong Ay (Tasal) granitic intrusive complex. Location: 104° 05'E, 11° 36'N.

The marble occurs in the form of bands and lenses of less than 1 m thickness, and often much less than 1 m. The sedimentary series in the area (Devono – Carboniferous schists and quartzites, Permian limestones, Triassic sandstones) have been strongly hornfelsed or marblized near the contacts of the intrusives. Strong skarn reaction and mobilization of the marbles is evident.

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The fine-grained marble appears to be of good quality, but the dimension of the occurrences limits its use to small handicraft and decorative uses (21).

The occurrence area lies immediately (1-2 km) to the northeast of the Knong Ay tin occurrences (q.v.).

3. **Phnom Chean** (Cheam) occurrences are situated 55 km WNW of the town of Kompong Speu, in Kompong Speu province. The showings are a few kilometres WNW of the Stung Kleach marble showings (q.v.). Location: 103° 54'E, 11° 41'N.

The general setting is the same as in the Stung Kleach area. The Permian limestones series southwest of the Kchol massif have been strongly folded and recrystallized by the same intrusion.

The marblized limestones are seen as four small outcrop areas located south of the village of Cheam (latterly known as Roliak Kang Chung). The marble has been known and used for a long time, worked occasionally by local artisanal miners, but is now reported to be abandoned.

Phosphate accumulations are found in pockets and crevasses in the karsted limestones (see Section 6.19 on phosphate occurrences) (21).

D. FUEL MINERALS

24. Lignite and Coal

(a) Coal

Numerous indications of coal are known in Cambodia, but no immediately exploitable seams have been discovered. During the period 1958-1961, a mineral survey mission from China (10) investigated reported indications of coal in the following localities:

(i) Permo-Carboniferous of Bos Dambang (Kampot province);
(ii) Triassic (Rhaetic) of Chhep, Sokream and Phnom Ker (Kompong Thom province);
(iii) Triassic (Rhaetic) of Sre Roneam and Phnom Khonpuh (Kratie province);
(iv) Liassic of the Voeun-Nhung and Phum Talat districts (Stung Treng province);
(v) Permian of Phnom Sampou (Battambang province).

Special emphasis was placed on a search for coking coal suitable for metallurgical applications. At all except one of the above mentioned localities, the irregular quality of the coal and/or the small size of the deposit ruled out any possibility of exploitation. The one exception in the deposit of Phum Talat, in the Tonle Sap valley 30 km southwest of Voeune Sai.

1. Phum Talat, Stung Treng province

The deposit occurs near the village of this name on the east side of the Tonle Sap river. Location: 106°36' E, 13° 47'N.

The coal seams occur in the Lower-Middle Jurassic part of a thick Triassic-Jurassic sequence of shales and sandstones. Structurally this formation is in the form of a shallow basin which covers an area of some 25,000 sq km in north-eastern Cambodia, mainly on the eastern side of the Mekong in Stung Treng province.

It is reported that 27 individual coal seams, mostly thin, have been identified in the Jurassic of the Stung Treng basin (42) and there are said to be measured reserves of 7 million tons, but it is not certain that this is coal in seams of exploitable thickness and geometry.

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Table 16. Analysis of Ban Talat coal

Calorific value	7,600 – 8,100 cal/gm.
Volatiles	average 30.2%
Ash	average 37.5%
Sulphur	0.71 – 3.44%
(Number of samples not known)	

Although the Chinese mission considered that the Talat seams were the one deposit for which exploitation might be feasible, the deposit is not without problems. The sedimentary assemblage is locally folded and faulted, and is intruded by stocks and dykes of dioritic rocks of the Late Jurassic-Early Cretaceous intrusive cycle. The coal has a variable continuity and thickness, rarely exceeding 0.5 m. Although the calorific value appears adequate, the ash content is high. Such coal may find application in cement manufacture and as local domestic fuel.

2. Voeune Nhung, Stung Treng province

This site is located on the Tonle Kong river, approximately 35 km WNW of Phum Talat. Location: 106° 18'E, 13° 54'N.

The geological setting here is analogous to that at Phum Talat, according to available reports. Details and dimensions are unknown.

(b) Lignite

1. Kompong Som – Koh Kong districts

A number of small showings of lignite occur along the southwest and south flanks of the Cardomom Highlands between these two towns at various sites. Cited location: 103° 34'E, 10° 34'N, 8 km SSE of Kompong Som.

The lignite, which is more or less pyritic, occurs in a large number of outcrops as small decimetre-thick discontinuous seams intercalated in the Middle Jurassic to Lower Cretaceous series of "upper sandstones" and subhorizontal continental conglomerates. No analysis results are available.

2. Kampot – Tuk Meas district, Kampot province

Showings are located 55 km east-northeast of the town of Kampot in the province of the same name; the lignite, which is more or less pyritic, occurs as a decimetre-thick discontinuous seam intercalated in the Permian fusuline marine limestone, seen in the numerous isolated phnoms (inselbergs) which occur in this area (e.g. Bos Dambang). No analysis results or dimensional data are available. Location: 104° 30'E, 10° 40'N.

3. Phnom Puck (Rumpuck), Kratie province

At a site 35 km southeast of the town of Kratie and 30 km northwest of Snoul, lignite has been encountered in drill cuttings. Location: 106° 16'E, 12° 12'N. (Also called Svay Chreak).

The lignite occurs as their beds in the Liassic red bed series of red, slightly folded marls and sandstones of the 'terrains ranges' of eastern Cambodia. The showing appears to be of no commercial promise.

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4. Rovieng district, Preah Vihear province

Lignite occurrences have been noted at several sites to the east of Rovieng around Phnom Kev. Location: 105° 10'E, 13° 20'N.

The lignite occurs as several seams of unknown thickness intercalated in a slightly folded Late Triassic (Norian to Rhaetian) continental sandstone series. The showings are apparently of no commercial promise.

References: 16, 21, 41, 42, ESCAP data files.

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VII. SUMMARY OF MINERAL DEPOSITS AND OCCURRENCES OF CAMBODIA

	No. Name	Province	Longitude, Latitude	Class	Page
	Metallic Minerals				
1. Antimony (Sb)	1 Sre Preang	Pursat	103°15'E, 12°22'N	4	
2. Base Metals (Cu, Pb, Zn)	1 Ban Chai	Ratanakiri	107°15'E, 14°09'N	4	
	2 Sam Rong	Kompong Speu	103°55'E, 11°50'N	3	
	3 O Chhung – Richomme	Mondolkiri	106°36'E, 12°42'N 106°32'E, 12°33'N	4	
	4 Phum Pring	Preah Vihear	105°04'E, 13°24'N	5	
	5 Knong Ay	Kompong Speu	104°05'E, 11°30'N	4	
	6 Lomphat	Ratanakiri	106°58'E; 13°34'N	5	
	7 Kroch Chhma	Kratie	105°46'E, 12°31'N	4	
	8 Phnom Pel	Preah Vihear	104°42'E, 13°26'N	5	
	9.1 Phnom Ke	Preah Vihear	105°10'E, 13°20'N	5	
	9.2 Phnom Sekahom	Preah Vihear	105°22'E, 13°44'N	5	
	9.3 Chamker Keu	Stung Treng	105°49'E, 13°32'N	5	
3. Bauxite (Al)	1 Battambang	Rattambang	103°02'E, 13°04'N	3	
	2 Haute Chhlong	Mondolkiri	107°12'E, 12°23'N	3	
4. Chromium (Cr)	1 Sre Peang	Pursat	103°14'E, 12°22'N	4	
5. Gold (Au)	1 Bo Sup Trup	Odder Meanchey (Siem Reap)	103°06'E, 14°04'N	1	
	2 Phnom Kambor (Bar)	Oddar Meanchey	103°45'E, 14°04'N	3	
	3 Phom Lung	Preah Vihear	104°57'E, 13°22'N	3	
	4 Rom Dey	Preah Vihear	104°57'E, 13°21'N	2	
	5 Phnom Deck	Preah Vihear	105°01'E, 13°19'N	3	
	6.1 Bo Kham	Ratanakiri	107°21'E, 13°51'N	4	
	6.2 Voeune Sai	Ratanakiri	106°45'E, 14°05'N	4	
	7 Phnom Thmar Meas	Battambang	102°21'E, 13°18'N	4	
	8 Pailin	Battambang	102°35'E, 12°52'N	4	
	9 Phnom Chi-Sud	Kompong Thom	105°39'E, 12°45'N	4	
	10 Memung	Mondolkiri	106°55'E, 12°35'N	4	
	11 Romchek	Kompong Cham	106°11'E, 11°54'N	4	
	12 Phnom Lok	Kampot	104°28'E, 10°25'N	5	

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	No. Name	Province	Longitude, Latitude	Class	Page
6. Iron (Fe)	1 Phnom Deck	Preah Vihear	105°03'E, 13°14'N	3	
	2 Phnom Deck 2	Preah Vihear	105°00'E, 13°19'N	3	
	3.1 Phnom Deck 3	Preah Vihear	105°00'E, 13°20'N	4	
	3.2 Phnom Thmak				
	3.3 Kat Keo				
	3.4 Popun				
	4 Phnom Kbol	Preah Vihear	105°20'E, 13°35'N	4	
	5 Authmor Sreal	Preah Viher	105'03'E, 13'23'N		
	6 Koh Keo	Preah Vihear	105°11'E, 13°28'N	4	
	7 Phnom Prolean	Preah Vihear	105°03'E, 13°19'N	4	
	8.1 Chhep 1	Preah Vihear	105°23'E, 13°33'N	4	
	8.2 Chhep 2	Preah Vihear	105°22'E, 13°47'N	4	
	9 Chvang	Stung Treng	105°49'E, 13°43'N	4	
	10 Sam Ang	Stung Treng	105°53'E, 13°38'N	4	
	11 An Long Chey	Stung Treng	105°52'E, 13°38'N	4	
	12 Stung Treng	Stung Treng	105°54'E, 13°31'N	4	
	13 Phnom Khbal Stung	Oddar Meanchey	104°19'E, 13°35'N	4	
	14 Phnom Rumdek	Oddar Meanchey	103°50'E, 14°05'N	5	
	15 Phnom Pours	Oddar Meanchey	104°00'E, 13°30'N	3	
	16 Phnom Prean(Preach)	Battambang	102°52'E, 12°46'N	3	
	17 Kampot	Kampot	104°40'E, 10°44'N	5	
7. Manganese (Mn)	1 Chhep 1	Preah Vihear	105°21'E, 13°26'N	2	
	2 Chhep 2	Preah Vihear	105°28'E, 13°47'N	3	
	3 Chhep 3	Preah Vihear	105°20'E, 13°46'N	5	
8. Molybdenum (Mo)	1 Phnom Basset	Kandal	104°45'E, 11°41'N	4	
	2 Phnom Den	Takeo	104°53'E, 10°36'N	2	
	3 Phnom Thong	Preah Vihear	105°00'E, 13°20'N	5	
9. Silver (Ag)	1 Bo Sup Trup	Oddar Meanchey	103°06'E, 14°04'N	5	
	2 Phnom Kambor	Oddar Meanchey	103°45'E, 14°04'N	5	
	3 Samrong	Kompong Speu	103°55'E, 11°50'N	5	
10. Tin (Sn)	1 Knong Ay	Kompong Speu	104°06'E, 11°35'N	4	
11. Tungsten (W)	1 Knong Ay	Kompong Speu	104°06'E, 11°35'N	5	

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	No.	Name	Province	Longitude, Latitude	Class	Page
Industrial Minerals and Construction Materials						
12. Clay Minerals (Cly)	1	Phnom Krom	Siem Reap	104°02'E, 13°23'N	3	
	2	Andong Srey	Kompong Chhnang	103°49'E, 12°09'N	3	
	3	Phnom Traok	Kompong Chhnang	103°38'E, 12°13'N	3	
	4	Prek Kak	Kompong Cham	105°52'E, 12°15'N	3	
	5	Trasey	Pursat	103°45'E, 12°00'N	2	
	6	Phnom Sar	Kampot	104°16'E, 10°39'N	2	
13. Dimension Stone – See text						
14. Fluorite (F)	1	Ba Phnom	Prey Veng	105°22'E, 11°14'N	5	
15. Graphite (Gr)	1	Phnom Chac Khley	Takeo	104°30'E, 10°40'N	4	
	2	Phnom Kaoh Sla	Kampot	104°15'E, 10°54'N	4	
16. Gypsum and Anthydrite – See text						
17. Heavy Mineral Sands – See text						
18. Limestone and Dolomite (Ls, Do)						
A. Limestone (Ls)	1	Battambang	Battambang	102°55;E, 13°00'N	3	
	2	Sisophon	Battambang	102°50'E, 13°45'N	3	
	3	Kampot 1	Kampot	104°16'E, 10°39'N	3	
	4	Kampot 2	Kampot	104°21'E, 10°44'N	3	
	5	Phnom Loang	Kampot	104°32'E, 10°40'N	3	
	6	Stung Treng	Stung Treng	Various		
B. Dolomite (Do)	1	Chrang	StungTreng	105°46'E, 13°44'N	3	
19. Phosphate (P)						
	1	Phnom Bak	Battambang	102°57'E, 13°36'N	4	
	2	Phnom Chung Chiang	Battambang	102°57'E, 13°36'N	4	
	3	Phnom Ban Teay Neang	Battambang	103°01'E, 13°30'N	3	
	4	Phnoms Thom and Prasat	Battambang	103°01'E, 13°28'N	4	
	5	Phnom Thoch	Battambang	103°01'E, 13°24'N	4	
	6	Phnom Sampeou	Battambang	103°06'E, 13°01'N	3	
	7	Phnom Totung	Kampot	104°32'E, 10°41'N	1	
	8	Khnom Kanlang	Kampot	104°34'E, 10°38'N	3	
	9	Phnom Cheam	Pursat	103°54'E, 11°41'N	4	
	10	Phnom Loang	Kampot	104°23'E, 10°46'N	4	
	11	Phnom Bak	Kampot	104°28'E, 10°35'N	4	

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	No.	Name	Province	Longitude, Latitude	Class	Page
20. Silica Sands (Sis)	1	Tuk Sap	Kampot	103°41'E, 10°35'N	3	
	2	O Treas	Kampot	103°37'E, 10°32'N	3	
	3	O Chrou	Kampot	103°48'E, 10°33'N	3	
	4	Phum Sralau	Koh Kong	103°42'E, 11°05'N	3	
	5	Phum Nesath	Koh Kong	103°40'E, 11°00'N	3	
	6	Phum Krapeu	Koh Kong	103°42'E, 10°56'N	3	
	7	Phum Stoeng Thma	Koh Kong	103°43'E, 10°53'N	3	
	8	Phum Thma Rong	Koh Kong	103°41'E, 10°46'N	3	
	9	Tram Khnar	Taker	104°45'E, 11°17'N	3	
21. Zirconia (Zr)	1	Bokeo	Ratanakiri	107°10'E, 13°40'N	3	
Precious and Ornamental Stones						
22. Gemstones (Gem)	I – Sapphire –Ruby –Zircon –Spinel Association					
	1	Pailin	Battambang	102°36'E, 12°50'N	3	
	2	Samlot	Battambang	102°41'E, 12°40'N	3	
	3	Chamnop	Koh Kong	103°51'E, 11°50'N	3	
	4	Phnom Chnuon	Preah Vihear	105°15'E, 13°26'N	3	
	5	Phnom Thmei 1	Preah Vihear	105°15'E, 13°39'N	4	
	6	Phnom Thmei 2	Preah Vihear	105°17'E, 13°38'N	4	
	7	Bokeo	Ratanakiri	107°10'E, 13°40'N	3	
II – Quartz – Amethyst – Opal Association						
	1	Khone	Stung treng	105°57'E, 13°56'N	4	
	2	Phnom Chi	Kompong Thom	105°39'E, 12°54'N	4	
23. Ornamental and Craft Stones (Os)						
	1	Trasey	Pursat	103°45'E, 12°00'N	2	
	2	Kompong Sralau 1	Preah Vihear	105°17'E, 14°02'N	4	
	3	Kompong Sralau 2	Preah Vihear	105°13'E, 13°57'N	4	
	4	Kompong Smach	Koh Kong	102°58'E, 11°41'N	4	
	5	Khone	Stung Treng	105°55'E, 13°55'N	5	
Fuel Minerals						
24. Coal and Lignite (C)	I – Coal (C_1)					
	1	Phum Talat	Stung Treng	106°36'E, 13°47'N	2	
	2	Voeune Nhung	Stung Treng	106°18'E, 13°54'N	4	

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	No. Name	Province	Longitude, Latitude	Class	Page
		II – Lignite (C_2)			
	1 Kompong Som – Koh Kong	Koh Kong	103°34'E, 10°34'N	5	
	2 Kampot – Tuk Meas	Kampot	104°30'E, 10°40'N	5	
	3 Phnom Puck	Kratie	106°16'E, 12°12'N	5	
	4 Rovieng	Preah Vihear	105°10'E, 13°20'N	5	

Class: 1 – Mine, present or past

2 – Deposit, tonnage known

3 – Deposit, tonnage not known

4 – Prospect site or occurrence

5 – Mineralogical interest only

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Annex

LIST OF THE PAPERS PRESENTED AT THE FIRST EXPERT WORKING GROUP MEETING ON ASSESSMENT OF MINERAL RESOURCES OF CAMBODIA PHNOM PENH, 5 – 6 MAY 1993

1. Present State of Development of the Mineral Sector of Cambodia – Souv Chiv Kun, Director of the Department of Geology and Mines.
2. General Background Data on Mines and Quarries of Cambodia - Presented by Jon L. Rau, United Nations Transitional Authority of Cambodia (UNTAC).
3. Overview of Geology and Tectonic Setting of Cambodia – Lynn R. Moxham, ESCAP, Natural Resources Division.
4. Metallogenic Provinces and Mineral Potential of Cambodia – Lynn R. Moxham, ESCAP, Natural Resources Division.
5. Iron and Bauxite Deposits of Cambodia – Lynn R. Moxham, ESCAP.
6. Molybdenum Resources at Phnom Basset and other locations of Cambodia – A.N. Ejov, Head, Department of Geology, Phnom Penh Institute of Technology.
7. Research on Gold Occurrences in Cambodia – Phok Salern and Khun Panhasith, DGM.
8. Research on Clay - Kaolin Deposits of Cambodia – Yin Sidet, DGM.
9. Non-Metallic Minerals in Cambodia – A.N. Ejov, Phnom Penh Institute of Technology.
10. General Characteristics of the Phosphorites in the Battambang Area –Thoch Saophan, DGM.
11. Gemstones in Cambodia – Thach Soval Say, DGM.
12. The Source and Precious Stone Resources in Cambodia – Sok Leng, DGM.
13. Kaolinite in Cambodia –Thach Soval Say, DGM.
14. Phosphorite Deposits at Tuk Meas – Jon L. Rau, UNTAC.
15. Sand and Gravel Resources – Khun Sokhunthea, Phom Penh Institute of Technology.
16. Environment and Sustainable Development in Cambodia – Michael Ward, UNTAC.

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LIST OF PAPERS PRESENTED AT THE SECOND EXPERT WORKING GROUP MEETING ON ASSESSMENT OF HYDROCARBON POTENTIAL OF CAMBODIA, PHNOM PENH, 17–18 JUNE 1993

1. Present State of Development of Hydrocarbon Sector (Status of Exploration, PetroleumAgreements and Potential) – Ith Praing, Vice-Minister of Industry of Cambodia.

2. Khmer Basin Offshore and Onshore – Men Den, DGM.

3. Geology of the Sedimentary Basins of Cambodia – Mak Ngorn Ly, DGM.

4. Geology and Potential of Selected Parts of the Offshore Areas of Cambodia – Barrington P. Rogers, Enterprise Oil Ltd., United Kingdom.

5. Geology and Potential of Selected Parts of the Offshore Areas of Cambodia – T. Kude, S. Ikeda and Tugua Kobori, Cambodian Petroleum Exploration (Campex), Singapore.

6. Geology and Potential of Selected Parts of the Offshore Areas of Cambodia – Frank Inoune, Premier Oil Pacific Ltd.

7. Hydrocarbon Assessment of the Gulf of Thailand – Prisdapunt Pojanapreecha, Petroleum Authority of Thailand – PTT Exploration and Public Production Co Ltd.

8. Hydrocarbon Resources Assessment – Steinar Maehle, STATOIL (Bangkok).

9. Assessment of the Hydrocarbon Resources Potential of Cambodia – Ing Kieth, Civil Engineer, ENPC.

10. The Role of Hydrocarbon Exploration in the National Economy – Steven Torok, ESCAP, Natural Resources Division.

11. Some Important Issues of Mining Regulations and Mining Law in Cambodia – Jack Garnett, United Nations Development Programme (UNDP)/World Bank Consultant.

12. Coal Resources of Cambodia – Alexander Ejov, Head, Department of Geology, Phnom Penh Institute of Technology.

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ห้ามทำซ้ำหรือดัดแปลงและแก้ไขโดยไม่ได้รับอนุญาต

LIST OF THE PAPERS PRESENTED AT THE THIRD EXPERT WORKING GROUP MEETING ON THE DEVELOPMENT OF CAMBODIA'S MINERAL AND HYDROCARBON RESOURCES: INSTITUTIONAL, LEGAL AND ENVIRONMENTAL ISSUES PHNOM PENH, 21-22 JULY 1993

1. Hydrocarbon, Energy and Mineral Development of Cambodia – Ith Praing, Vice Minister of Energy and Mines of Cambodia.

2. Some Aspects of the Economics of Natural Resources Development of Cambodia – Michael Ward, Senior Economist, World Bank/UNTAC.

3. Issues Related to Regulation and Administration of the Mineral Sector in Cambodia – Jack Garnett,UNDP Consultant.

4. The Gemstone Industry: The Sri Lankan Example – Dulip Jayawardena, Economic Affairs Officer,ESCAP Natural Resourves Division.

5. Mineral Development Policies for Cambodia: Guidelines for Legislation, Regulation and Administration – Paper for discussion.

6. Issues Related to Energy Planning in Cambodia – Garth Harris, World Bank Consultant.

7. Issues Related to the Environment with Regard to the Mining Sector – Brief statement by the Secretariat.

8. International Petroleum Agreements: A View from the Private Sector – Barrington Rogers, Enterprise Oil Ltd. (London).

9. Learning from the Thai Experience – Chiridacha Phungsunthorn, Attorney, Baker & McKenzie.

10. Production Sharing Contracts: Examples from Indonesia, Thailand, Australia, Viet Nam, Lao People's Democratic Republic, Myanmar and Malaysia – Steve Torok, Economic Affairs Officer, ESCAP Natural Resources Division.

11. International Petroleum Agreements: A view from the State – Mbella-Ikomi Ngongi, UNTAC Legal Officer.

12. Training for Geoscientists from Developing Countries – Mike Katz, Associate Director, Key Centre of Mines, Australia, University of New South Wales.

13. Assessment of Training Needs in Cambodia: (i) Department of Geology and Mines, (ii) Department of Energy, (iii) Phnom Penh Institute of Technology and (iv) New Training Centre (Petroleum Technology and Petroleum Geology) – Discussion paper.

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SOMSAK POTISAT

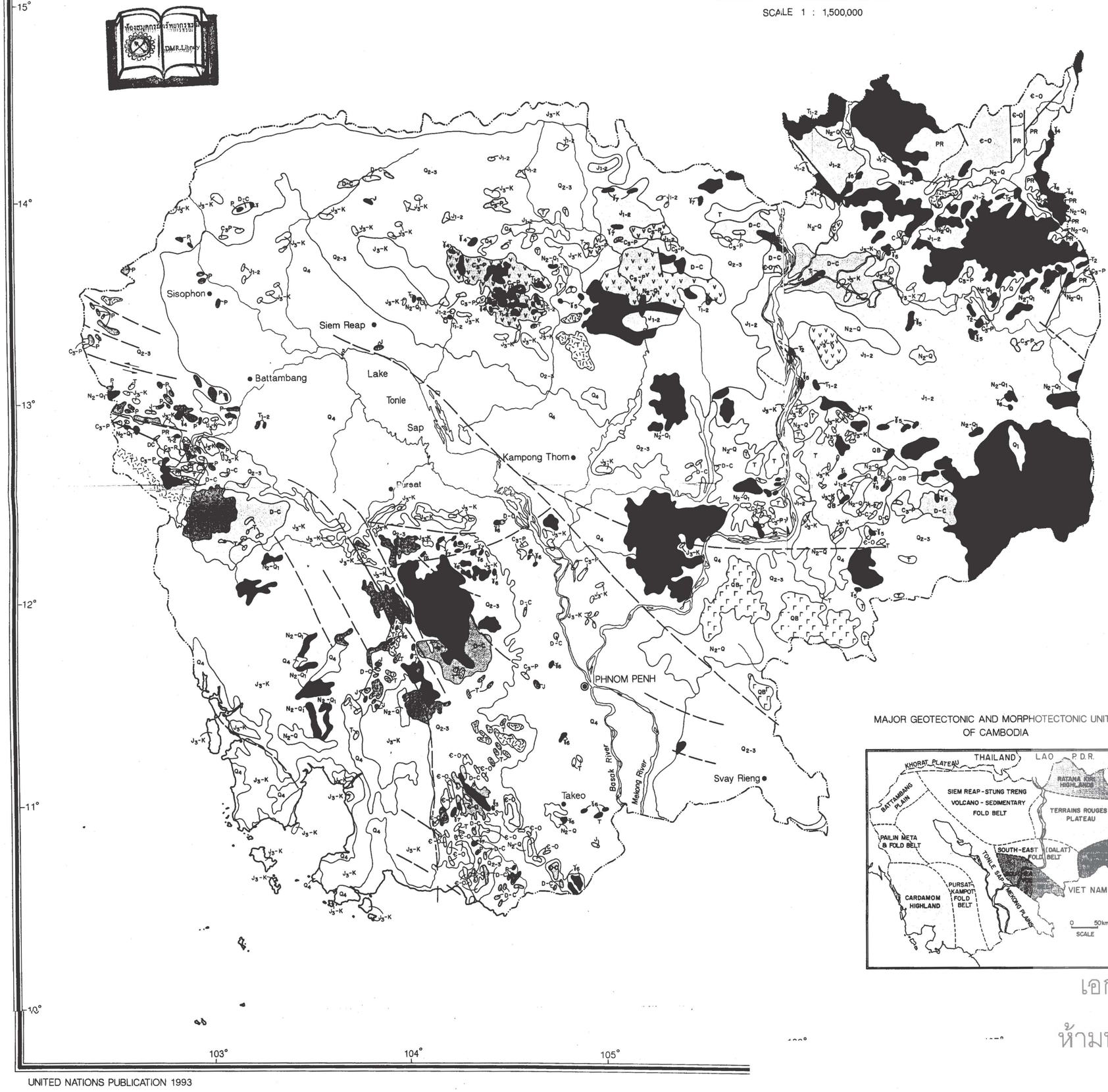
ATLAS OF MINERAL RESOURCES
OF THE ESCAP REGION

GEOLOGICAL MAP OF CAMBODIA

Published by the United Nations Economic and Social Commission for Asia and the Pacific in cooperation with the Department of Geology and Mines of Cambodia

0 50 100 km

SCALE 1 : 1,500,000



EXPLANATION

STRATIGRAPHIC UNITS

Q_4	Holocene	Silts, sands, organic material. Tonlesap Formation.
Q_{2-3}	Quaternary (Middle-Upper Pleistocene)	Fluvial sediments of Battambang Formation.
QB	Quaternary (Middle)	Basalts, alkaline olivine affinity.
Q_2	Quaternary (Middle Pleistocene)	Sands, grits, clays, mainly of fluvial origin.
Q_1	Quaternary (Lower Pleistocene)	Sands, silts, clays of fluvial paralic and marine origin.
N_2-Q_1	Pliocene-Pleistocene	Basalts. mainly tholeiitic.
N_2-Q	Upper Neogene-Quaternary	Mainly Pliocene-Pleistocene. Sands, silts, gravels, claystone; the "Older Alluvium" laterites at top of Barnieu Formation in eastern Cambodia.
J_3-K	Upper Jurassic-Cretaceous	"Gres Superieures." East of Mekong: Cato Formation: sandstone, andesite, dacite, tuffs. Western regions: Cardamom and Phuquoc Formations: red conglomerates, sandstone, siltstone, thin coal beds, marls.
J_{1-2}	Jurassic (Lower-Middle)	"Terrain rouge" continental red beds. Marine calcareous shales, marine silts, sands grading upward to continental shales, sandstones. Bandon Formation.
J	Jurassic (Undiff)	Acid volcanics of Jurassic(?) age in Western regions.
T_2	Triassic (Middle)	Greywacke, sandstone, limestone plus dacite and tuffs; Rovieng Formation
T_{1-2}	Triassic (Lower-Middle)	Conglomerate, sandstone, shale; some rhyolite, dacite, tuffs.
T_1	Triassic (Lower)	Calcareous shales, gray shales and sandstones of mainly marine origin.
T	Triassic (Undiff)	Siltstones, marls, shales.sandstones, party continental or paralic. Western Cambodia: conglomerates, sandstones, mudstones plus volcanoclastics. Eastern Cambodia: marine facies. green shales, sandstones, limestones. More continental in later Triassic.
P	Permian	White-gray oolitic and organo-detrital limestones in Pursat, Sisophon and Kampot regions; lesser sandstones and marls. Limestone. cherty shales in Svay Rieng region.
C_3-P	Upper Carboniferous-Permian	Fossiliferous limestones in western regions with andesitic to rhyolitic volcanism in Later Permian. Intermediate effusive rocks in Preah Vihear and Siem Reap provinces.
C	Carboniferous	White dolomitic limestones northeast of Stung Treng plus andesite, rhyolite, tuffs.
D-C	Devonian-Carboniferous	Shales, sandstones, cherts, marls; lesser limestone mainly in lower Carboniferous.
€-O	Cambro-Ordovician	Also extends to Silurian. White-pink quartzites of Pailin area. Quartzites and schists of Kampot province. Spotted schists, mica schists, quartzites in Ratanakiri province.
€	Cambrian	Argillites (Stung Treng), fossiliferous; sandy shales.
PR	Proterozoic	Quartzo-feldspathic gneisses, migmatites, schist; crystalline limestones and schists of Bokham Series.

INTRUSIVE UNITS

- γ_7 – Gabbro, norite, olivine gabbro, pyroxenite, diorite in Western Cambodia, "Bam nak" gabbros, diorites of Pursat-Kompong Chhnong districts (Late Cretaceous- Early Tertiary).
- γ_6 – Diorite, granite, monzonite, of "Knong Ay Kchon" type (Cretaceous-Early Tertiary).
- γ_5 – Diorite, granodiorite, hornblende-biotite granites of "Dinhquan Complex" (Late Jurassic-Cretaceous).
- γ_4 – Gabbro, diorite, granodiorite of "Phnomtasson" type (Triassic).
- γ_3 – Garnet cordierite two-mica granites, "Preas" granite (Early-Mid Paleozoic).

(Units adapted from Phan Cu Tien, 1988)

VOLCANIC ROCKS

Basic Basalts.

Intermediate. Andesites, dacites.

Acid and Intermediate. Rhyolites, dacites.

Age of volcanic rocks is shown mainly by the appropriate colour assigned for corresponding age of sedimentary rocks or its shade. Grey colour of Neogene-Quaternary basalts is predetermined by the same colour pattern of basalts in Viet Nam.

OTHER SYMBOLS

GEOLOGICAL BOUNDARIES

a. Established
b. Conjectural

FAULTS AND THRUSTS

a. Established
b. Conjectural

MAJOR GEOTECTONIC AND MORPHOTECTONIC UNITS OF CAMBODIA

Data Source: 1. Geological Map of Kampuchea, Laos and Viet Nam, Scale 1:1,000,000, Hanoi, 1988. Compiler-in-Chief: Phan Cu Tien.
2. Carte Miniere Du Cambodge, Scale 1:1,000,000 Bureau De Recherches Geologiques Et Minieres, D.R.M.E/A.O Mission, Cambodge, 1970.

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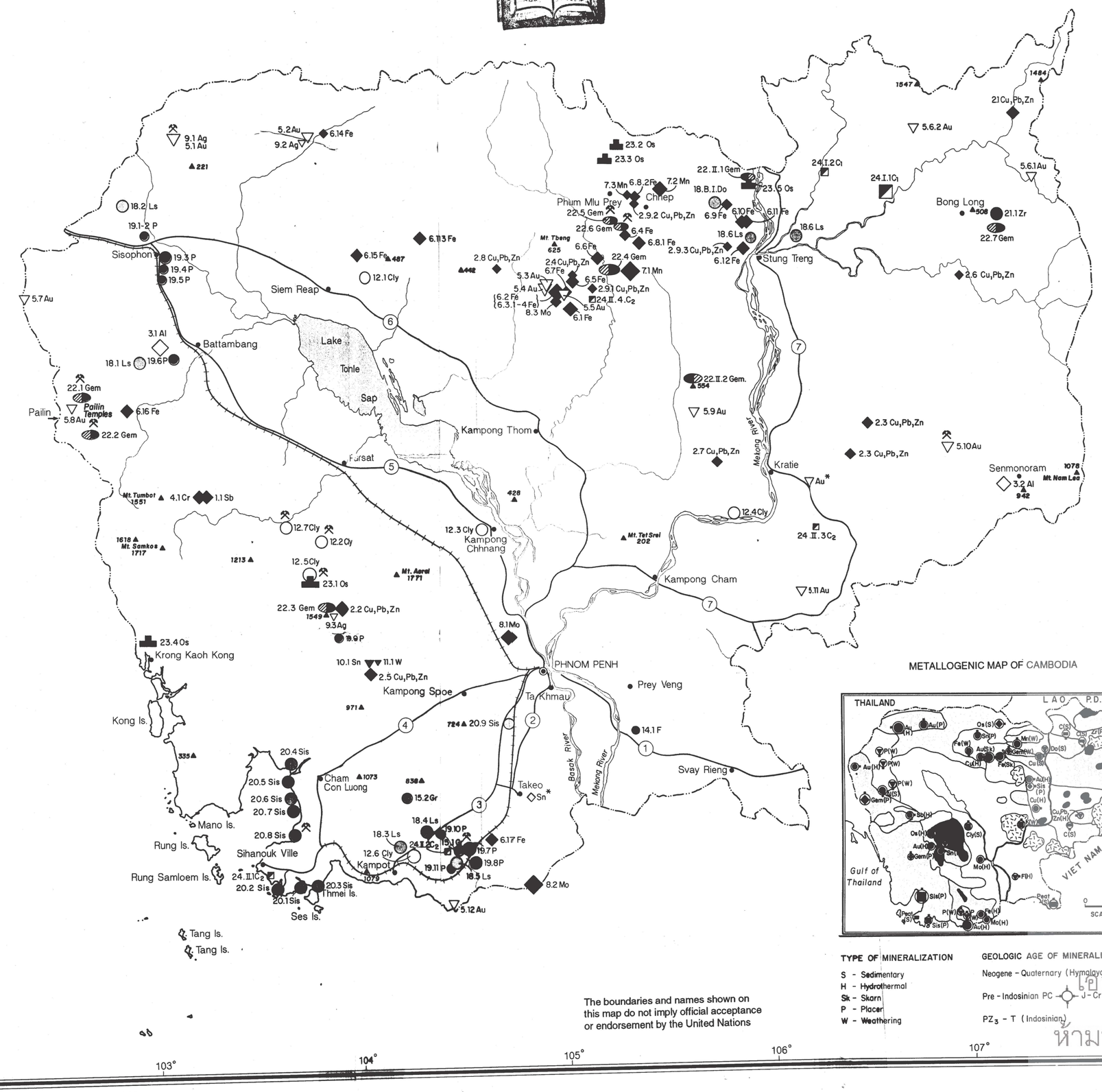


EXPLANATION

METALLIC MINERALS

1. Antimony	Sb	◆
2. Base Metals	Cu,Pb,Zn	◆
3. Bauxite	Al	◇
4. Chromium	Cr	◆
5. Gold	Au	▽
6. Iron	Fe	◆
7. Manganese	Mn	◆
8. Molybdenum	Mo	◆
9. Silver	Ag	▽
10. Tin	Sn	◆
11. Tungsten	W	◆

INDUSTRIAL MINERALS AND CONSTRUCTION MATERIALS

12. Clay Minerals	Cly	○
13. Dimension Stone	Ds	■
14. Fluorite	F	●
15. Graphite	Gr	●
16. Gypsum and Anhydrite	Gy	○
17. Heavy Minerals Sands	Mis	▽
18. Limestone and Dolomite	Ls.Do	○
19. Phosphate	P	●
20. Silica Sands	Sis	●
21. Zircon	Zr	●

PRECIOUS AND ORNAMENTAL STONES

22. Gemstones	Gem	
I. Sapphire-Ruby-Zircon-Spinel Association		⬬
II. Quartz-Amethyst-Opal Association		⬬
23. Ornamental and Craft Stones	Os	▟

FUEL MINERALS

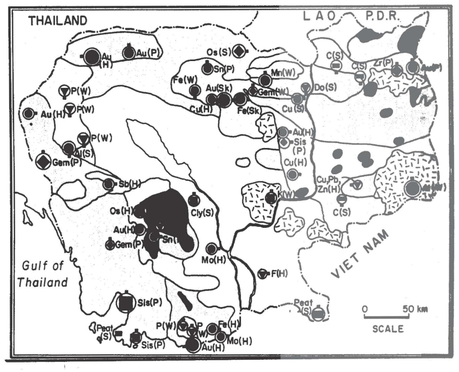
24. Coal and Lignite	C	
I. Coal	IC1	◪
II. Lignite	IC2	◪

RESOURCES CLASSIFICATION

Operating and abandoned mines	⚒ ⚒
Deposit (tonnage known)	□
Deposit (tonnage unknown)	□
Prospect or Occurrance	□
Mineralogical interest only	□

(Number beside symbols refer to text section and occurrence list.)
(* Recent discoveries not described in the text.)

METALLOGENIC MAP OF CAMBODIA



TYPE OF MINERALIZATION

S - Sedimentary
H - Hydrothermal
Sk - Skarn
P - Placer
W - Weathering

GEOLOGIC AGE OF MINERALIZATION

Neogene - Quaternary (Hymalayan)
Pre - Indosinian PC - J - Cr (Kimmerian)
PZ_3 - T (Indosinian)

METALLOGENIC EPOCHS		MAJOR MINERALIZATION
Cenozoic Epoch	(Neogene) Quaternary alluvial sediments. Late Neogene-Quaternary basalts.	Placer Au, Sn and gems; lateritic bauxite, manganese, kaolinite, phosphate; sedimentary glass sands, peat, clays, construction materials
Mezozoic Epoch	Upper Jurassic-Lower Cretaceous continental red sediments and rhyolite. Upper Triassic (Norian-Rhaetian) - Middle Jurassic continental sediments with intercalations of marine sediments, rhyolite and dacite.	Hydrothermal Au, Sn, W, Mo, Cu, Pb, Zn, F, Fe (Pyrite); Skarn Fe (magnetite), Cu, Pb, Zn, Au, Bi, Te, Ba, Mo, F.
Indosinian Epoch	Triassic terrigerous sediments, rhyolite. Carboniferous-Permian limestone, dolomite.	Sedimentary iron and bauxite (diasporic) copper, coal, limestone, gypsum.
Pre-Indosinian folded basement	Lower-Middle Paleozoic terrigenous and siliceous sediments and rhyolite (in the South). Metasediments and crystalline rocks.	Small vein gold deposits, mica, chromium, graphite.
	Granites	

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